

TEST Shiny GOE Monitoring DataViz

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Load Libraries

```
library(shiny)
library(tidyverse)

## -- Attaching packages ----- tidyverse 1.3.2 --
## v ggplot2 3.4.0      v purrr  0.3.5
## v tibble  3.1.8      v dplyr  1.0.10
## v tidyr   1.2.1      v stringr 1.4.1
## v readr   2.1.3      v forcats 0.5.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()

library(plotly)

##
## Attaching package: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##   last_plot
##
## The following object is masked from 'package:stats':
##
##   filter
##
## The following object is masked from 'package:graphics':
##
##   layout

library(Diagrammer)
```

Data

- Note Data is in `data_raw/TEST-GOE-Monitor.xlsx` file, with one worksheet for Data, and one worksheet for Site info
- Save each worksheet to 2 separate csv files `data/SiteDetails-TEST-GOE-Monitor.csv` & `data/SiteData-TEST-GOE-Monitor.csv`

Load Data

- Data Cleaning to ensure no space at beginning of any text values and column headings Note: data/SiteDetails-TEST-GOE-Monitor.csv has space in front of values NEED TO change data_raw/TEST-GOE-Monitor.xlsx excel file first, then resave as csv

```
# Site Data
goe_eccc <- read.csv("data/SiteData-TEST-GOE-Monitor.csv", header = TRUE, sep = ",")
# Site Details
Sites_eccc <- read.csv("data/SiteDetails-TEST-GOE-Monitor.csv", header = TRUE, sep = ",")
```

Data Discovery

Column Names Data

```
colnames(goe_eccc)

## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"
```

Column Names Sites

```
colnames(Sites_eccc)

## [1] "Subregion"          "Site"          "Land.Manager"
## [4] "Main.Restoration.Type" "Restoration.Intensity" "Area.ha"
## [7] "Lat"                "Lng"
```

Data Manipulation

proportion to Percentage_of_native_species

```
# using scales creates % character instead of number
# Percentage_of_native_species <- scales::percent(goe_eccc$Proportion_of_native_species)
# Percentage_of_native_species
```

```
Percentage_ns <- goe_eccc$Proportion_of_native_species * 100
Percentage_ns
```

```
## [1] 94 55 77 66 75 53 45 65 62 58 45 67 62 61 32 64 49 41 47 34 34 44 54 37
```

add Percentage_ns data to column

Click to see results of data

```
# https://sparkbyexamples.com/r-programming/add-column-to-dataframe-in-r/
# goe_eccc_1 <- cbind(goe_eccc, Percentage_of_native_species)
# goe_eccc_1
```

```
goe_eccc <- cbind(goe_eccc, Percentage_ns)
goe_eccc
```

##	Subregion	Site	Proportion_of_native_species		
## 1	Gulf Islands	Anniversary Island	0.94		
## 2	Gulf Islands	AVNR	0.55		
## 3	Gulf Islands	Brackman Island	0.77		
## 4	Gulf Islands	Crows Nest	0.66		
## 5	Gulf Islands	Dock Islet N	0.75		
## 6	Gulf Islands	Eagle Islet	0.53		
## 7	Gulf Islands	Mt Maxwell	0.45		
## 8	Gulf Islands	Owl Islet	0.65		
## 9	Gulf Islands	Reay Islet	0.62		
## 10	Gulf Islands	Retreat Island	0.58		
## 11	Gulf Islands	Rum Islet	0.45		
## 12	Gulf Islands	SISCENEM	0.67		
## 13	Saanich Peninsula	Bear Hill Park	0.62		
## 14	Saanich Peninsula	Camas Hill	0.61		
## 15	Saanich Peninsula	Gonzales Hill Park	0.32		
## 16	Saanich Peninsula	Gore Park	0.64		
## 17	Saanich Peninsula	Highrock Park	0.49		
## 18	Saanich Peninsula	Mill Hill Park	0.41		
## 19	Saanich Peninsula	Oak Haven Park	0.47		
## 20	Saanich Peninsula	Scafe Hill	0.34		
## 21	Saanich Peninsula	Seymour Hill	0.34		
## 22	Saanich Peninsula	Stewart Hill	0.44		
## 23	Saanich Peninsula	Trial Island	0.54		
## 24	Saanich Peninsula	Uplands Park	0.37		
##	Cultural_species_richness	Exotic_species	Trampling	Herbivory	Composite_Index
## 1	2.71	0.17	5.00	1.14	1.8
## 2	1.30	57.19	4.70	36.50	0.6
## 3	2.80	11.17	5.40	0.00	2.0
## 4	1.07	31.51	1.64	26.71	1.0
## 5	3.60	8.80	12.20	0.10	1.6
## 6	0.80	28.25	16.60	0.00	0.8
## 7	0.42	64.30	10.00	6.00	0.2
## 8	2.00	27.72	10.67	0.67	1.4
## 9	3.60	11.51	11.00	8.00	1.0
## 10	2.33	33.00	32.33	0.00	1.0
## 11	1.00	73.21	22.20	70.80	0.0
## 12	3.40	16.12	12.80	0.50	1.6
## 13	1.86	24.46	7.57	9.50	1.0
## 14	2.43	27.19	7.71	4.71	1.4
## 15	1.00	83.16	10.40	1.20	0.4
## 16	4.40	7.35	6.20	1.60	1.6
## 17	1.60	37.10	8.20	1.20	1.0
## 18	1.56	46.33	14.33	25.67	0.4
## 19	1.50	34.60	1.88	12.75	1.0
## 20	1.40	55.35	10.20	1.40	0.4
## 21	2.20	65.99	9.20	0.20	0.8
## 22	1.00	79.56	4.80	0.80	0.6
## 23	3.57	30.73	2.00	0.14	1.6
## 24	1.45	49.38	2.36	5.82	1.0
##	Year	Percentage_ns			
## 1	2023	94			
## 2	2023	55			
## 3	2023	77			

```
## 4 2023 66
## 5 2023 75
## 6 2023 53
## 7 2023 45
## 8 2023 65
## 9 2023 62
## 10 2023 58
## 11 2023 45
## 12 2023 67
## 13 2023 62
## 14 2023 61
## 15 2023 32
## 16 2023 64
## 17 2023 49
## 18 2023 41
## 19 2023 47
## 20 2023 34
## 21 2023 34
## 22 2023 44
## 23 2023 54
## 24 2023 37
```

Check data structure of new column

```
str(goe_eccc)
```

```
## 'data.frame': 24 obs. of 10 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness : num 2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species : num 0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling : num 5 4.7 5.4 1.64 12.2 ...
## $ Herbivory : num 1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index : num 1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns : num 94 55 77 66 75 53 45 65 62 58 ...
```

Data Structure Data

- make sure each column is same data type in both data file

```
str(goe_eccc)
```

```
## 'data.frame': 24 obs. of 10 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness : num 2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species : num 0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling : num 5 4.7 5.4 1.64 12.2 ...
## $ Herbivory : num 1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index : num 1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns : num 94 55 77 66 75 53 45 65 62 58 ...
```

Data Structure Sites

- make sure each column is same data type

```
str(Sites_eccc)
```

```
## 'data.frame':   24 obs. of  8 variables:
## $ Subregion      : chr  "Saanich Peninsula" "Saanich Peninsula" "Saanich Peninsula" "Saanich Peninsula" ...
## $ Site           : chr  "Gore Park" "Highrock Park" "Mill Hill Park" "Oak Haven Park" ...
## $ Land.Manager   : chr  "Central Saanich" "City of Esquimalt" "CRD" "CRD" ...
## $ Main.Restoration.Type: chr  "invasive removal" "invasive removal" "invasive removal" "invasive removal" ...
## $ Restoration.Intensity: chr  "moderate" "low" "high" "high" ...
## $ Area.ha        : num  0.74 2.88 16.66 3.16 3.36 ...
## $ Lat            : num  48.5 48.4 48.5 48.6 48.5 ...
## $ Lng            : num  -123 -123 -123 -123 -123 ...
```

Combine Dataframes

- first make sure they have the same number of rows

```
length(unique(goe_eccc$Site))
```

```
## [1] 24
```

```
length(unique(Sites_eccc$Site))
```

```
## [1] 24
```

Arrange Order

- Site details Site names are in different order than Site Data, so put each site in alphabetical order first

```
# tidyverse
# https://stackoverflow.com/questions/68989228/sort-dataframe-by-column-value-r
Sites_eccc_1 <- Sites_eccc %>% arrange(Sites_eccc$Site)
goe_eccc_1 <- goe_eccc %>% arrange(goe_eccc$Site)
```

cbind arranged df works

- (merge doesn't work, leaves columns empty)

```
goe_eccc_sites <- cbind(Sites_eccc_1, goe_eccc_1)
names(goe_eccc_sites)
```

```
## [1] "Subregion"      "Site"
## [3] "Land.Manager"   "Main.Restoration.Type"
## [5] "Restoration.Intensity" "Area.ha"
## [7] "Lat"            "Lng"
## [9] "Subregion"      "Site"
## [11] "Proportion_of_native_species" "Cultural_species_richness"
## [13] "Exotic_species"    "Trampling"
## [15] "Herbivory"         "Composite_Index"
## [17] "Year"              "Percentage_ns"
```

remove duplicate columns

- now there are 2 columns with same name for, Subregion, Site

```
# https://stackoverflow.com/questions/7072159/how-do-you-remove-columns-from-a-data-frame
goe_eccc_sites_1 <- goe_eccc_sites[c(-9,-10)]
names(goe_eccc_sites_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Land.Manager"       "Main.Restoration.Type"
## [5] "Restoration.Intensity" "Area.ha"
## [7] "Lat"                "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Tramplng"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"
```

Rename columns with . using base R

```
names(goe_eccc_sites_1)[names(goe_eccc_sites_1) == "Land.Manager"] <- "Land_Manager"
names(goe_eccc_sites_1)[names(goe_eccc_sites_1) == "Main.Restoration.Type"] <- "Main_Restoration_Type"
names(goe_eccc_sites_1)[names(goe_eccc_sites_1) == "Restoration.Intensity"] <- "Restoration_Intensity"
names(goe_eccc_sites_1)[names(goe_eccc_sites_1) == "Area.ha"] <- "Area_ha"
```

```
names(goe_eccc_sites_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Land_Manager"       "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Tramplng"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"
```

Create and save new csv file with both datasets

```
write.csv(goe_eccc_sites_1, "data/goe_eccc_all_site_data.csv", row.names = FALSE)
```

Randomly Change Numbers to Create New Data Set for Year 2

```
goe_eccc_random_2024 <- read.csv("data/TEST-GOE-Monitor-randomChange-diffYear-TEST.csv", header = TRUE,
```

```
library(dplyr)
```

```
# https://stackoverflow.com/questions/73773734/adding-a-random-value-to-each-value-in-the-data-condition
```

```
# this adds some values that are negative
```

```
# goe_eccc_random_2024 <- goe_eccc_random_2024 %>%
```

```
# mutate(across(where(is.numeric), \(x) x + rnorm(n(), mean = 0, sd = 0.5 * mean(x))))
```

```
# need to avoid one column year
```

```
str(goe_eccc_random_2024)
```

```
## 'data.frame': 24 obs. of 9 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness : num 2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species : num 0.17 57.19 11.17 31.51 8.8 ...
```



```
## $ Trampling      : num  5 4.7 5.4 1.64 12.2 ...
## $ Herbivory      : num  1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index : num  1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year           : int   2024 2024 2024 2024 2024 2024 2024 2024 2024 2024 ...
```

```
# change year to date for the moment
```

```
# goe_eccc_random_2024$Year1 <- goe_eccc_random_2024$Year
```

```
# https://stackoverflow.com/questions/69555486/how-can-i-convert-a-column-of-date-time-data-from-numeri
```

```
goe_eccc_random_2024$Year <- as.character(goe_eccc_random_2024$Year)
str(goe_eccc_random_2024)
```

```
## 'data.frame': 24 obs. of 9 variables:
```

```
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site           : chr  "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num  0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness  : num  2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species   : num  0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling        : num  5 4.7 5.4 1.64 12.2 ...
## $ Herbivory        : num  1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index   : num  1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year              : chr  "2024" "2024" "2024" "2024" ...
```

```
goe_eccc_random_2024[] <- lapply(goe_eccc_random_2024, function(x) {
  if(!is.numeric(x)) x else x + rnorm(length(x), 0, 0.1 * mean(x))
})
goe_eccc_random_2024
```

```
##           Subregion      Site Proportion_of_native_species
## 1      Gulf Islands Anniversary Island      0.9467108
## 2      Gulf Islands      AVNR      0.5529387
## 3      Gulf Islands      Brackman Island      0.8094509
## 4      Gulf Islands      Crows Nest      0.7065883
## 5      Gulf Islands      Dock Islet N      0.7259031
## 6      Gulf Islands      Eagle Islet      0.5255178
## 7      Gulf Islands      Mt Maxwell      0.4622967
## 8      Gulf Islands      Owl Islet      0.5586031
## 9      Gulf Islands      Reay Islet      0.5259315
## 10     Gulf Islands      Retreat Island      0.6490815
## 11     Gulf Islands      Rum Islet      0.5089213
## 12     Gulf Islands      SISCENEM      0.6836979
## 13 Saanich Peninsula      Bear Hill Park      0.6340467
## 14 Saanich Peninsula      Camas Hill      0.5955381
## 15 Saanich Peninsula      Gonzales Hill Park      0.2899193
## 16 Saanich Peninsula      Gore Park      0.6239178
## 17 Saanich Peninsula      Highrock Park      0.4292146
## 18 Saanich Peninsula      Mill Hill Park      0.4136627
## 19 Saanich Peninsula      Oak Haven Park      0.4928555
## 20 Saanich Peninsula      Scafe Hill      0.2750271
## 21 Saanich Peninsula      Seymour Hill      0.3029491
## 22 Saanich Peninsula      Stewart Hill      0.4883193
## 23 Saanich Peninsula      Trial Island      0.5366848
## 24 Saanich Peninsula      Uplands Park      0.3923356
## Cultural_species_richness Exotic_species Trampling Herbivory
## 1      2.9564564      -1.308547 5.0958523 1.07706746
```

```
## 2      1.3218616      59.286231  4.1274813 35.85365527
## 3      2.5815140      11.221791  5.7315242 -0.03606702
## 4      1.4649281      35.607299  0.8625730 25.58508055
## 5      3.3520352       2.399800 11.8794176  0.68259127
## 6      1.1832564      29.260787 15.0072425  0.63690178
## 7      0.4751135      66.147552 10.0210570  5.15457987
## 8      2.1102128      26.066028  9.8792213  0.32950742
## 9      3.6627540      13.535775 10.8598771  8.38570077
## 10     2.5662267      29.866629 31.9885022  0.45176979
## 11     0.9854669      70.868483 23.3764837 70.84970558
## 12     3.6983202       8.729106 13.2005064  0.28499730
## 13     1.5589495      23.044092  7.7618631  9.36996655
## 14     2.2758000      23.745643  7.9309954  5.13366805
## 15     0.9544472      85.488151 11.2977201  0.11543181
## 16     4.4542667       4.917437  6.3066777  2.38716222
## 17     1.3897242      35.591653  8.5784277  2.07362849
## 18     1.4673761      50.505655 13.4112047 25.70132500
## 19     1.2504675      29.025006  2.1034388 11.15422881
## 20     1.5612472      60.657687 11.3640092  1.60064223
## 21     2.4719415      61.554466  8.5528812  0.20516663
## 22     0.9257509      73.516665  4.7587809  2.33230471
## 23     3.8026920      36.349627  0.9041726  2.37401318
## 24     1.2744249      47.957335  2.5461868  4.25108332
##      Composite_Index Year
## 1      1.91165701 2024
## 2      0.52130481 2024
## 3      1.85975592 2024
## 4      0.77035756 2024
## 5      1.73894700 2024
## 6      0.79010204 2024
## 7      0.13204308 2024
## 8      1.50472562 2024
## 9      1.01866172 2024
## 10     0.75824352 2024
## 11    -0.02584849 2024
## 12     1.77443198 2024
## 13     0.89725033 2024
## 14     1.43580904 2024
## 15     0.51417078 2024
## 16     1.58747032 2024
## 17     0.97679104 2024
## 18     0.31737548 2024
## 19     0.85950475 2024
## 20     0.39182248 2024
## 21     0.80666984 2024
## 22     0.52520769 2024
## 23     1.54291626 2024
## 24     0.95756449 2024
```

Data Cleaning and Transformation

Click open to see R code for Data Cleaning and Transformation

```
# check each column for negative value
# https://stackoverflow.com/questions/50078917/replace-negative-values-with-positive-values-in-r
```

```

# z<--pmin(-z, z)
goe_eccc_random_2024$Proportion_of_native_species <--pmin(-goe_eccc_random_2024$Proportion_of_native_sp

goe_eccc_random_2024$Cultural_species_richness <--pmin(-goe_eccc_random_2024$Cultural_species_richness,

goe_eccc_random_2024$Exotic_species <--pmin(-goe_eccc_random_2024$Exotic_species, goe_eccc_random_2024$

goe_eccc_random_2024$Tramplng <--pmin(-goe_eccc_random_2024$Tramplng, goe_eccc_random_2024$Tramplng)

goe_eccc_random_2024$Herbivory <--pmin(-goe_eccc_random_2024$Herbivory, goe_eccc_random_2024$Herbivory)

goe_eccc_random_2024$Composite_Index <--pmin(-goe_eccc_random_2024$Composite_Index, goe_eccc_random_2024$

names(goe_eccc_random_2024)

## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Tramplng"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"

str(goe_eccc_random_2024)

## 'data.frame': 24 obs. of 9 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.947 0.553 0.809 0.707 0.726 ...
## $ Cultural_species_richness : num 2.96 1.32 2.58 1.46 3.35 ...
## $ Exotic_species : num 1.31 59.29 11.22 35.61 2.4 ...
## $ Tramplng : num 5.096 4.127 5.732 0.863 11.879 ...
## $ Herbivory : num 1.0771 35.8537 0.0361 25.5851 0.6826 ...
## $ Composite_Index : num 1.912 0.521 1.86 0.77 1.739 ...
## $ Year : chr "2024" "2024" "2024" "2024" ...

# https://www.reddit.com/r/rstats/comments/wfn1rf/forcing\_visualizations\_in\_r\_to\_have\_only\_2/
# round numbers to 2 digits
library(dplyr)

goe_eccc_random_2024 %>% mutate_if(is.numeric, round, digits=2) -> goe_eccc_random_2024

str(goe_eccc_random_2024)

## 'data.frame': 24 obs. of 9 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.95 0.55 0.81 0.71 0.73 0.53 0.46 0.56 0.53 0.65 ...
## $ Cultural_species_richness : num 2.96 1.32 2.58 1.46 3.35 1.18 0.48 2.11 3.66 2.57 ...
## $ Exotic_species : num 1.31 59.29 11.22 35.61 2.4 ...
## $ Tramplng : num 5.1 4.13 5.73 0.86 11.88 ...
## $ Herbivory : num 1.08 35.85 0.04 25.59 0.68 ...
## $ Composite_Index : num 1.91 0.52 1.86 0.77 1.74 0.79 0.13 1.5 1.02 0.76 ...
## $ Year : chr "2024" "2024" "2024" "2024" ...

# create new column with proportions changed to percentage as a whole number
# As this Makes Anniversary Island Proportion >1 change to 1
# data.frame[row_number, column_number] = new_value

```

```

# goe_eccc_random_2024[1, 3] <- 1
# but if more than one row ends up being more than 1
# https://stackoverflow.com/questions/66398854/r-set-column-to-maximum-of-current-entry-and-specified-v
# this made them all to 1
# goe_eccc_random_2024$Proportion_of_native_species <- pmax(goe_eccc_random_2024$Proportion_of_native_s
# https://stackoverflow.com/questions/66398854/r-set-column-to-maximum-of-current-entry-and-specified-v
# data.table version data[foo < 1, foo := 1])
# doesn't work
# goe_eccc_random_2024[goe_eccc_random_2024$Proportion_of_native_species > 1.0000, goe_eccc_random_2024

# change Year back to number
goe_eccc_random_2024$Year <- as.numeric(as.character(goe_eccc_random_2024$Year))
str(goe_eccc_random_2024)

```

```

## 'data.frame':    24 obs. of  9 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site           : chr  "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num  0.95 0.55 0.81 0.71 0.73 0.53 0.46 0.56 0.53 0.65 ...
## $ Cultural_species_richness  : num  2.96 1.32 2.58 1.46 3.35 1.18 0.48 2.11 3.66 2.57 ...
## $ Exotic_species           : num  1.31 59.29 11.22 35.61 2.4 ...
## $ Trampling                : num  5.1 4.13 5.73 0.86 11.88 ...
## $ Herbivory                 : num  1.08 35.85 0.04 25.59 0.68 ...
## $ Composite_Index           : num  1.91 0.52 1.86 0.77 1.74 0.79 0.13 1.5 1.02 0.76 ...
## $ Year                      : num  2024 2024 2024 2024 2024 ...

```

WHAT TO DO IF PROPORTION OF NATIVE SPECIES is >1

```
goe_eccc_random_2024
```

```

##           Subregion      Site Proportion_of_native_species
## 1      Gulf Islands Anniversary Island                0.95
## 2      Gulf Islands              AVNR                0.55
## 3      Gulf Islands      Brackman Island                0.81
## 4      Gulf Islands        Crows Nest                0.71
## 5      Gulf Islands      Dock Islet N                0.73
## 6      Gulf Islands      Eagle Islet                0.53
## 7      Gulf Islands      Mt Maxwell                0.46
## 8      Gulf Islands      Owl Islet                0.56
## 9      Gulf Islands      Reay Islet                0.53
## 10     Gulf Islands      Retreat Island                0.65
## 11     Gulf Islands      Rum Islet                0.51
## 12     Gulf Islands      SISCENEM                0.68
## 13 Saanich Peninsula      Bear Hill Park                0.63
## 14 Saanich Peninsula      Camas Hill                0.60
## 15 Saanich Peninsula Gonzales Hill Park                0.29
## 16 Saanich Peninsula      Gore Park                0.62
## 17 Saanich Peninsula      Highrock Park                0.43
## 18 Saanich Peninsula      Mill Hill Park                0.41
## 19 Saanich Peninsula      Oak Haven Park                0.49
## 20 Saanich Peninsula      Scafe Hill                0.28
## 21 Saanich Peninsula      Seymour Hill                0.30
## 22 Saanich Peninsula      Stewart Hill                0.49
## 23 Saanich Peninsula      Trial Island                0.54
## 24 Saanich Peninsula      Uplands Park                0.39

```

##	Cultural_species_richness	Exotic_species	Trampling	Herbivory	Composite_Index
## 1	2.96	1.31	5.10	1.08	1.91
## 2	1.32	59.29	4.13	35.85	0.52
## 3	2.58	11.22	5.73	0.04	1.86
## 4	1.46	35.61	0.86	25.59	0.77
## 5	3.35	2.40	11.88	0.68	1.74
## 6	1.18	29.26	15.01	0.64	0.79
## 7	0.48	66.15	10.02	5.15	0.13
## 8	2.11	26.07	9.88	0.33	1.50
## 9	3.66	13.54	10.86	8.39	1.02
## 10	2.57	29.87	31.99	0.45	0.76
## 11	0.99	70.87	23.38	70.85	0.03
## 12	3.70	8.73	13.20	0.28	1.77
## 13	1.56	23.04	7.76	9.37	0.90
## 14	2.28	23.75	7.93	5.13	1.44
## 15	0.95	85.49	11.30	0.12	0.51
## 16	4.45	4.92	6.31	2.39	1.59
## 17	1.39	35.59	8.58	2.07	0.98
## 18	1.47	50.51	13.41	25.70	0.32
## 19	1.25	29.03	2.10	11.15	0.86
## 20	1.56	60.66	11.36	1.60	0.39
## 21	2.47	61.55	8.55	0.21	0.81
## 22	0.93	73.52	4.76	2.33	0.53
## 23	3.80	36.35	0.90	2.37	1.54
## 24	1.27	47.96	2.55	4.25	0.96
##	Year				
## 1	2024				
## 2	2024				
## 3	2024				
## 4	2024				
## 5	2024				
## 6	2024				
## 7	2024				
## 8	2024				
## 9	2024				
## 10	2024				
## 11	2024				
## 12	2024				
## 13	2024				
## 14	2024				
## 15	2024				
## 16	2024				
## 17	2024				
## 18	2024				
## 19	2024				
## 20	2024				
## 21	2024				
## 22	2024				
## 23	2024				
## 24	2024				

```
Percentage_ns <- goe_eccc_random_2024$Proportion_of_native_species * 100
```

```
# add-new column for Percentages
```

```

goe_eccc_random_2024 <- cbind(goe_eccc_random_2024, Percentage_ns)

# arrange order of site names
Sites_eccc_1 <- Sites_eccc %>% arrange(Sites_eccc$Site)
goe_eccc_random_2024_1 <- goe_eccc_random_2024 %>% arrange(goe_eccc_random_2024$Site)

# bind 2 dataframes together
# this requires Sites rows from both data frames to be in alphabetical order
## to assign the data rows correctly
goe_eccc_random_2024_sites <- cbind(Sites_eccc_1, goe_eccc_random_2024_1)

# Remove duplicate columns resulting from binding 2 data frames
goe_eccc_random_2024_sites_1 <- goe_eccc_random_2024_sites[c(-9,-10)]

# Rename columns to remove dot-format
names(goe_eccc_random_2024_sites_1)[names(goe_eccc_random_2024_sites_1) == "Land.Manager"] <- "Land_Man"
names(goe_eccc_random_2024_sites_1)[names(goe_eccc_random_2024_sites_1) == "Main.Restoration.Type"] <- "Main_Restoration_Type"
names(goe_eccc_random_2024_sites_1)[names(goe_eccc_random_2024_sites_1) == "Restoration.Intensity"] <- "Restoration_Intensity"
names(goe_eccc_random_2024_sites_1)[names(goe_eccc_random_2024_sites_1) == "Area.ha"] <- "Area_ha"

# write csv file with combined data
## to create new data file which is used for creating the visualizations
write.csv(goe_eccc_random_2024_sites_1, "data/goe_eccc_random_2024_all_site_data.csv", row.names = FALSE)

```

Plots

Site by Percentage Native Species

```

site_natsp <- ggplot(goe_eccc,
  aes(x = Site, y = Percentage_ns)) +
  geom_point(shape = 18, size = 2) +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Site by Percentage Native Species',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")

site_natsp

```

Plotting Site by Percentage Native Species ECCC

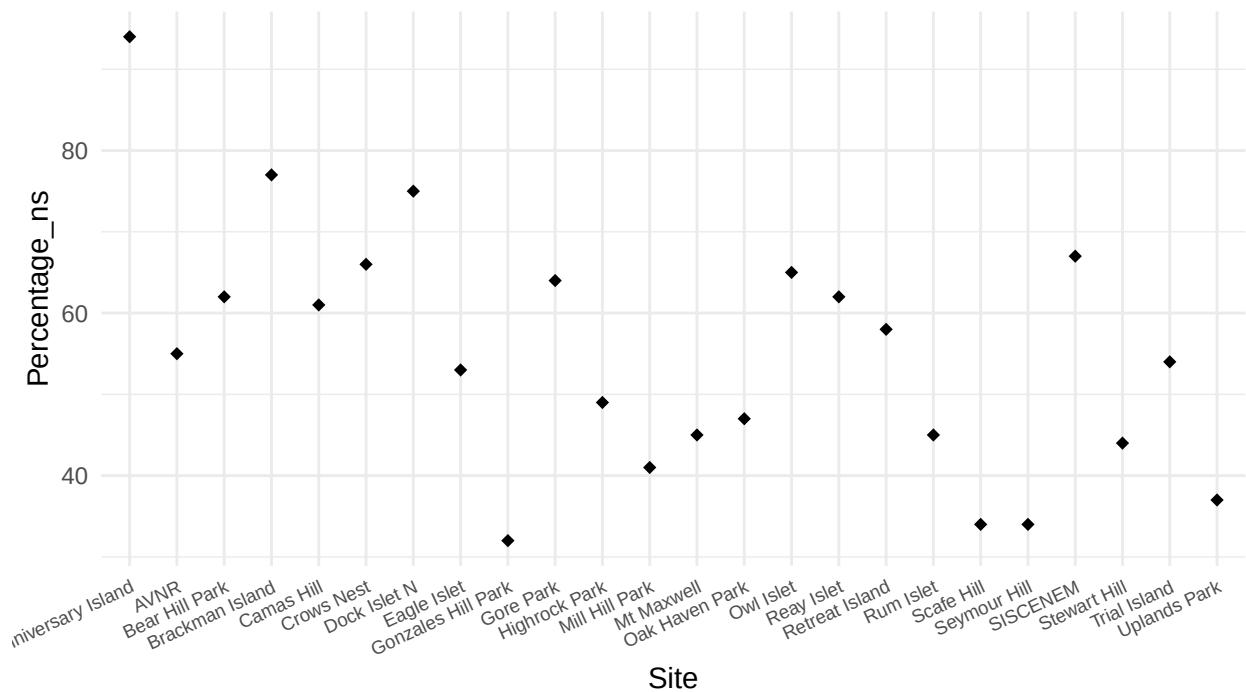


Chart by Wendy Anthony
2025-08-04

Site by Composite_Index

```
site_natsp <- ggplot(goe_eccc,
  aes(x = Site, y = Composite_Index)) +
  geom_point(shape = 18, size = 2) +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Site by Composite Index',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")

site_natsp
```

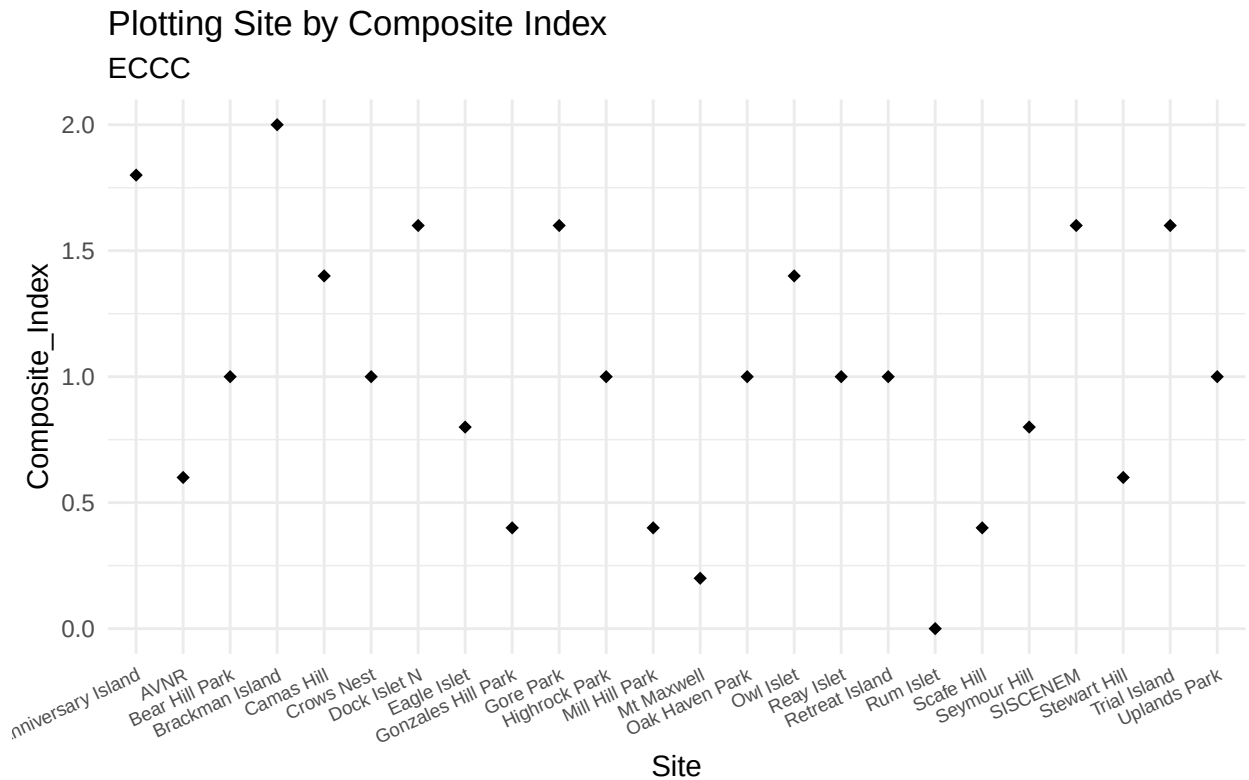


Chart by Wendy Anthony
2025-08-04

TEST DOESn't Reorder Site by Restoration_Intensity

- note Reay Islet was showing a separate High, so in original csv I copied High value from another site, aslo Scafe Hill had a different low value ...
- also all high values have a space in front ... change all ...
- Having trouble ordering Restoration_Intensity values ... see next example

Click to see results of TEST

```
# names(goe_eccc_sites_1)
# str(goe_eccc_sites_1)
# unique(goe_eccc_sites_1$Restoration_Intensity)
#
# # change character to factor
# goe_eccc_sites_1$Restoration_Intensity <- as.factor(goe_eccc_sites_1$Restoration_Intensity)
# # change factor to character
# goe_eccc_sites_1 <- data.frame(lapply(goe_eccc_sites_1, as.character), stringsAsFactors=FALSE)
#
# # WORKS !!!
# Restoration_Intensity_order <- c("high", "moderate", "low", "minimal")
# arranged_goe_eccc_sites_1 <- goe_eccc_sites_1 %>%
#   arrange(factor(Restoration_Intensity, levels = Restoration_Intensity_order))
# arranged_goe_eccc_sites_1
# str(arranged_goe_eccc_sites_1)
#
# # still not showing the order
# site_restInt <- ggplot(arranged_goe_eccc_sites_1,
```



```

#       aes(x = Site, y = Restoration_Intensity, color = Subregion)) +
#       geom_point(shape = 18, size = 2) +
#       #geom_bar(stat="identity") +
#       theme_minimal() + #get rid of grey background and tick marks
#       theme(legend.position="bottom") + #remove legend
#       theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
#       labs(title='Plotting Site by Restoration_Intensity',
#            subtitle='ECCC',
#            caption = "Chart by Wendy Anthony \n 2025-08-20")
# site_restInt

# ggplotly(site_restInt)

```

ORDER NOT WORKING TEST Site by Restoration_Intensity and Ex- otic_species

```
names(goe_eccc_sites_1)
```

```

## [1] "Subregion"           "Site"
## [3] "Land_Manager"        "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                 "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Trampling"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"

```

```
str(goe_eccc_sites_1)
```

```

## 'data.frame':   24 obs. of  16 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site           : chr  "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island" ...
## $ Land_Manager    : chr  "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr  "herbivore reduction" "herbivore reduction" "invasive removal" ...
## $ Restoration_Intensity : chr  "high" "minimal" "low" "high" ...
## $ Area_ha        : num  4.39 20.54 3.8 4.41 10.1 ...
## $ Lat            : num  48.8 48.8 48.5 48.7 48.4 ...
## $ Lng            : num  -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num  0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num  2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species : num  0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling       : num  5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory       : num  1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index : num  1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year            : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns   : num  94 55 62 77 61 66 75 53 32 64 ...

```

```

# Step 1
desired_levels <- c("high", "moderate", "low", "minimal")
# create new df to create desired levels
goe_eccc_sites_2 <- goe_eccc_sites_1
# this changes data type from character to factor
goe_eccc_sites_2$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = desired_levels)
print(levels(goe_eccc_sites_2$Restoration_Intensity)) # New order of levels

```

```
## [1] "high"      "moderate" "low"      "minimal"
```

```
str(goe_eccc_sites_2)
```

```
## 'data.frame':  24 obs. of  16 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands"
## $ Site           : chr  "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island"
## $ Land_Manager   : chr  "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr  "herbivore reduction" "herbivore reduction" "invasive removal"
## $ Restoration_Intensity : Factor w/ 4 levels "high","moderate",...: 1 4 3 1 1 1 1 3 2 ...
## $ Area_ha        : num  4.39 20.54 3.8 4.41 10.1 ...
## $ Lat            : num  48.8 48.8 48.5 48.7 48.4 ...
## $ Lng            : num  -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num  0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness  : num  2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species   : num  0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling        : num  5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory        : num  1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index   : num  1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year              : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns     : num  94 55 62 77 61 66 75 53 32 64 ...
```

```
# step 2
```

```
goe_eccc_sites_2 <- goe_eccc_sites_2[order(goe_eccc_sites_2$Restoration_Intensity),]
```

```
site_restInt_exotic <- ggplot(goe_eccc_sites_2,
  aes(x = Exotic_species, y = Restoration_Intensity, color = Site)) +
  geom_point(shape = 18, size = 2) +
  #geom_bar(stat="identity") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend legen won't stick to bottom with ggp
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Site by Restoration_Intensity & Exotic Species',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-20")
site_restInt_exotic
```

Plotting Site by Restoration_Intensity & Exotic Species ECCC

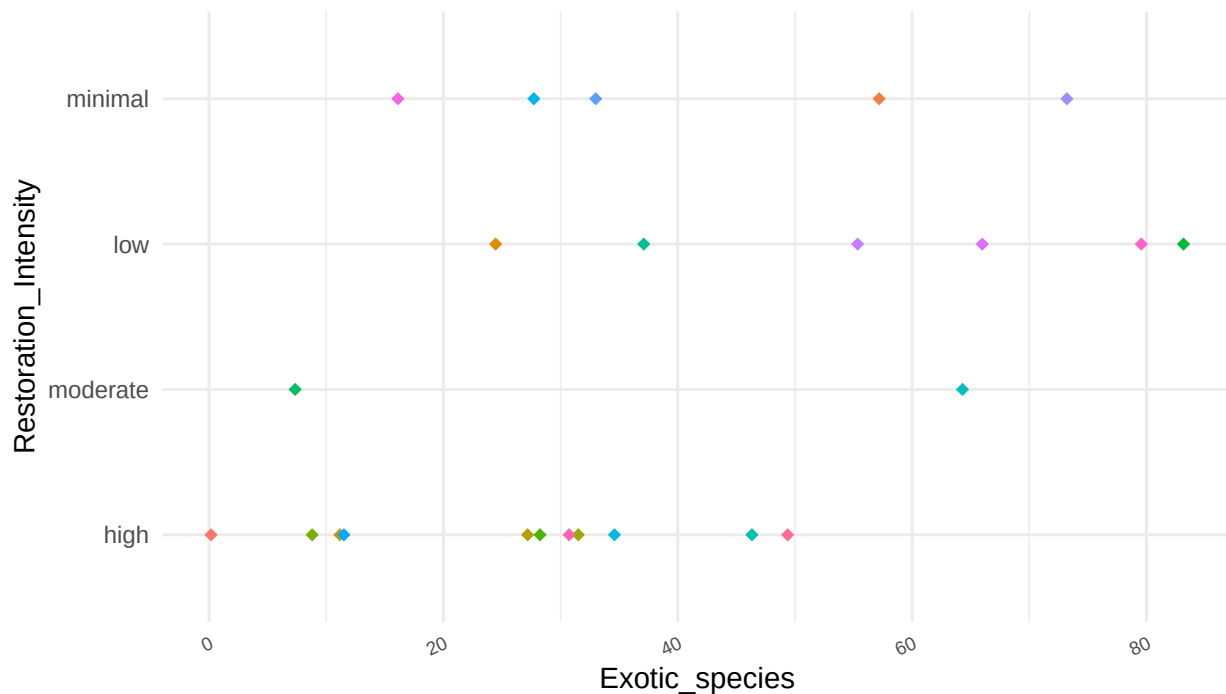


Chart by Wendy Anthony
2025-08-20

```
# ggplotly(site_restInt_exotic)
# change ggplotly legend to bottom
# %>%
#   plotly::layout(legend=list(x=0,
#                               xanchor='left',
#                               yanchor='bottom',
#                               orientation='h'))
```

ORDER WORKs Point Plot Site by Restoration_Intensity

```
names(goe_eccc_sites_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Land_Manager"       "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"     "Trampling"
## [13] "Herbivory"          "Composite_Index"
## [15] "Year"               "Percentage_ns"
```

```
# Reorder following a precise order
# https://r-graph-gallery.com/267-reorder-a-variable-in-ggplot2.html
# https://stackoverflow.com/questions/18413756/re-ordering-factor-levels-in-data-frame
# mydf$task <- factor(mydf$task, levels = c("up", "down", "left", "right", "front", "back"))
# goe_eccc_sites_1$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = #c("
str(goe_eccc_sites_1)
```

```
## 'data.frame':   24 obs. of  16 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands"
## $ Site           : chr  "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island"
## $ Land_Manager    : chr  "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr  "herbivore reduction" "herbivore reduction" "invasive removal"
## $ Restoration_Intensity : chr  "high" "minimal" "low" "high" ...
## $ Area_ha        : num  4.39 20.54 3.8 4.41 10.1 ...
## $ Lat            : num  48.8 48.8 48.5 48.7 48.4 ...
## $ Lng            : num  -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num  0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num  2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species   : num  0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling        : num  5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory        : num  1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index  : num  1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year             : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns    : num  94 55 62 77 61 66 75 53 32 64 ...
```

```
# change character to factor
# str(goe_eccc_sites_1)
# goe_eccc_sites_1$Restoration_Intensity <- as.factor(goe_eccc_sites_1$Restoration_Intensity)
# str(goe_eccc_sites_1)
# # change factor to character
# goe_eccc_sites_1 <- data.frame(lapply(goe_eccc_sites_1, as.character), stringsAsFactors=FALSE)
# str(goe_eccc_sites_1)
```

```
# Step 1
desired_levels <- c("high", "moderate", "low", "minimal")
# create new df to create desired levels
goe_eccc_sites_2 <- goe_eccc_sites_1
# this changes data type from character to factor
goe_eccc_sites_2$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = desired_levels)
print(levels(goe_eccc_sites_2$Restoration_Intensity)) # New order of levels
```

```
## [1] "high"      "moderate" "low"       "minimal"
```

```
str(goe_eccc_sites_2)
```

```
## 'data.frame':   24 obs. of  16 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands"
## $ Site           : chr  "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island"
## $ Land_Manager    : chr  "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr  "herbivore reduction" "herbivore reduction" "invasive removal"
## $ Restoration_Intensity : Factor w/ 4 levels "high","moderate",...: 1 4 3 1 1 1 1 1 3 2 ...
## $ Area_ha        : num  4.39 20.54 3.8 4.41 10.1 ...
## $ Lat            : num  48.8 48.8 48.5 48.7 48.4 ...
## $ Lng            : num  -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num  0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num  2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species   : num  0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling        : num  5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory        : num  1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index  : num  1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year             : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns    : num  94 55 62 77 61 66 75 53 32 64 ...
```

```

# step 2
goe_eccc_sites_2 <- goe_eccc_sites_2[order(goe_eccc_sites_2$Restoration_Intensity),]

# values in order now -- ?? how to change to reverse order
site_restInt <- ggplot(goe_eccc_sites_2,
  aes(x = Site, y = reorder(Restoration_Intensity, desc(Restoration_Intensity)), color = Subregion,
  # reorder y variable axis
  # https://stackoverflow.com/questions/28391850/reverse-order-of-discrete-y-axis-in-ggplot2s
  # y = Restoration_Intensity
    geom_point(shape = 18, size = 2) +
  #geom_bar(stat="identity") +
  # Replace default palette of pink & blue
  scale_fill_brewer(palette = "Dark2") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="bottom") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Sites by Restoration Intensity',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-22", x = "Site", y = "Restoration Intensity")

site_restInt

```

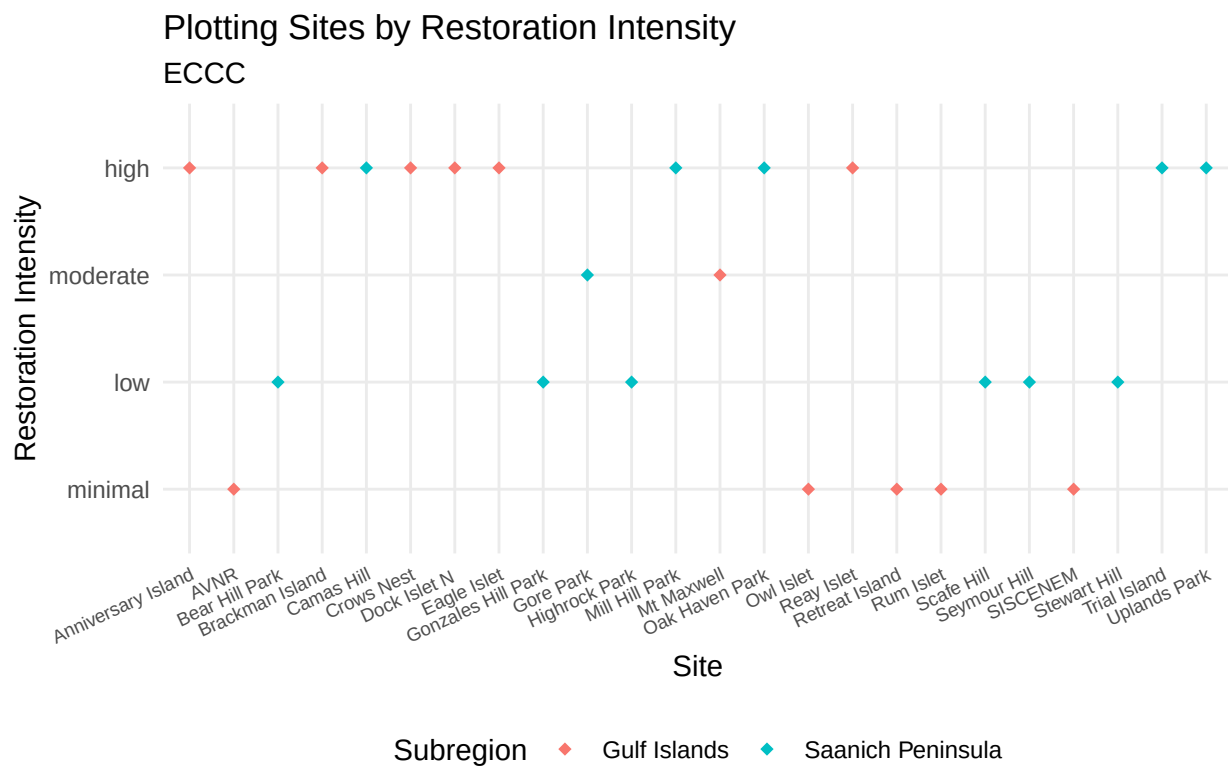
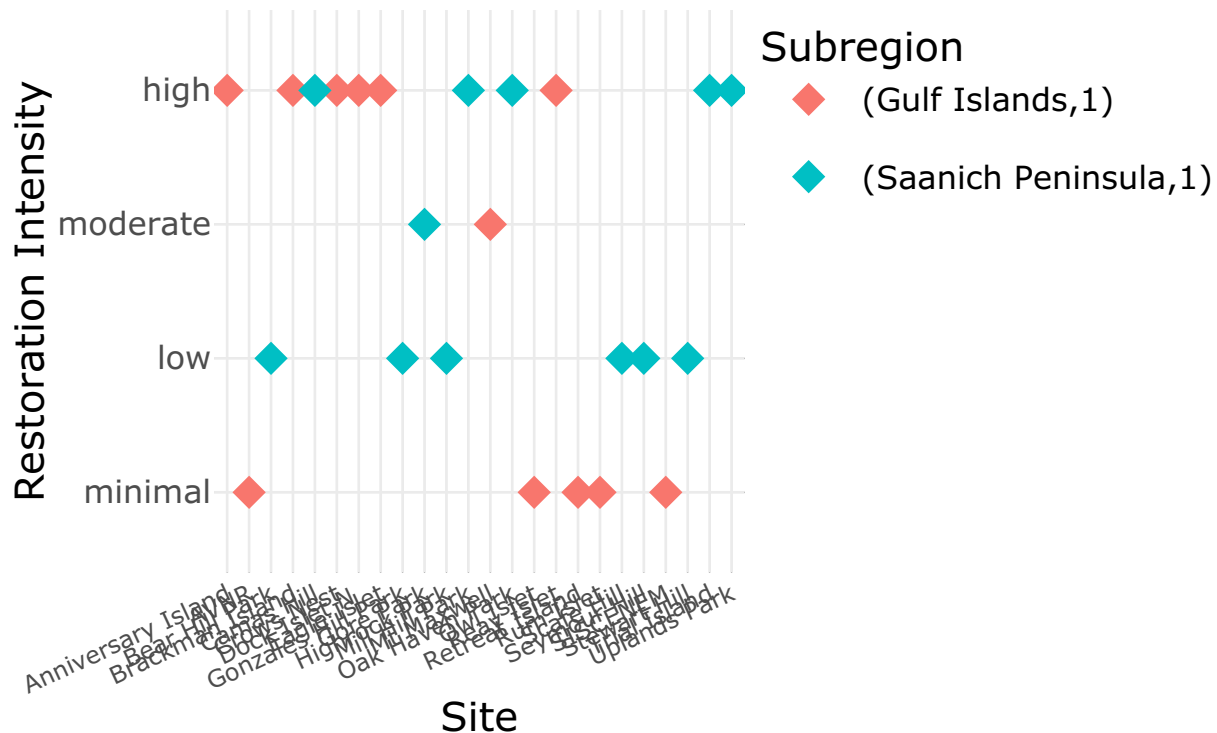


Chart by Wendy Anthony
2025-08-22

```
ggplotly(site_restInt)
```

Plotting Sites by Restoration Intensity



Change x axis ORDER WORKs Point Plot Site by Restoration_Intensity

```
names(goe_eccc_sites_1)
```

```
## [1] "Subregion" "Site"
## [3] "Land_Manager" "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat" "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species" "Trampling"
## [13] "Herbivory" "Composite_Index"
## [15] "Year" "Percentage_ns"
```

```
# Reorder following a precise order
```

```
# https://r-graph-gallery.com/267-reorder-a-variable-in-ggplot2.html
```

```
# https://stackoverflow.com/questions/18413756/re-ordering-factor-levels-in-data-frame
```

```
# mydf$task <- factor(mydf$task, levels = c("up", "down", "left", "right", "front", "back"))
```

```
#goe_eccc_sites_1$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = #c("high", "moderate", "low", "minimal"))
str(goe_eccc_sites_1)
```

```
## 'data.frame': 24 obs. of 16 variables:
```

```
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site : chr "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island" ...
## $ Land_Manager : chr "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr "herbivore reduction" "herbivore reduction" "invasive removal" ...
## $ Restoration_Intensity : chr "high" "minimal" "low" "high" ...
## $ Area_ha : num 4.39 20.54 3.8 4.41 10.1 ...
## $ Lat : num 48.8 48.8 48.5 48.7 48.4 ...
```

```
## $ Lng : num -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num 0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num 2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species : num 0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling : num 5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory : num 1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index : num 1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns : num 94 55 62 77 61 66 75 53 32 64 ...

# change character to factor
# str(goe_eccc_sites_1)
# goe_eccc_sites_1$Restoration_Intensity <- as.factor(goe_eccc_sites_1$Restoration_Intensity)
# str(goe_eccc_sites_1)
# # change factor to character
# goe_eccc_sites_1 <- data.frame(lapply(goe_eccc_sites_1, as.character), stringsAsFactors=FALSE)
# str(goe_eccc_sites_1)

# Step 1
desired_levels <- c("high", "moderate", "low", "minimal")
# create new df to create desired levels
goe_eccc_sites_2 <- goe_eccc_sites_1
# this changes data type from character to factor
goe_eccc_sites_2$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = desired_levels)
print(levels(goe_eccc_sites_2$Restoration_Intensity)) # New order of levels

## [1] "high" "moderate" "low" "minimal"

str(goe_eccc_sites_2)

## 'data.frame': 24 obs. of 16 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site : chr "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island" ...
## $ Land_Manager : chr "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr "herbivore reduction" "herbivore reduction" "invasive removal" ...
## $ Restoration_Intensity : Factor w/ 4 levels "high","moderate",...: 1 4 3 1 1 1 1 1 3 2 ...
## $ Area_ha : num 4.39 20.54 3.8 4.41 10.1 ...
## $ Lat : num 48.8 48.8 48.5 48.7 48.4 ...
## $ Lng : num -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num 0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num 2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species : num 0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling : num 5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory : num 1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index : num 1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns : num 94 55 62 77 61 66 75 53 32 64 ...

# step 2
goe_eccc_sites_2 <- goe_eccc_sites_2[order(goe_eccc_sites_2$Restoration_Intensity),]

# values in order now -- ?? how to change to reverse order
site_restInt_x <- ggplot(goe_eccc_sites_2,
  aes(x = reorder(Site,
    # -desc reverses descending
    -desc(Restoration_Intensity))),
```

```

    y = reorder(Restoration_Intensity, desc(Restoration_Intensity)), color = Subregion)) +
  # reorder y variable axis
  # https://stackoverflow.com/questions/28391850/reverse-order-of-discrete-y-axis-in-ggplot2s
  # y = Restoration_Intensity
    geom_point(shape = 18, size = 2) +
  #geom_bar(stat="identity") +
  # Replace default palette of pink & blue
  # makes ggplotly legend show (Saanich Peninsula,1)
  # scale_fill_brewer(palette = "Dark2") +
    theme_minimal() + #get rid of grey background and tick marks
    theme(legend.position="bottom") + #remove legend
    theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
    labs(title='Plotting Site by Restoration_Intensity',
         subtitle='ECCC',
         caption = "Chart by Wendy Anthony \n 2025-08-26",
         x = "Site", y = "Restoration_Intensity")

```

site_restInt_x

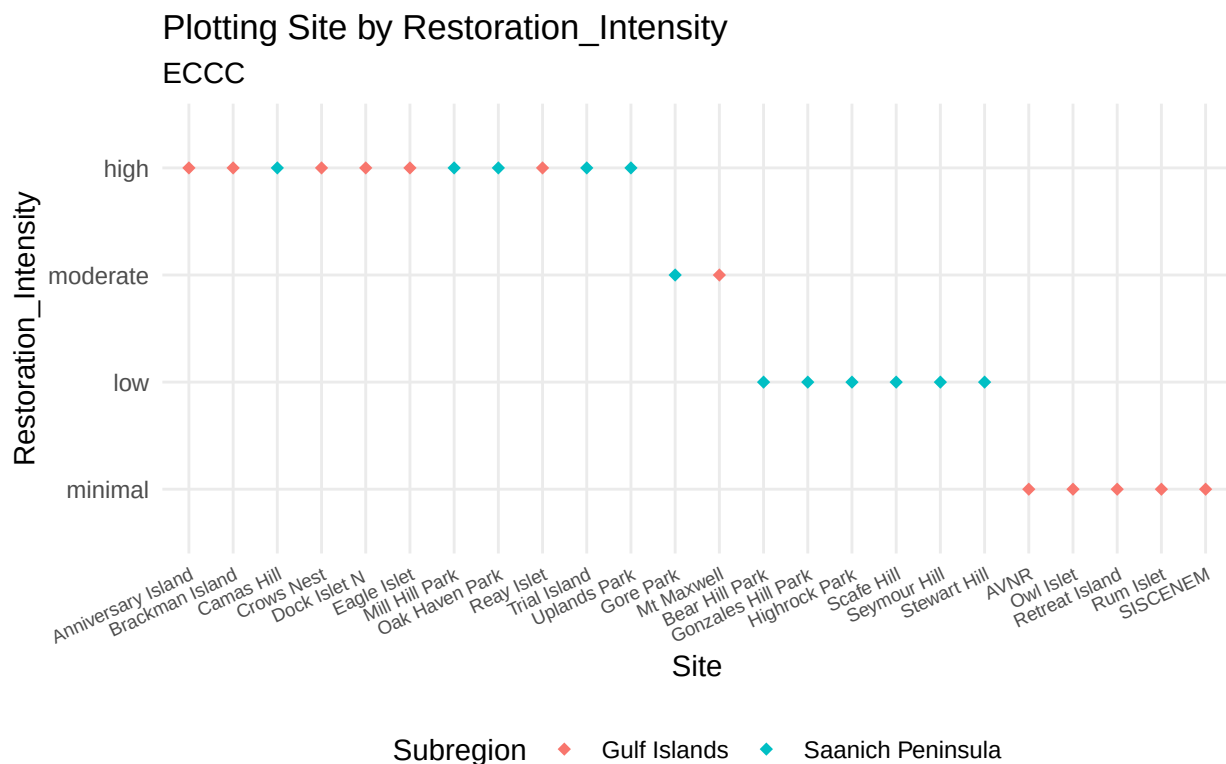
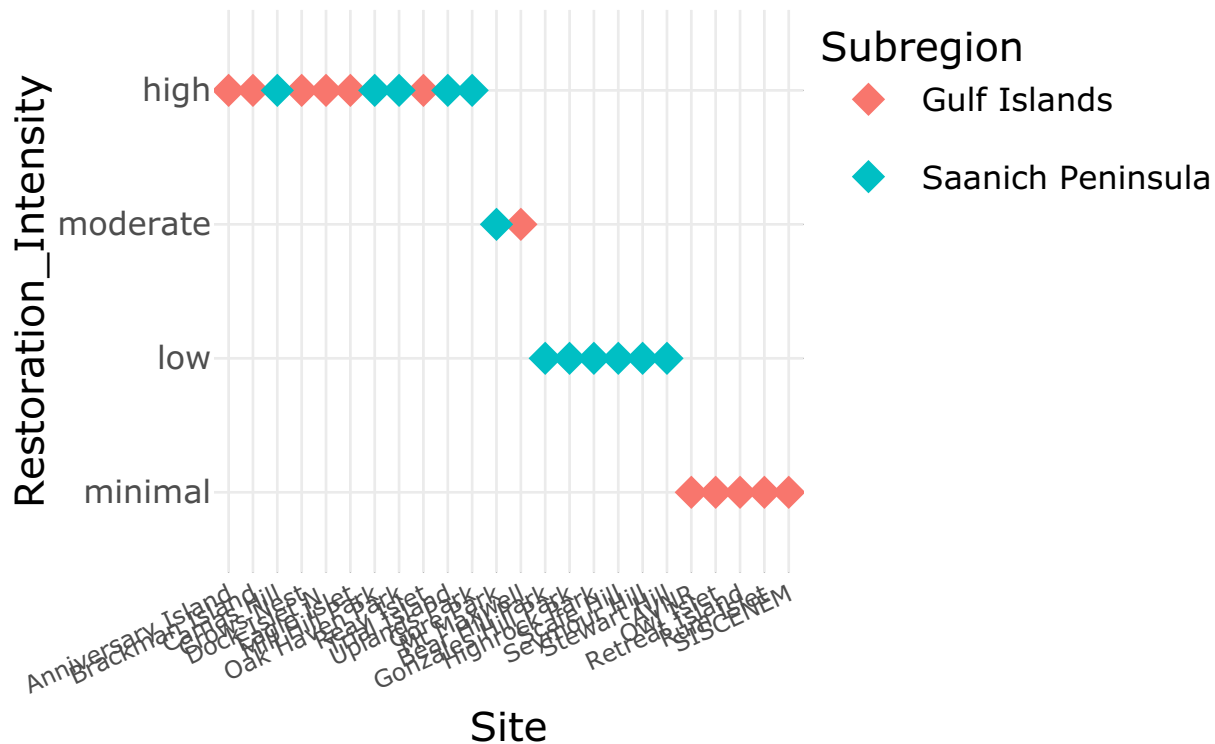


Chart by Wendy Anthony
2025-08-26

ggplotly(site_restInt_x)

Plotting Site by Restoration_Intensity



BAR Plot WORKS Site by Restoration_Intensity

```
names(goe_eccc_sites_1)
```

```
## [1] "Subregion"           "Site"
## [3] "Land_Manager"        "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                 "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Trampling"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"
```

```
# Reorder following a precise order
```

```
# https://r-graph-gallery.com/267-reorder-a-variable-in-ggplot2.html
```

```
# https://stackoverflow.com/questions/18413756/re-ordering-factor-levels-in-data-frame
```

```
# mydf$task <- factor(mydf$task, levels = c("up", "down", "left", "right", "front", "back"))
```

```
#goe_eccc_sites_1$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = #c("high", "moderate", "low", "minimal"))
str(goe_eccc_sites_1)
```

```
## 'data.frame': 24 obs. of 16 variables:
```

```
## $ Subregion           : chr  "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site                : chr  "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island" ...
## $ Land_Manager        : chr  "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr  "herbivore reduction" "herbivore reduction" "invasive removal" ...
## $ Restoration_Intensity : chr  "high" "minimal" "low" "high" ...
## $ Area_ha             : num  4.39 20.54 3.8 4.41 10.1 ...
```

```
## $ Lat : num 48.8 48.8 48.5 48.7 48.4 ...
## $ Lng : num -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num 0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num 2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species : num 0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling : num 5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory : num 1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index : num 1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns : num 94 55 62 77 61 66 75 53 32 64 ...

# change character to factor
# str(goe_eccc_sites_1)
# goe_eccc_sites_1$Restoration_Intensity <- as.factor(goe_eccc_sites_1$Restoration_Intensity)
# str(goe_eccc_sites_1)
# # change factor to character
# goe_eccc_sites_1 <- data.frame(lapply(goe_eccc_sites_1, as.character), stringsAsFactors=FALSE)
# str(goe_eccc_sites_1)

# Step 1
desired_levels <- c("high", "moderate", "low", "minimal")
# this changes data type from character to factor
goe_eccc_sites_2$Restoration_Intensity <- factor(goe_eccc_sites_1$Restoration_Intensity, levels = desired_levels)
print(levels(goe_eccc_sites_2$Restoration_Intensity)) # New order of levels

## [1] "high" "moderate" "low" "minimal"

str(goe_eccc_sites_2)

## 'data.frame': 24 obs. of 16 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site : chr "Anniversary Island" "Brackman Island" "Camas Hill" "Crows Nest" ...
## $ Land_Manager : chr "GINPR" "GINPR" "HAT" "Trinity Western University" ...
## $ Main_Restoration_Type : chr "herbivore reduction" "invasive removal" "invasive removal" "h" ...
## $ Restoration_Intensity : Factor w/ 4 levels "high","moderate",...: 1 4 3 1 1 1 1 1 3 2 ...
## $ Area_ha : num 4.39 4.41 10.1 15.34 0.54 ...
## $ Lat : num 48.8 48.7 48.4 48.8 48.7 ...
## $ Lng : num -123 -123 -124 -123 -123 ...
## $ Proportion_of_native_species: num 0.94 0.77 0.61 0.66 0.75 0.53 0.41 0.47 0.62 0.54 ...
## $ Cultural_species_richness : num 2.71 2.8 2.43 1.07 3.6 0.8 1.56 1.5 3.6 3.57 ...
## $ Exotic_species : num 0.17 11.17 27.19 31.51 8.8 ...
## $ Trampling : num 5 5.4 7.71 1.64 12.2 ...
## $ Herbivory : num 1.14 0 4.71 26.71 0.1 ...
## $ Composite_Index : num 1.8 2 1.4 1 1.6 0.8 0.4 1 1 1.6 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns : num 94 77 61 66 75 53 41 47 62 54 ...

# step 2
goe_eccc_sites_2 <- goe_eccc_sites_2[order(goe_eccc_sites_2$Restoration_Intensity),]

# values in order now -- ?? how to change to reverse order
site_restInt_bar <- ggplot(goe_eccc_sites_2,
  aes(y = reorder(Restoration_Intensity, desc(Restoration_Intensity)), color = Site, fill = Site))
# reorder y variable axis
# https://stackoverflow.com/questions/28391850/reverse-order-of-discrete-y-axis-in-ggplot2s
# y = Restoration_Intensity
```

```

geom_bar() +
#geom_bar(stat="identity") +
theme_minimal() + #get rid of grey background and tick marks
theme(legend.position="bottom") + #remove legend
theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
labs(title='Plotting Site by Restoration_Intensity',
      subtitle='ECCC',
      caption = "Chart by Wendy Anthony \n 2025-08-26", y = "Restoration_Intensity")
site_restInt_bar

```

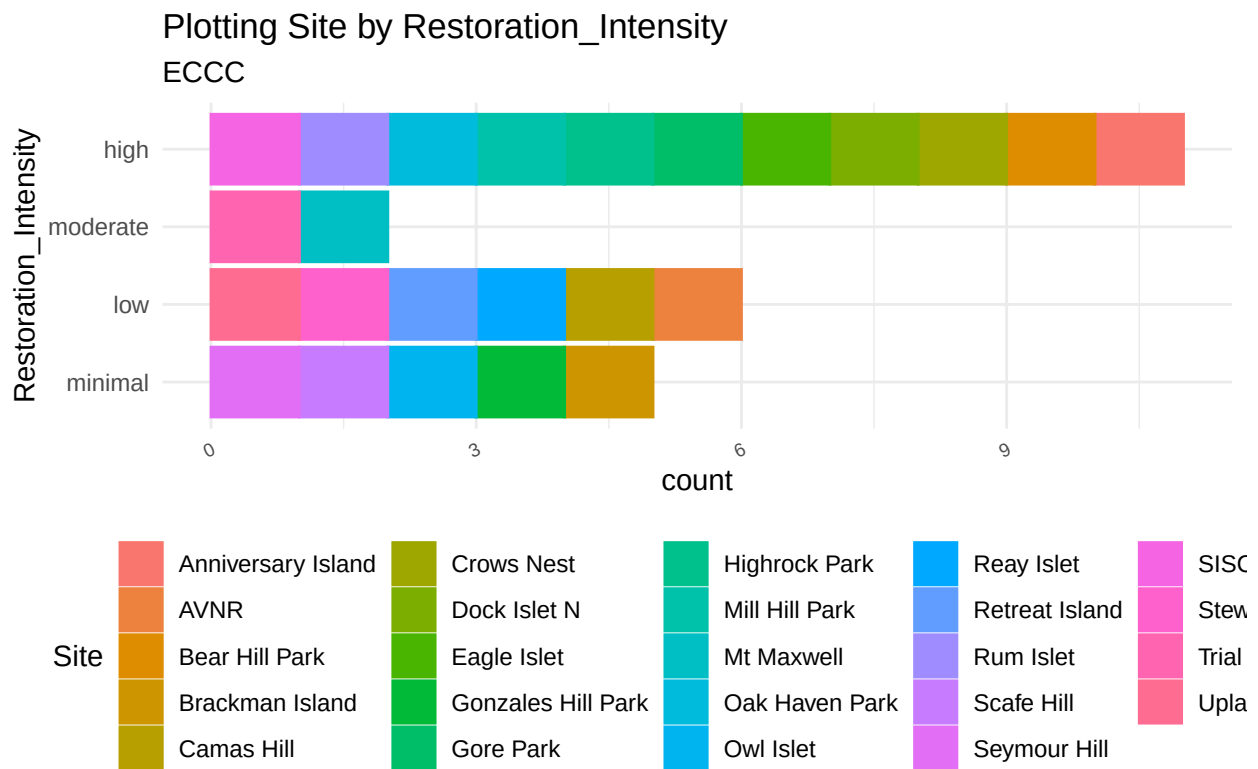
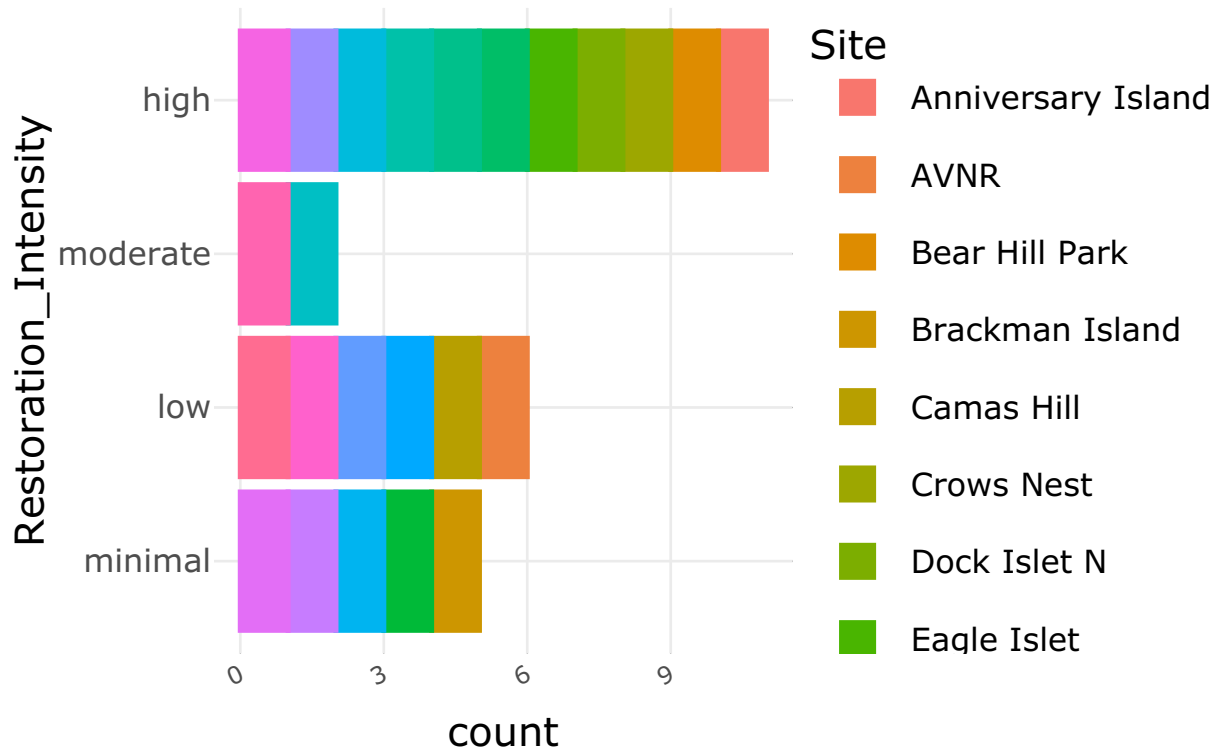


Chart by Wendy Anthony
2025-08-26

```
ggplotly(site_restInt_bar)
```

Plotting Site by Restoration_Intensity



Violin Subregion by Exotic_species

```
subregion_exsp <- ggplot(goe_eccc,
  aes(x = Subregion, y = Exotic_species)) +
  geom_violin(fill = "seagreen2") +
  geom_boxplot(width = 0.1, fill = "sandybrown") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(vjust = 1, size = 9)) +
  labs(title='Plotting Subregion by Exotic_species',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")

subregion_exsp
```

Plotting Subregion by Exotic_species ECCC



Chart by Wendy Anthony
2025-08-04

plotly won't work here

Violin Subregion by Percentage_ns

```
names(goe_eccc)
```

```
## [1] "Subregion"           "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"                "Percentage_ns"
```

```
subregion_nsp <- ggplot(goe_eccc,
  aes(x = Subregion, y = Percentage_ns)) +
  geom_violin(fill = "seagreen2") +
  geom_boxplot(width = 0.1, fill = "sandybrown") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(vjust = 1, size = 9)) +
  labs(title='Plotting Subregion by Percentage of Native Species',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-27")
```

```
subregion_nsp
```

Plotting Subregion by Percentage of Native Species ECCC



Chart by Wendy Anthony
2025-08-27

plotly won't work here

Cowplot Violin Subregion plot_grid

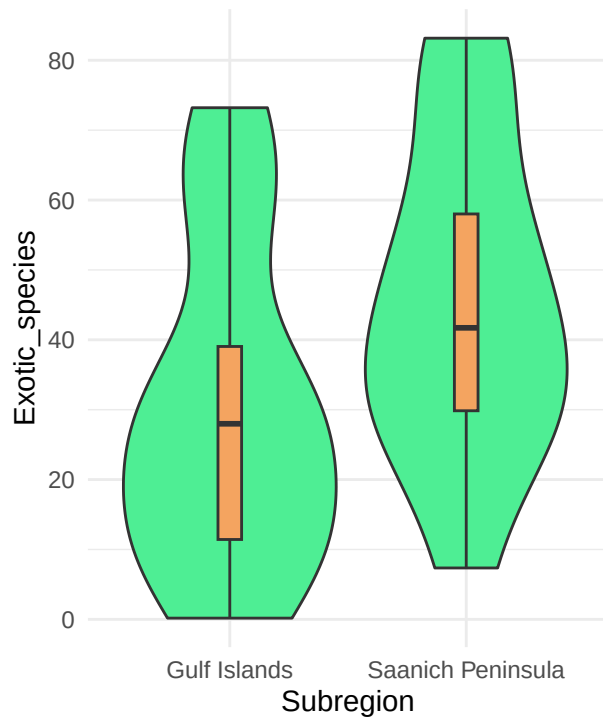
```
library(cowplot)

subregion_exsp_1 <- subregion_exsp + labs(title='Exotic Species',
                                           subtitle='ECCC',
                                           caption = "2025-08-27")

subregion_nsp_1 <- subregion_nsp + labs(title='Native Species',
                                          subtitle='ECCC',
                                          caption = "2025-08-27")

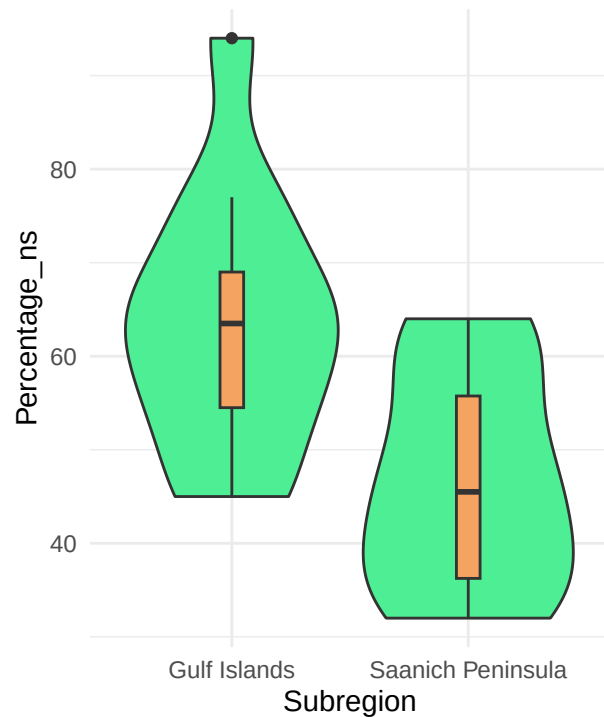
plot_grid(subregion_exsp_1, subregion_nsp_1, labels = "AUTO")
```

A Exotic Species
ECCC



2025-08-27

B Native Species
ECCC



2025-08-27

TEST NOTWorking Bar Chart Subregion by Restoration_Intensity

```
names(goe_eccc_sites_1)
```

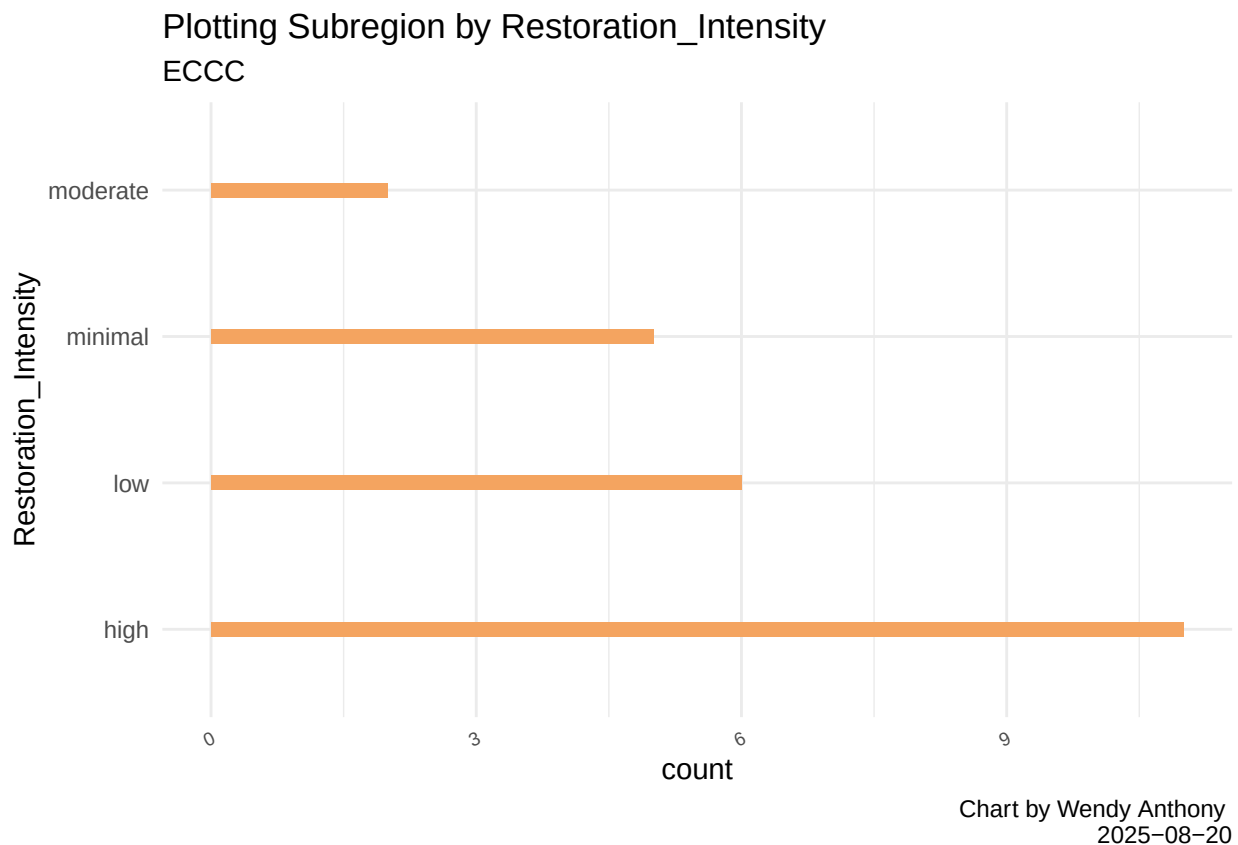
```
## [1] "Subregion"           "Site"
## [3] "Land_Manager"        "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                 "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Trampling"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"
```

```
str(goe_eccc_sites_1)
```

```
## 'data.frame': 24 obs. of 16 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site : chr "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island" ...
## $ Land_Manager : chr "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr "herbivore reduction" "herbivore reduction" "invasive removal" ...
## $ Restoration_Intensity : chr "high" "minimal" "low" "high" ...
## $ Area_ha : num 4.39 20.54 3.8 4.41 10.1 ...
## $ Lat : num 48.8 48.8 48.5 48.7 48.4 ...
## $ Lng : num -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num 0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num 2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species : num 0.17 57.19 24.46 11.17 27.19 ...
```

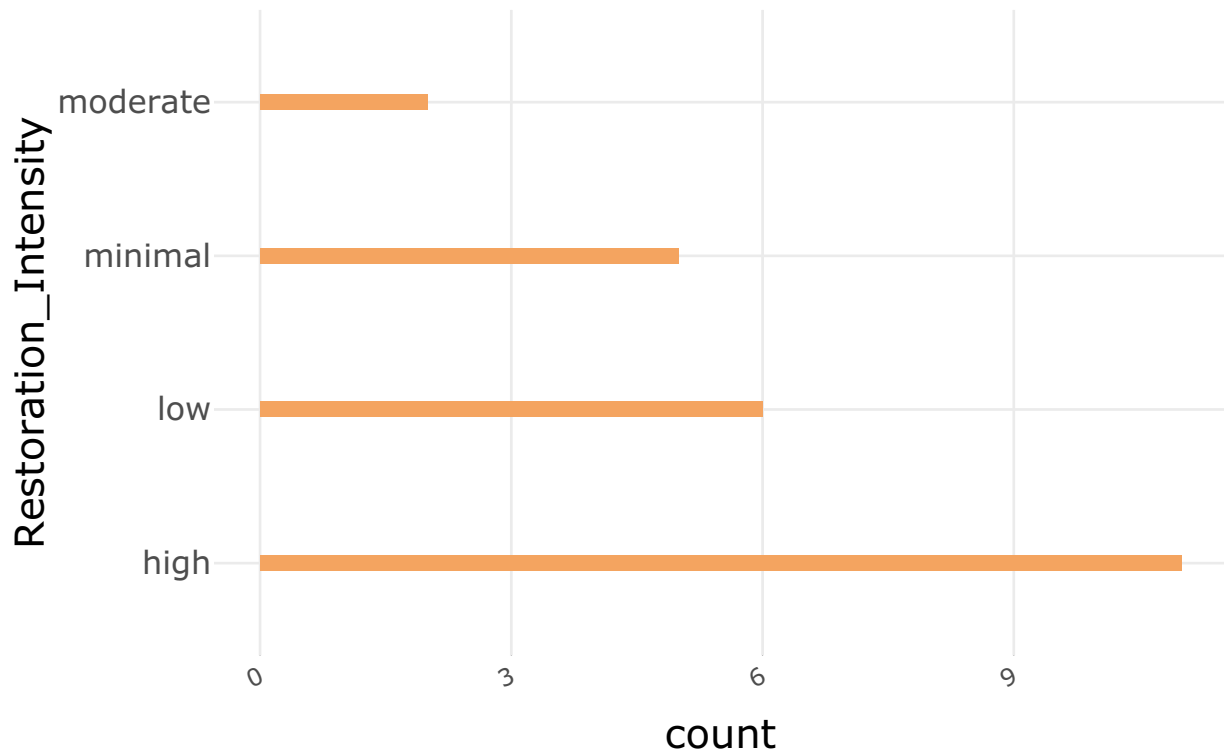
```
## $ Trampling      : num  5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory      : num  1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index : num  1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year            : int   2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns   : num  94 55 62 77 61 66 75 53 32 64 ...
```

```
subregion_restInt <- ggplot(goe_eccc_sites_1,
  aes(y = Restoration_Intensity)) +
#   geom_violin(fill = "seagreen2") +
  geom_bar(width = 0.1, fill = "sandybrown") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Subregion by Restoration_Intensity',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-20")
subregion_restInt
```



```
ggplotly()
```


Plotting Subregion by Restoration_Intensity



plotly won't work here

Subregion by Proportion_of_native_species

```
subregion_natsp <- ggplot(goe_eccc_1,
  aes(x = Subregion, y = Percentage_ns)) +
  geom_violin(fill = "seagreen2") +
  geom_boxplot(width = 0.1, fill = "sandybrown") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Subregion by Proportion_of_native_species',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")

subregion_natsp
```

Plotting Subregion by Proportion_of_native_species ECCC

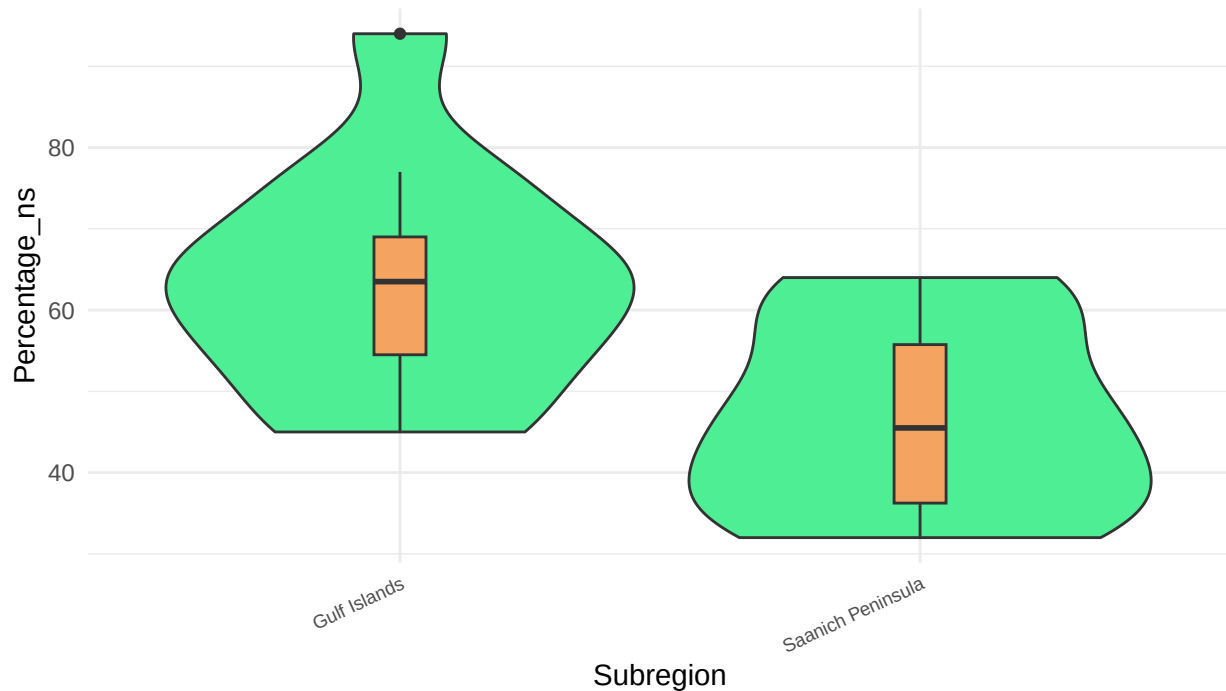


Chart by Wendy Anthony
2025-08-04

Plots by Separate Sites

- use `pivot_longer()` for single row with multiple columns convert to long

pivot longer one row to long df

```
colnames(goe_eccc_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"                "Percentage_ns"
```

```
# choose index row from data sorted alphabetically
# delete first 2 columns as they can't be characters ? also Proportion to leave Percentage
# delete year column
```

```
# Anniversary_Island
```

```
Anniversary_Island <- goe_eccc_1[1, ]
Anniversary_Island
```

```
##      Subregion      Site Proportion_of_native_species
## 1 Gulf Islands Anniversary Island                0.94
## Cultural_species_richness Exotic_species Trampling Herbivory Composite_Index
```

```
## 1          2.71          0.17          5          1.14          1.8
##   Year Percentage_ns
## 1 2023          94
```

```
Anniversary_Island <- Anniversary_Island[, -c(1,2,9)]
Anniversary_Island_long <- Anniversary_Island %>% pivot_longer(everything(), names_to = "variable", value_to = "value")
Anniversary_Island_long
```

```
## # A tibble: 7 x 2
##   variable          value
##   <chr>          <dbl>
## 1 Proportion_of_native_species 0.94
## 2 Cultural_species_richness    2.71
## 3 Exotic_species              0.17
## 4 Trampling                   5
## 5 Herbivory                  1.14
## 6 Composite_Index             1.8
## 7 Percentage_ns              94
```

```
# Gonzales
Gonzales <- goe_eccc_1[15, ]
Gonzales
```

```
##           Subregion      Site Proportion_of_native_species
## 15 Gulf Islands Owl Islet          0.65
##           Cultural_species_richness Exotic_species Trampling Herbivory Composite_Index
## 15                2          27.72    10.67    0.67          1.4
##           Year Percentage_ns
## 15 2023          65
```

```
Gonzales <- Gonzales[, -c(1,2,9)]
Gonzales
```

```
##           Proportion_of_native_species Cultural_species_richness Exotic_species
## 15                0.65                2          27.72
##           Trampling Herbivory Composite_Index Percentage_ns
## 15    10.67    0.67          1.4          65
```

```
Gonzales_long <- Gonzales %>% pivot_longer(everything(), names_to = "variable", values_to = "value")
Gonzales_long
```

```
## # A tibble: 7 x 2
##   variable          value
##   <chr>          <dbl>
## 1 Proportion_of_native_species 0.65
## 2 Cultural_species_richness    2
## 3 Exotic_species              27.7
## 4 Trampling                   10.7
## 5 Herbivory                   0.67
## 6 Composite_Index             1.4
## 7 Percentage_ns              65
```

```
# Uplands
Uplands <- goe_eccc_1[24, ]
Uplands
```

```
##           Subregion      Site Proportion_of_native_species
## 24 Saanich Peninsula Uplands Park          0.37
```

```
##      Cultural_species_richness Exotic_species Trampling Herbivory Composite_Index
## 24                1.45          49.38      2.36      5.82                1
##      Year Percentage_ns
## 24 2023                37

Uplands <- Uplands[, -c(1,2,9)]
Uplands

##      Proportion_of_native_species Cultural_species_richness Exotic_species
## 24                0.37                1.45          49.38
##      Trampling Herbivory Composite_Index Percentage_ns
## 24      2.36      5.82                1                37

Uplands_long <- Uplands %>% pivot_longer(everything(), names_to = "variable", values_to = "value")
Uplands_long

## # A tibble: 7 x 2
##   variable      value
##   <chr>      <dbl>
## 1 Proportion_of_native_species 0.37
## 2 Cultural_species_richness    1.45
## 3 Exotic_species              49.4
## 4 Trampling                   2.36
## 5 Herbivory                   5.82
## 6 Composite_Index             1
## 7 Percentage_ns               37
```

Point Plot Long Uplands

```
Uplands_long_plot <- ggplot(Uplands_long,
  aes(x = variable, y = value)) +
  # https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
  ylim(0, 100) +
  geom_point() +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Uplands Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")

Uplands_long_plot
```

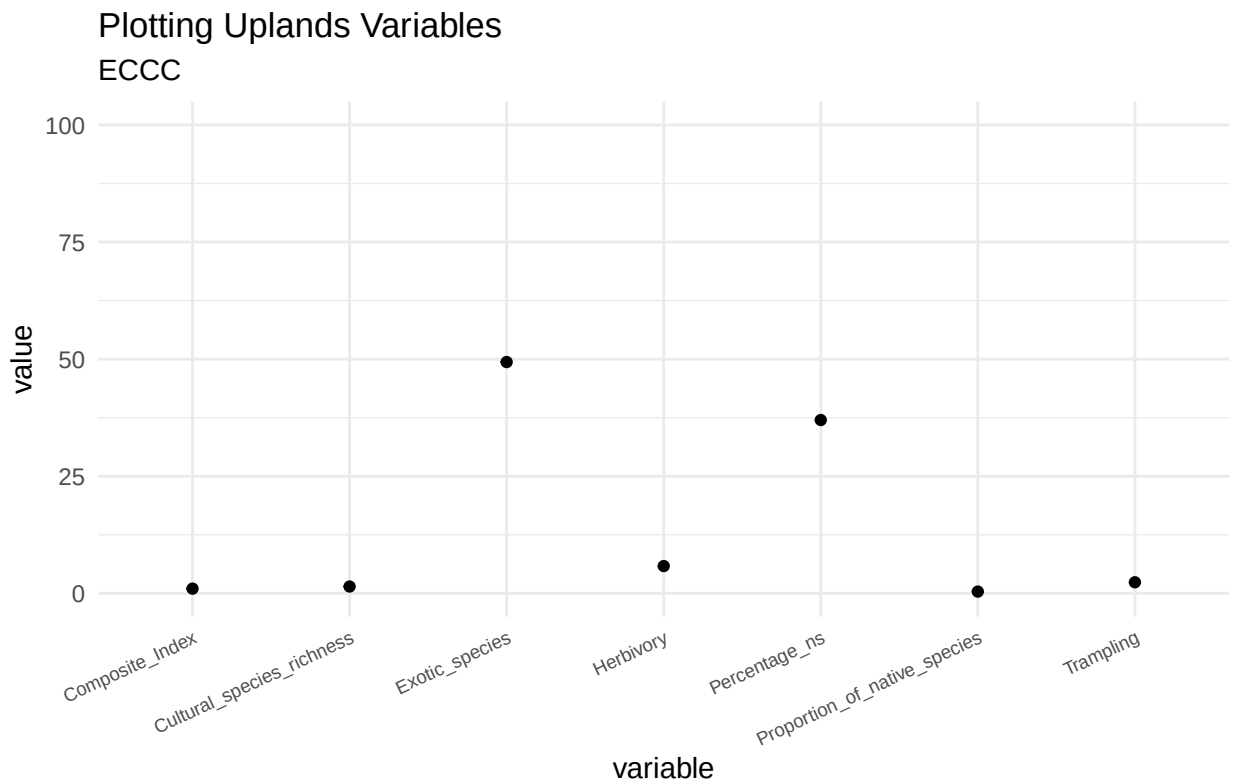


Chart by Wendy Anthony
2025-08-04

Point Plot to Compare Gonzales Variables

```
Gonzales_long_plot <- ggplot(Gonzales_long,
  aes(x = variable, y = value)) +
  # https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
  ylim(0, 100) +
  geom_point() +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Gonzales Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")
Gonzales_long_plot
```

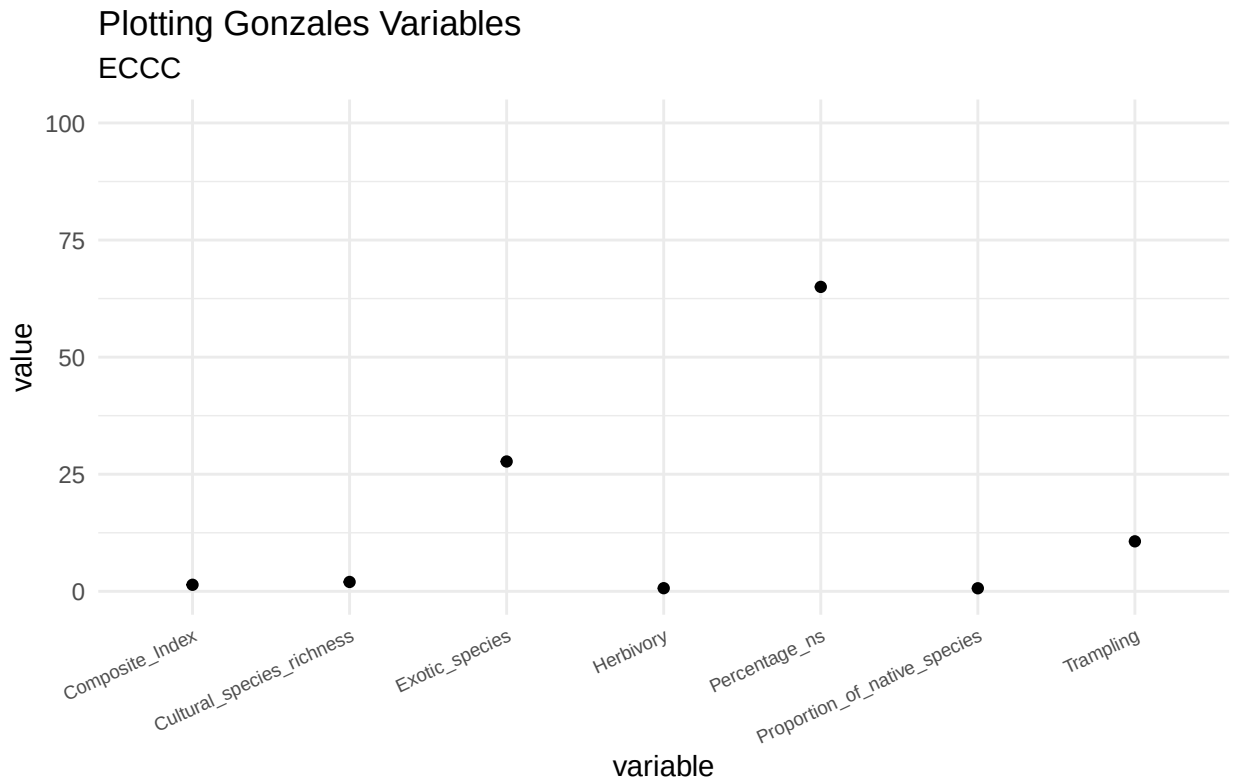


Chart by Wendy Anthony
2025-08-04

Point Plot Long Anniversary_Island_long

```
Anniversary_Island_long_plot <- ggplot(Anniversary_Island_long,
  aes(x = variable, y = value)) +
  # https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
  ylim(0, 100) +
  geom_point() +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Plotting Anniversary Island Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-04")
Anniversary_Island_long_plot
```

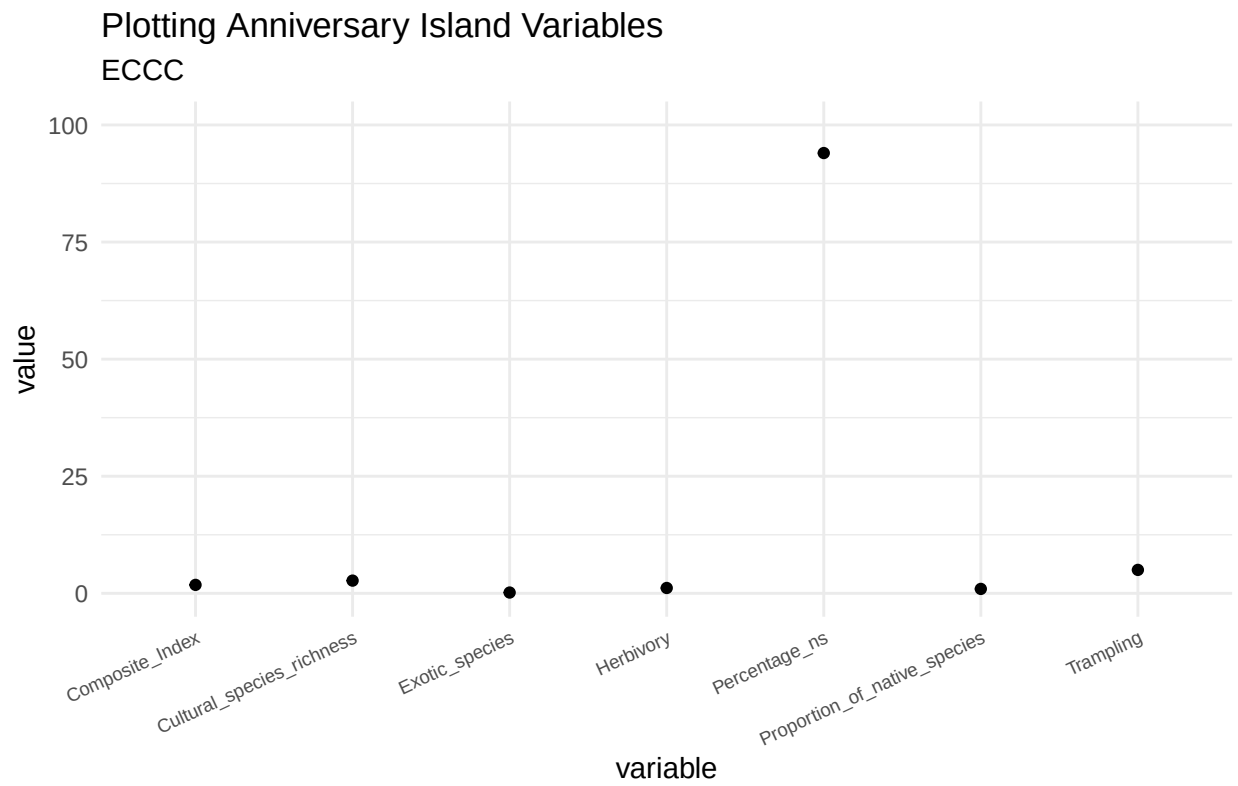


Chart by Wendy Anthony
2025-08-04

Point Plot: Individual Site - Uplands Variables of Degradation

Plotting Uplands Degradation Variables

ECCC (Malloff & Shackelford, 2024)

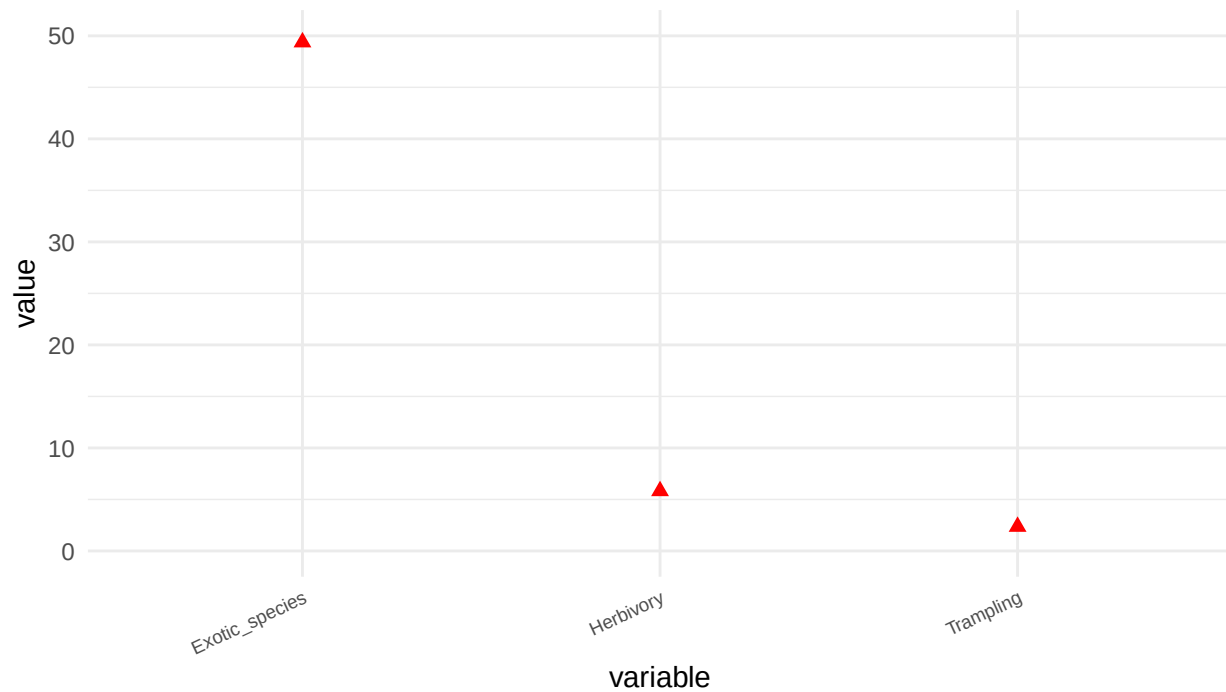


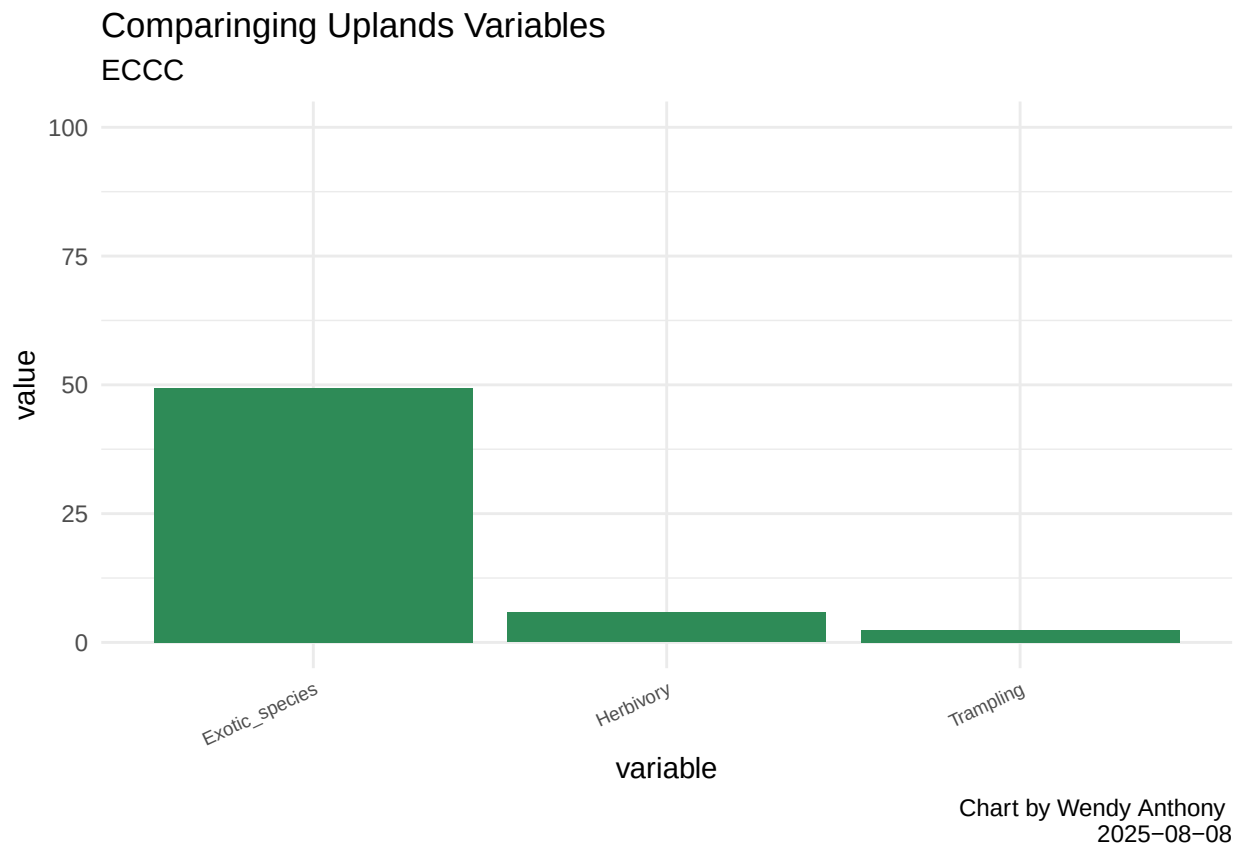
Chart by Wendy Anthony
2025-08-30

Bar Variables Compare Uplands

```
compare_bar_uplands <- ggplot(Uplands_long,
  aes(x = variable, y = value)) +

# Exotic_species
# https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
  ylim(0, 100) +
  geom_bar(stat = "identity", fill = "seagreen") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Comparing Uplands Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-08")

compare_bar_uplands
```

Compare Bar Variables multiple sites

```
names(goe_eccc_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"          "Composite_Index"
## [9] "Year"              "Percentage_ns"

compare_bar_multiple_sites <- ggplot(goe_eccc_1,
  aes(x = Site, y = Percentage_ns, fill = Subregion)) +

# Exotic_species
# https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
  ylim(0, 100) +
  geom_bar(stat = "identity") +
# Replace default palette of pink & blue
  scale_fill_brewer(palette = "Dark2") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Comparing Site Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-08")
compare_bar_multiple_sites
```

Comparing Site Variables

ECCC

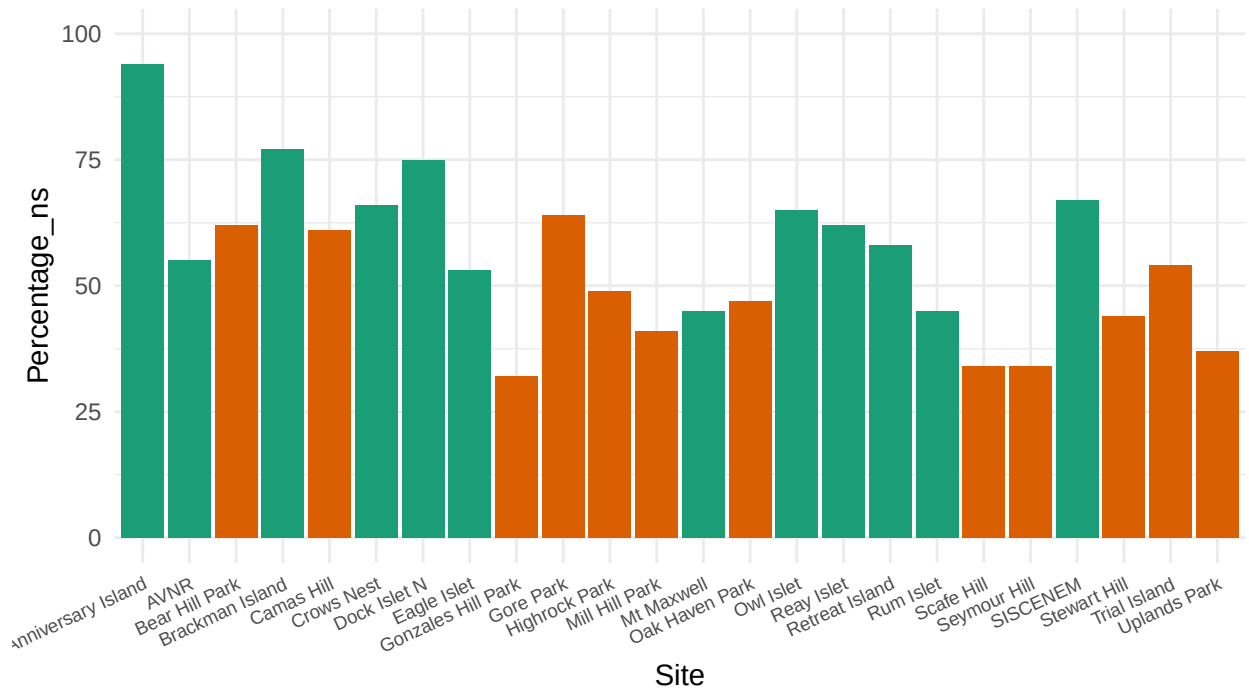


Chart by Wendy Anthony
2025-08-08

Arrange Compare Bar Variables multiple sites

<https://r-graph-gallery.com/267-reorder-a-variable-in-ggplot2.html>

```
names(goe_eccc_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"          "Composite_Index"
## [9] "Year"               "Percentage_ns"
```

```
# arrange values
```

```
#str(arr)
```

```
# goe_eccc_1 %>% arrange(Percentage_ns) -> arr
```

```
goe_eccc_1 %>% arrange(desc(Percentage_ns)) %>%
```

```
# updates Site with new arrangement
```

```
mutate(Site = factor(Site, levels = Site)) %>%
```

```
ggplot(
```

```
  aes(x = Site, y = Percentage_ns, fill = Subregion)) +
```

```
# https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
```

```
  ylim(0, 100) +
```

```
  geom_bar(stat = "identity") +
```

```
# Replace default palette of pink & blue
```

```
scale_fill_brewer(palette = "Dark2") +
```

```
  theme_minimal() + #get rid of grey background and tick marks
```

```
  theme(legend.position="bottom") +
```

```
  theme(axis.text.x = element_text(angle = 35, vjust = 1, hjust=1, size = 7)) +
```

```
labs(title='Comparing Sites',
      subtitle='ECCC GOE Site Monitoring',
      caption = "Chart by Wendy Anthony \n 2025-08-19")
```

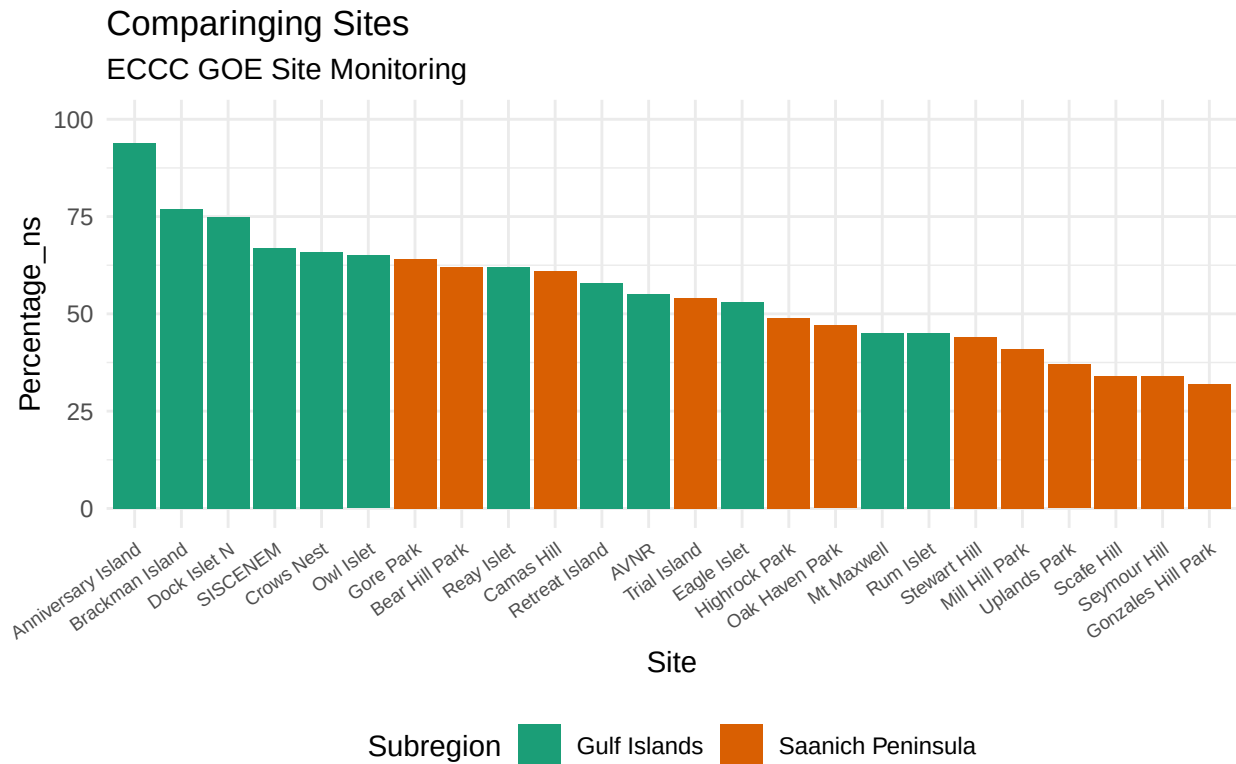


Chart by Wendy Anthony
2025-08-19

TEST Stacking Bar Variables multiple sites

```
# https://r-graph-gallery.com/267-reorder-a-variable-in-ggplot2.html
# View(mpg)
# g <- ggplot(goe_eccc_1, aes(Subregion))
# p <- g + geom_bar(aes(fill = Site))
# ggplotly(p)
#
# https://stackoverflow.com/questions/6693257/making-a-stacked-bar-plot-for-multiple-variables-ggplot2
# dfr <- data.frame(
#   V1 = c(0.1, 0.2, 0.3),
#   V2 = c(0.2, 0.3, 0.2),
#   V3 = c(0.3, 0.6, 0.5),
#   V4 = c(0.5, 0.1, 0.7),
#   row.names = LETTERS[1:3]
# )
#
# goe_eccc_2 <- subset(goe_eccc_1, , -c(Subregion))
#
# goe_eccc_2 %>% rownames_to_column("ID") %>% pivot_longer(!ID) %>%
#   ggplot() +
#   geom_col(aes(x = ID, y = value, fill = name), position = 'fill')
```

```

## Error in `pivot_longer_spec()`:
## ! Can't combine `Site` <character> and `Proportion_of_native_species` <double>.
#
# names(goe_eccc_1)
#   ggplot(goe_eccc_1,
#     aes(x = Site, y = Percentage_ns, fill = Subregion, color = Subregion)) +
#   # https://www.sthda.com/english/wiki/ggplot2-axis-scales-and-transformations
#     ylim(0, 100) +
#     geom_bar(stat = "identity") +
#     theme_minimal() + #get rid of grey background and tick marks
#     theme(legend.position="bottom") +
#     theme(axis.text.x = element_text(angle = 35, vjust = 1, hjust=1, size = 7)) +
#     labs(title='Comparing Sites',
#       subtitle='ECCC GOE Site Monitoring',
#       caption = "Chart by Wendy Anthony \n 2025-08-19")

```

??? NOTWORKING Compare Column Variables Uplands

```

colnames(goe_eccc_1)

## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"                "Percentage_ns"

goe_eccc_1_filter <- filter(goe_eccc_1, goe_eccc_1$Site %in% c("Uplands Park", "Trial Island"))

compare_col_uplands <- ggplot(goe_eccc_1_filter,
  aes(Site, fill = Exotic_species)) +
  geom_bar(position = "dodge", alpha = 0.5) +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Comparing Uplands Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-08")

compare_col_uplands

```

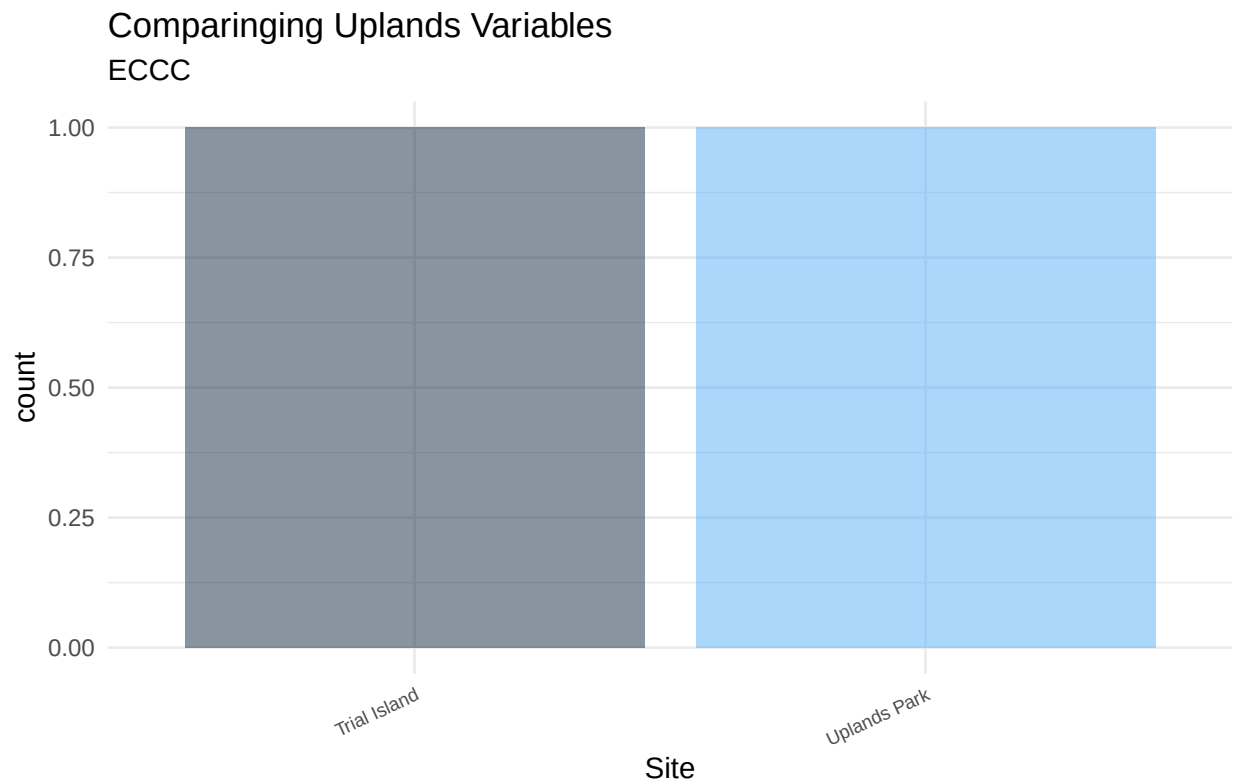


Chart by Wendy Anthony
2025-08-08

TEST NOT WORKING Facet wraps

`facet_wrap(~Subregion)`

- need to filter Subregion for each ???

```
names(goe_eccc_1)
```

```
## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"                "Percentage_ns"
```

```
ggplot(goe_eccc_1,
  aes(x = Site, y = Exotic_species)) +
  geom_col() +
  facet_wrap(~Subregion) +
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Comparing Uplands Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-19")
```

Comparing Uplands Variables

ECCC

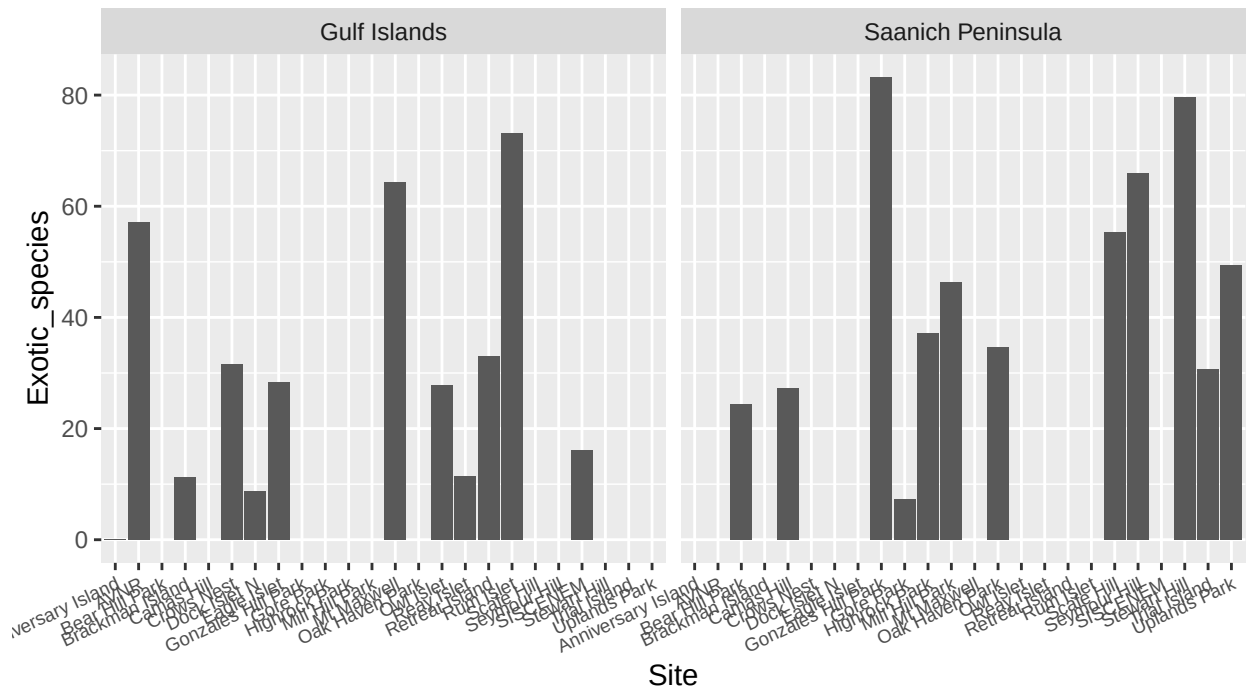


Chart by Wendy Anthony
2025-08-19

??? TEST FILTER SAME AS PREVIOUS NOT WORKINGfacet_wrap

- need to filter Subregion for each ???

```
names(goe_eccc_1)
```

```
## [1] "Subregion" "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species" "Trampling"
## [7] "Herbivory" "Composite_Index"
## [9] "Year" "Percentage_ns"
```

```
ggplot(goe_eccc_1,
  aes(x = Site, y = Exotic_species)) +
  geom_col() +
  facet_wrap(~Subregion) +
  theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  labs(title='Comparing Uplands Variables',
    subtitle='ECCC',
    caption = "Chart by Wendy Anthony \n 2025-08-19")
```

ECCC



Violin Chart Function

```
## 'data.frame':    24 obs. of  9 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site           : chr  "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num  0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness  : num  2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species           : num  0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling                : num  5 4.7 5.4 1.64 12.2 ...
## $ Herbivory                : num  1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index          : num  1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year                     : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
```

```
## [1] "Subregion" "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species" "Trampling"
## [7] "Herbivory" "Composite_Index"
## [9] "Year"
```

```
subregion_var_violin <- function(file, title, subtitle, colx, coly, xlab, ylab, caption){

  ggplot2::ggplot(file,
    ggplot2::aes(x = colx, y = coly)) +
  ggplot2::geom_violin(fill = "seagreen2") +
  ggplot2::geom_boxplot(width = 0.1, fill = "sandybrown") +
  ggplot2::theme_minimal() + #get rid of grey background and tick marks
  ggplot2::theme(legend.position="none") + #remove legend
  ggplot2::theme(
    axis.text.x = ggplot2::element_text(vjust = 1, hjust = .5, size = 8),
    axis.text.y = ggplot2::element_text(size = 5),
    plot.title = ggplot2::element_text(hjust = 0.5, size = 12),
    plot.subtitle = ggplot2::element_text(hjust = 0.5, size = 12)) +
  #theme(axis.text.x = element_text(angle = 25, vjust = 1, hjust=1, size = 7)) +
  ggplot2::labs(title=title,
    subtitle=subtitle,
    caption = caption,
    x = xlab, y = ylab)
  #subregion_exsp
}
```

```
subregion_var_violin(goe_eccc, 'Plotting Subregion by Exotic_species', 'ECCC Data Paper', goe_eccc$Subr
```

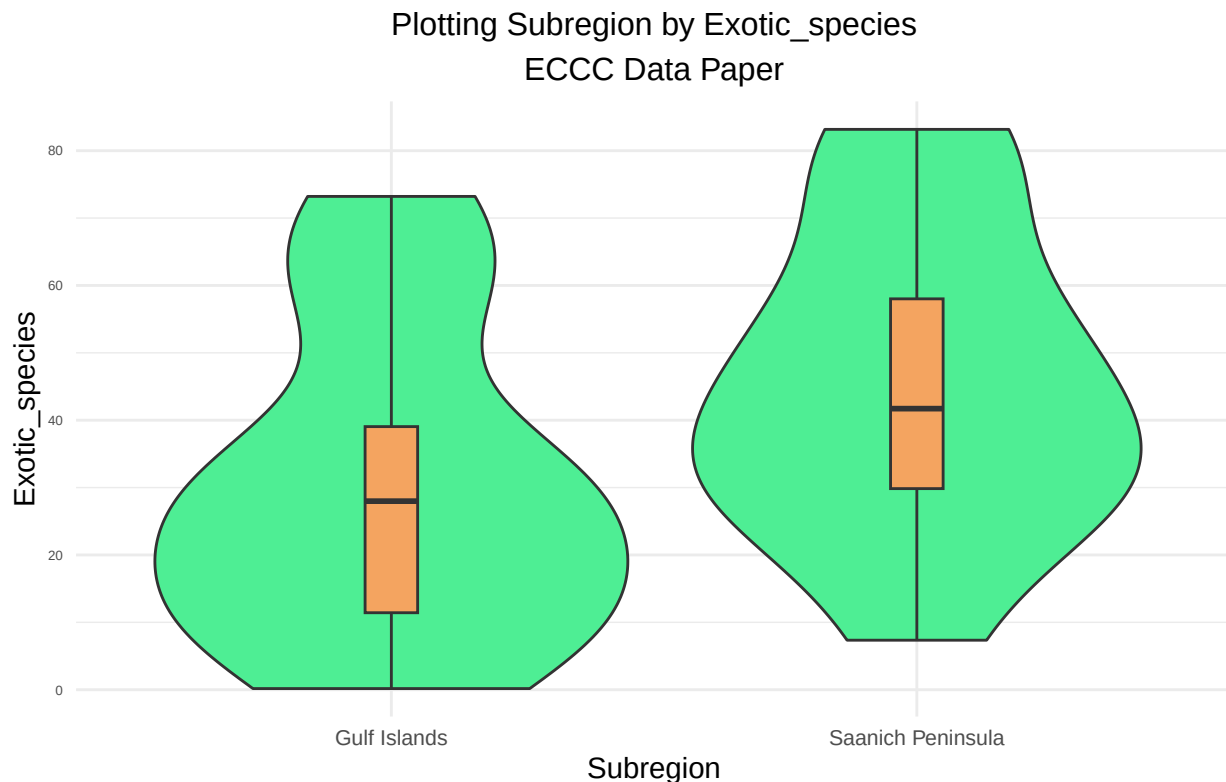


Chart by Wendy Anthony
2025-08-04

```
subregion_var_violin(goe_eccc, 'Plotting Subregion by Cultural_species_richness', 'ECCC Data Paper', goe.
```

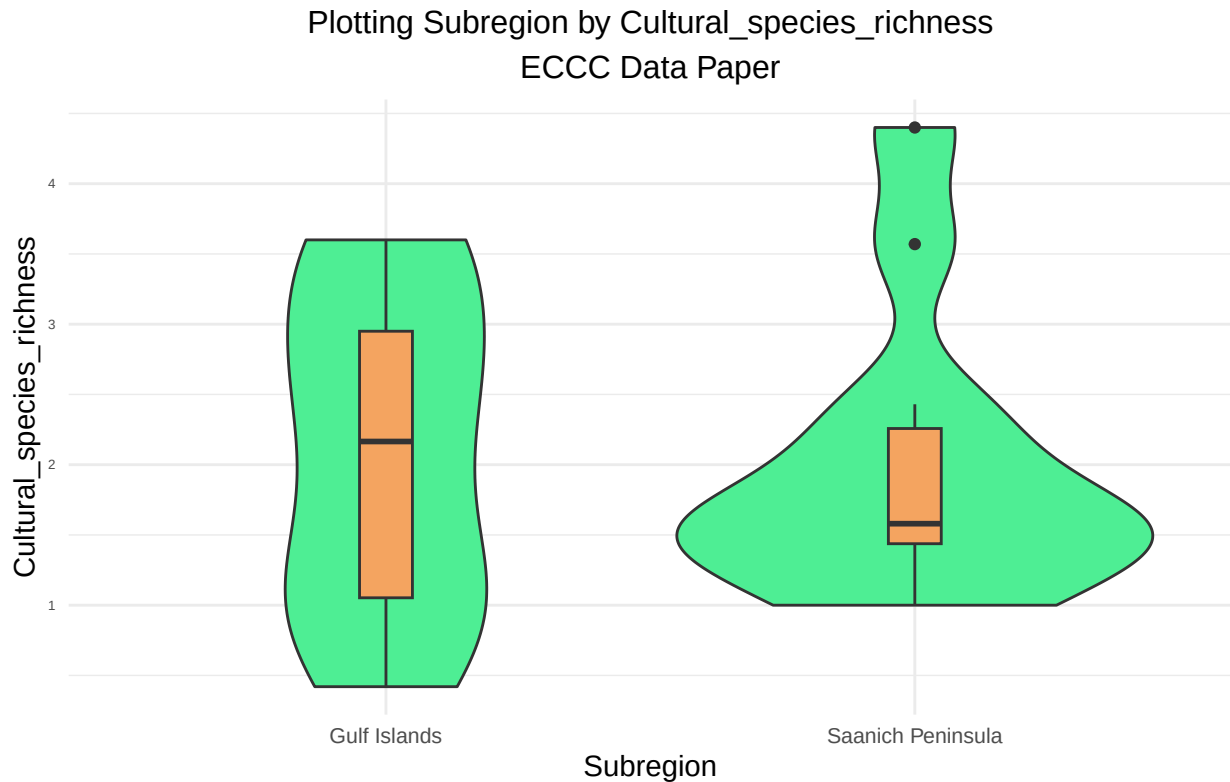



Chart by Wendy Anthony
2025-08-16

Point Chart Function

ggplot point chart

```
goe_eccc <- read.csv("data/SiteData-TEST-GOE-Monitor.csv", header = TRUE, sep = ",", stringsAsFactors=F)
str(goe_eccc)
```

```
## 'data.frame': 24 obs. of 9 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness : num 2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species : num 0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling : num 5 4.7 5.4 1.64 12.2 ...
## $ Herbivory : num 1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index : num 1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
```

```
names(goe_eccc)
```

```
## [1] "Subregion" "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species" "Trampling"
## [7] "Herbivory" "Composite_Index"
## [9] "Year"
```

```
subregion_var_point <- function(file, title, subtitle, colx, coly, xlab, ylab, caption){
  ggplot(file,
    aes(x = colx, y = coly, colour = Subregion)) +
    geom_point(shape = 18, size = 2) +
    # Replace default palette of pink & blue
    scale_fill_brewer(palette = "Dark2") +
    theme_minimal() + #get rid of grey background and tick marks
    theme(legend.position="bottom") +
    # theme(legend.position="none") + #remove legend
    theme(axis.text.x = element_text(angle = 35, vjust = 1, hjust=1, size = 7)) +
    labs(title=title,
      subtitle=subtitle,
      caption = caption,
      x = xlab, y = ylab)
}

subregion_var_point(goe_eccc, 'Plotting Site by Percentage Native Species', 'Malloff & Shackelford. (2024)',
  goe_eccc$Site, goe_eccc$Proportion_of_native_species, "Site", "Proportion of native species", "Chart by Wendy Anthony \n 2025-09-16")
```

Plotting Site by Percentage Native Species

Malloff & Shackelford. (2024).

Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems.
Restoration Futures Lab at the University of Victoria.

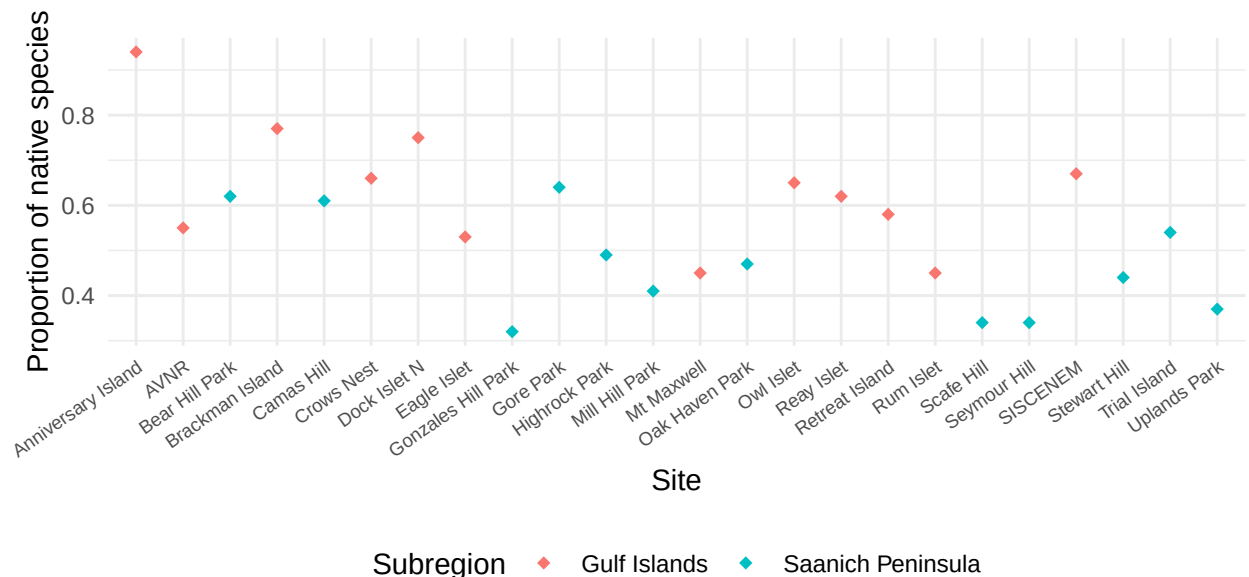


Chart by Wendy Anthony

2025-09-16

ECCC Data: (Malloff & Shackelford, 2024)

```
subregion_var_point(goe_eccc, 'Plotting Site by Exotic Species', 'Malloff & Shackelford. (2024). \nFeelin',
  goe_eccc$Site, goe_eccc$Exotic_species, "Site", "Exotic Species", "Chart by Wendy Anthony \n 2025-09-16")
```

Plotting Site by Exotic Species

Malloff & Shackelford. (2024).

Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems.
Restoration Futures Lab at the University of Victoria.

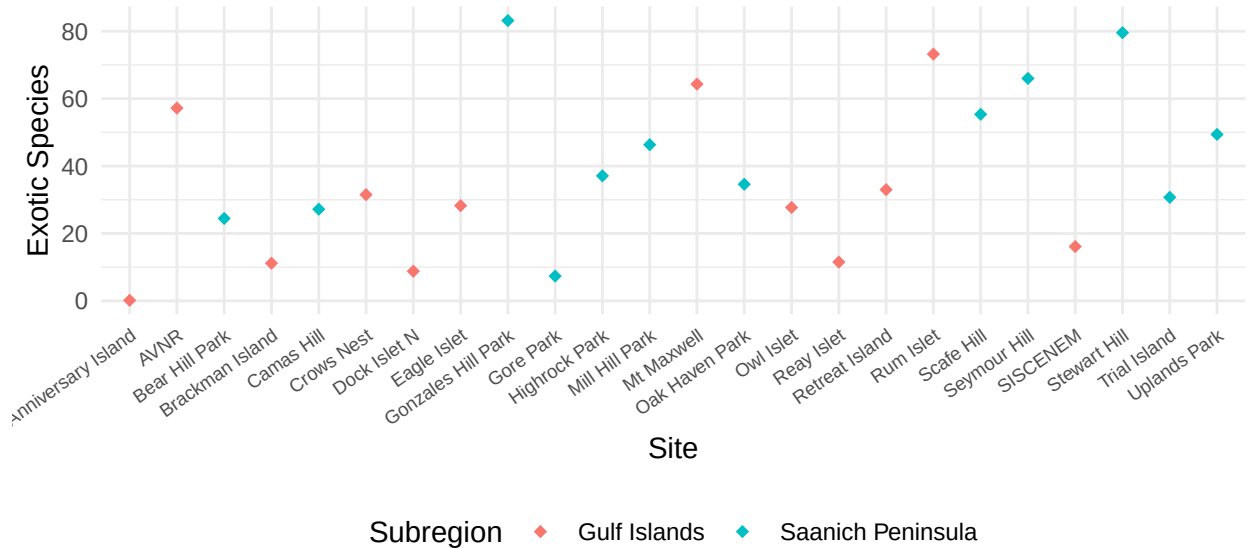


Chart by Wendy Anthony

2025-09-16

ECCC Data: (Malloff & Shackelford, 2024)

plotly ggplot point chart

```
goe_eccc <- read.csv("data/SiteData-TEST-GOE-Monitor.csv", header = TRUE, sep = ",", stringsAsFactors=F
```

```
str(goe_eccc)
```

```
## 'data.frame': 24 obs. of 9 variables:
## $ Subregion : chr "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site : chr "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num 0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness : num 2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species : num 0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling : num 5 4.7 5.4 1.64 12.2 ...
## $ Herbivory : num 1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index : num 1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year : int 2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
```

```
names(goe_eccc)
```

```
## [1] "Subregion" "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species" "Trampling"
## [7] "Herbivory" "Composite_Index"
## [9] "Year"
```

```
subregion_var_point <- function(file, title, subtitle, colx, coly, xlab, ylab, caption){
```

```
ggplot(file,
```

```

aes(x = colx, y = coly, colour = Subregion)) +
  geom_point(shape = 18, size = 2) +
  # Replace default palette of pink & blue
  scale_fill_brewer(palette = "Dark2") +
  theme_minimal() + #get rid of grey background and tick marks
  theme(legend.position="bottom") +
  # theme(legend.position="none") + #remove legend
  theme(axis.text.x = element_text(angle = 35, vjust = 1, hjust = 1, size = 7)) +
  labs(title=title,
       subtitle=subtitle,
       caption = caption,
       x = xlab, y = ylab)
}

p2 <- subregion_var_point(goe_eccc, 'Plotting Site by Percentage Native Species', 'Malloff & Shackelford
', goe_eccc$Site, goe_eccc$Proportion_of_native_species, "Site", "Proportion of native species", "Chart
subregion_var_point(goe_eccc, 'Plotting Site by Exotic Species', 'Malloff & Shackelford. (2024). \nFeelin
', goe_eccc$Site, goe_eccc$Exotic_species, "Site", "Exotic Species", "Chart by Wendy Anthony \n 2025-09-16

```

Plotting Site by Exotic Species

Malloff & Shackelford. (2024).

Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems.
Restoration Futures Lab at the University of Victoria.

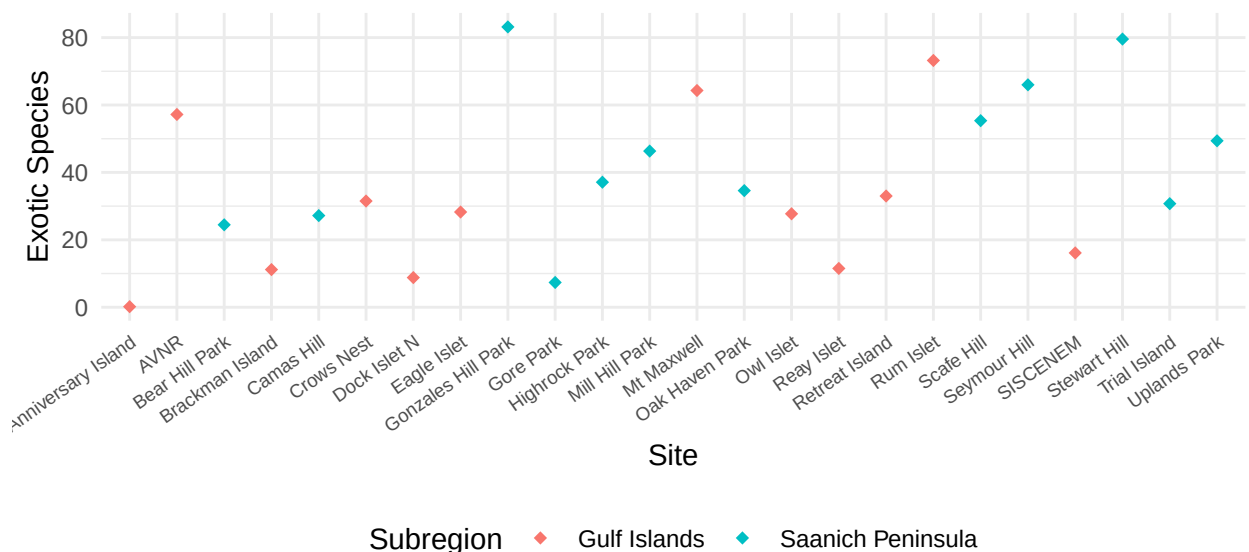
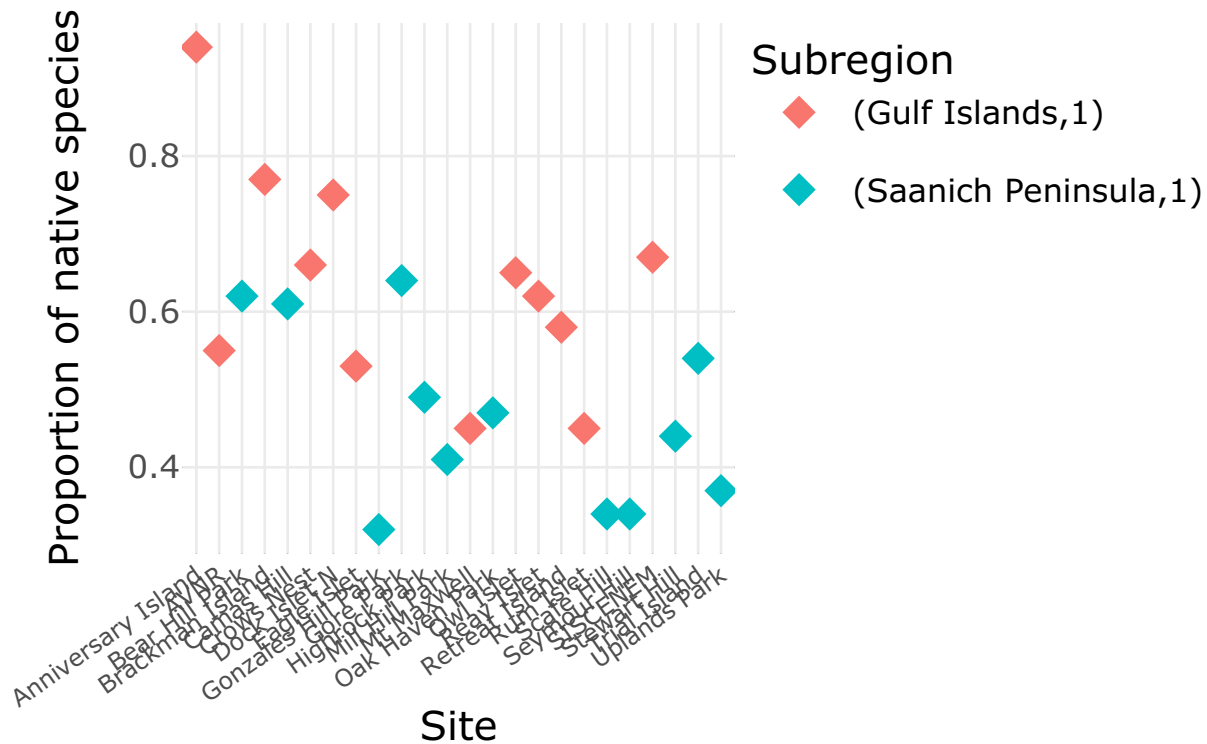


Chart by Wendy Anthony
2025-09-16
ECCC Data: (Malloff & Shackelford, 2024)

```
ggplotly(p2)
```

Plotting Site by Percentage Native Species



Leaflet Map

Custom Icons

```
Sites_eccc <- read.csv("data/SiteDetails-TEST-GOE-Monitor.csv", header = TRUE, sep = ",")

names(Sites_eccc)

## [1] "Subregion"           "Site"                 "Land.Manager"
## [4] "Main.Restoration.Type" "Restoration.Intensity" "Area.ha"
## [7] "Lat"                 "Lng"

library(dplyr)
library(leaflet)
library(htmlwidgets) # for responsive web map

# custom icon
# https://leafletjs.com/examples/custom-icons/
greenLeafIcon <- makeIcon(
  iconUrl = "https://leafletjs.com/examples/custom-icons/leaf-green.png",
  iconWidth = 38, iconHeight = 95,
  iconAnchorX = 22, iconAnchorY = 94,
  shadowUrl = "https://leafletjs.com/examples/custom-icons/leaf-shadow.png",
  shadowWidth = 50, shadowHeight = 64,
  shadowAnchorX = 4, shadowAnchorY = 62
)
# change LeafIcon size 1/2
```

```

greenLeafIconSm <- makeIcon(
  iconUrl = "https://leafletjs.com/examples/custom-icons/leaf-green.png",
  iconWidth = 19, iconHeight = 47,
  iconAnchorX = 11, iconAnchorY = 47,
  shadowUrl = "https://leafletjs.com/examples/custom-icons/leaf-shadow.png",
  shadowWidth = 25, shadowHeight = 32,
  shadowAnchorX = 2, shadowAnchorY = 31
)

# Title control
title <- tags$div(HTML('<b>ECC Site Maps</b><br>Malloff & Shackelford. (2024). Feeling the Pulse: Monit

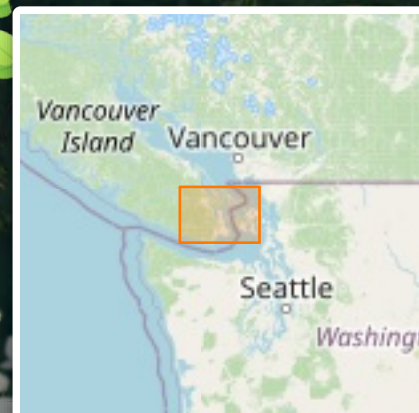
ECCC_map <- leaflet(Sites_eccc) %>%
  addProviderTiles("Esri.WorldImagery") %>%
  addMarkers(
    ~ Lng, ~ Lat, icon = greenLeafIconSm,
    popup = paste0("<b>Subregion:</b> ", Sites_eccc$Subregion, "<br>", "<b>Site:</b> ", Sites_e
  setView(-123.44799, 48.64919, 9) %>%
  addMiniMap(width = 150, height = 150, zoomLevelOffset = -4) %>%
  addControl(title, position = "topright")

# Display map
ECCC_map

```

ECC Site Maps

Malloff & Shackelford. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems. Restoration Futures Lab at the University of Victoria



Leaflet | Tiles © Esri — Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

```
# Save map
saveWidget(ECCC_map, "output/ECCC_map_custom_icons.html")
```

Circle Markers

```
# NOT WORKING - Sites not same
#goe_eccc_sites <- cbind(goe_eccc, Sites_eccc)
# try to add , "<br>", "<b>Composite Index:</b> ", goe_eccc_sites$Composite_Index
# names(goe_eccc_sites)
names(Sites_eccc)

## [1] "Subregion"          "Site"                "Land.Manager"
## [4] "Main.Restoration.Type" "Restoration.Intensity" "Area.ha"
## [7] "Lat"                "Lng"

library(dplyr)
library(leaflet)
library(htmlwidgets) # for responsive web map

# Title control
title <- tags$div(HTML('<b>ECC Site Maps</b><br>Malloff & Shackelford. (2024). Feeling the Pulse: Monit

# https://rstudio.github.io/leaflet/articles/markers.html
# Create a palette that maps factor levels to colors
pal <- colorFactor(c("green", "orange"), domain = c("Saanich Peninsula", "Gulf Islands"))

ECCC_map <- leaflet(Sites_eccc) %>%
  addProviderTiles("Esri.WorldImagery") %>%
  addCircleMarkers(
    ~ Lng, ~ Lat, radius = ~ifelse(Subregion == "Saanich Peninsula", 4, 5),
    color = ~pal(Subregion),
    stroke = FALSE, fillOpacity = 0.5,
    popup = paste0("<b>Site:</b> ", Sites_eccc$Site, "<br>", "<b>Subregion:</b> ", Sites_eccc$
  setView(-123.44799, 48.64919, 9) %>%
  addMiniMap(width = 150, height = 150, zoomLevelOffset = -4) %>%
  addControl(title, position = "topright")

# Display map
ECCC_map
```


ECC Site Maps

Malloff & Shackelford. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems. Restoration Futures Lab at the University of Victoria



Leaflet | Tiles © Esri — Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

```
# Save map
saveWidget(ECCC_map, "output/ECCC_map_circle_markers.html")
```

Circle Markers Sized to Area

```
names(Sites_eccc)
```

```
## [1] "Subregion"          "Site"                "Land.Manager"
## [4] "Main.Restoration.Type" "Restoration.Intensity" "Area.ha"
## [7] "Lat"                "Lng"
```

```
library(dplyr)
library(leaflet)
library(htmlwidgets) # for responsive web map
```

```
# Title control
```

```
# Malloff & Shackelford. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems
title <- tags$div(HTML('<p style="font-size:12px;font-weight:bold">ECC Sites Map</p>'))
```

```
# https://rstudio.github.io/leaflet/articles/markers.html
```

```
# Create a palette that maps factor levels to colors
```

```
pal <- colorFactor(c("green", "orange"), domain = c("Saanich Peninsula", "Gulf Islands"))
```

```
ECCC_map <- leaflet(Sites_eccc) %>%
  addProviderTiles("Esri.WorldImagery") %>%
```

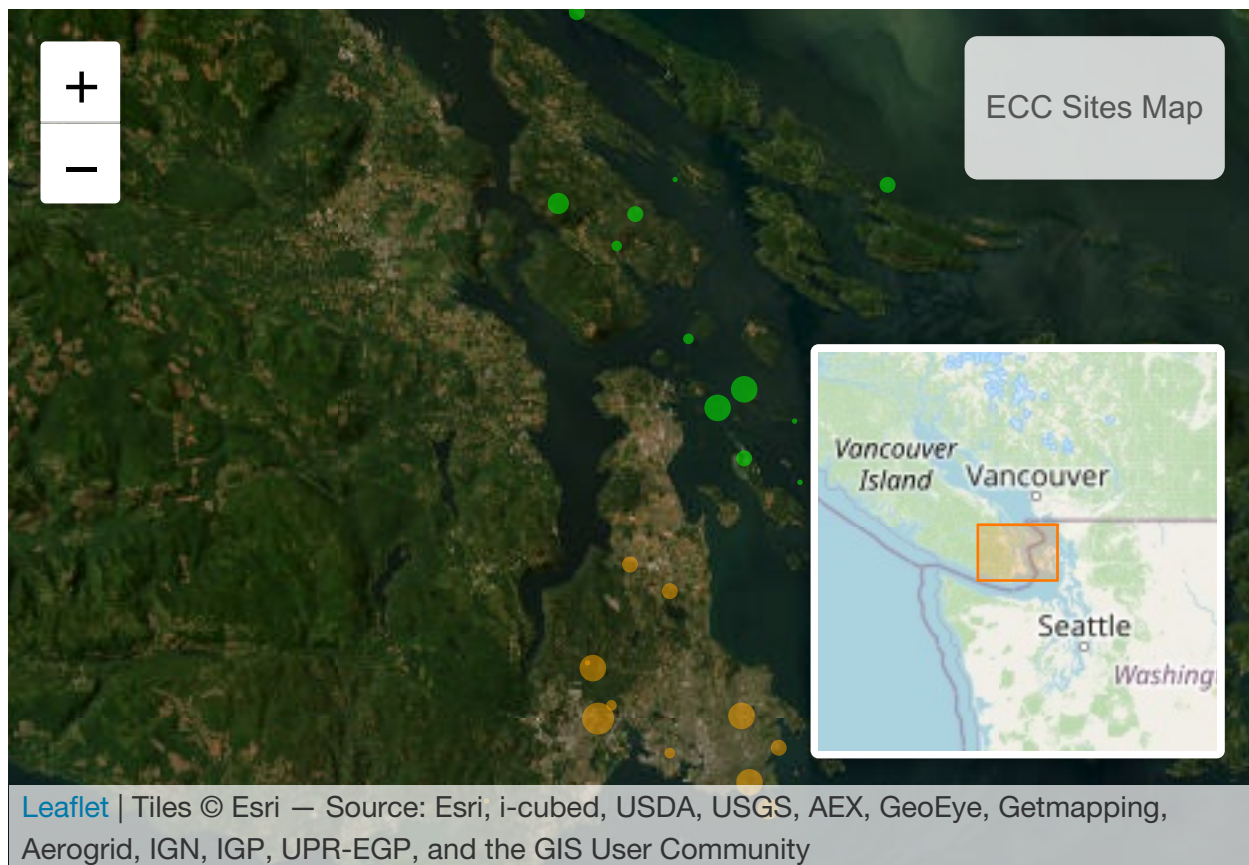


```

addCircleMarkers(
  ~ Lng, ~ Lat,
# https://stackoverflow.com/questions/40137212/variable-marker-size-feature-in-leaflet-r
  radius = ~goe_eccc$Composite_Index * 3,
  # radius = ~Sites_eccc$Area.ha,
  # radius = ~sqrt(Sites_eccc$Area.ha),
# size smaller is Saanich, else larger if Gulf
# radius = ~ifelse(Subregion == "Saanich Peninsula", 4, 5),
  color = ~pal(Subregion),
  stroke = FALSE, fillOpacity = 0.5,
  popup = paste0("<b>Site:</b> ", Sites_eccc$Site, "<br>", "<b>Subregion:</b> ", Sites_eccc$,
setView(-123.44799, 48.64919, 9) %>%
addMiniMap(width = 150, height = 150, zoomLevelOffset = -4) %>%
addControl(title, position = "topright")

# Display map
ECCC_map

```



```

# Save map
saveWidget(ECCC_map, "output/ECCC_map_area.html")

```

Circle Markers Sized to Composite_Index

goe_eccc_all_site_data.csv

```
goe_eccc_sites_1 <- read.csv("data/goe_eccc_all_site_data.csv")
names(goe_eccc_sites_1)
```

```
## [1] "Subregion"           "Site"
## [3] "Land_Manager"        "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                 "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Trampling"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"
```

```
library(dplyr)
library(leaflet)
library(htmlwidgets) # for responsive web map
```

```
# Title control
```

```
# Malloff & Shackelford. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak mead
```

```
title <- tags$div(HTML('<p style="font-size:12px;font-weight:bold">ECC Sites Map</p>'))
```

```
# https://rstudio.github.io/leaflet/articles/markers.html
```

```
# Create a palette that maps factor levels to colors
```

```
pal <- colorFactor(c("#d7191c", "#2c7bb6"), domain = c("Saanich Peninsula", "Gulf Islands"))
```

```
ECCC_map_1 <- leaflet(goe_eccc_sites_1) %>%
```

```
  addProviderTiles("Esri.WorldImagery") %>%
```

```
  addCircleMarkers(
```

```
    ~ Lng, ~ Lat,
```

```
# https://stackoverflow.com/questions/40137212/variable-marker-size-feature-in-leaflet-r
```

```
    radius = ~goe_eccc_sites_1$Composite_Index * 5,
```

```
    # radius = ~goe_eccc_sites_1$Area_ha,
```

```
    # radius = ~sqrt(goe_eccc_sites_1$Area_ha),
```

```
    # size smaller is Saanich, else larger if Gulf
```

```
#    radius = ~ifelse(Subregion == "Saanich Peninsula", 4, 5),
```

```
    fillColor = ~pal(Subregion),
```

```
    color = "black",
```

```
    weight = 3, # size of circle border
```

```
    stroke = TRUE, fillOpacity = 0.5,
```

```
    popup = paste0("<b>Site:</b> ", "<b>", "<b>", goe_eccc_sites_1$Site, "</b>", "</b>", "<br>
```

```
    "<b>Latitude:</b> ", goe_eccc_sites_1$Lat, "<br>", "<b>Longitude:</b> ", goe_eccc_sites_1$Lng,
```

```
    "<b>Land Manager:</b> ", goe_eccc_sites_1$Land_Manager, "<br>",
```

```
    "<b>Main Restoration:</b> ", goe_eccc_sites_1$Main_Restoration_Type, "<br>",
```

```
    "<b>Restoration Intensity:</b> ", goe_eccc_sites_1$Restoration_Intensity, "<br>
```

```
    "<b> Area:</b> ", goe_eccc_sites_1$Area_ha, " ha",
```

```
    "<br><br>",
```

```
    "<b>Proportion of native species:</b> ", goe_eccc_sites_1$Proportion_of_native_species,
```

```
    "<b>Cultural species richness:</b> ", goe_eccc_sites_1$Cultural_species_richness,
```

```
    "<b>Exotic species:</b> ", goe_eccc_sites_1$Exotic_species, "<br>",
```

```
    "<b>Trampling:</b> ", goe_eccc_sites_1$Trampling, "<br>",
```

```
    "<b>Herbivory:</b> ", goe_eccc_sites_1$Herbivory, "<br>",
```

```
    "<b>Composite Index:</b> ", goe_eccc_sites_1$Composite_Index, "<br>")) %>%
```

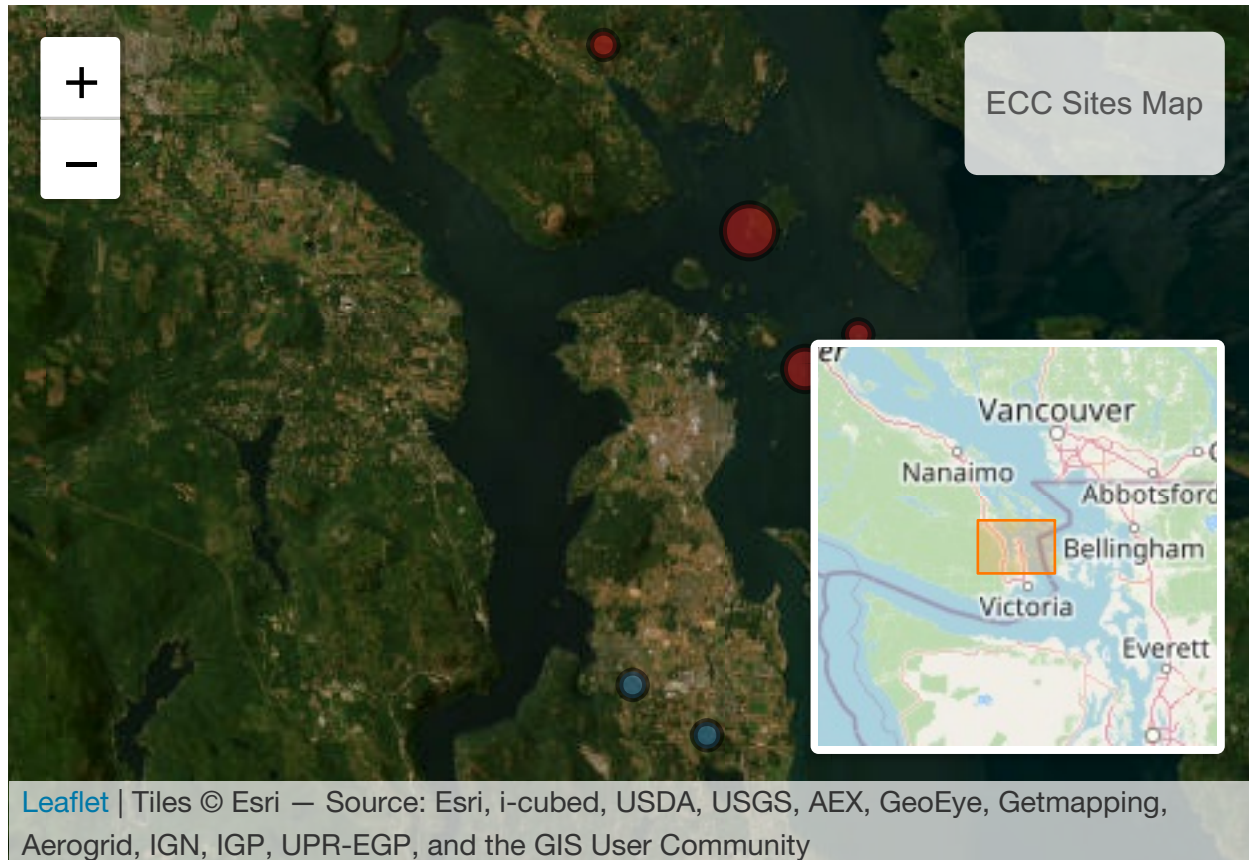
```
  setView(-123.44799, 48.64919, 10) %>%
```

```
  addMiniMap(width = 150, height = 150, zoomLevelOffset = -4) %>%
```

```
addControl(title, position = "topright")
```

```
# Display map
```

```
ECCC_map_1
```



Leaflet | Tiles © Esri — Source: Esri, i-cubed, USDA, USGS, AEX, GeoEye, Getmapping, Aerogrid, IGN, IGP, UPR-EGP, and the GIS User Community

```
# Save map
```

```
saveWidget(ECCC_map_1, "output/ECCC_map_CI.html")
```

Circle Markers Sized to Radius

Title code

```
goe_eccc_all_site_data.csv
```

```
goe_eccc_sites_1 <- read.csv("data/goe_eccc_all_site_data.csv")
```

```
names(goe_eccc_sites_1)
```

```
## [1] "Subregion"           "Site"
## [3] "Land_Manager"        "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                 "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Trampling"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"
```

```
library(dplyr)
library(leaflet)
```

```

library(htmlwidgets) # for responsive web map

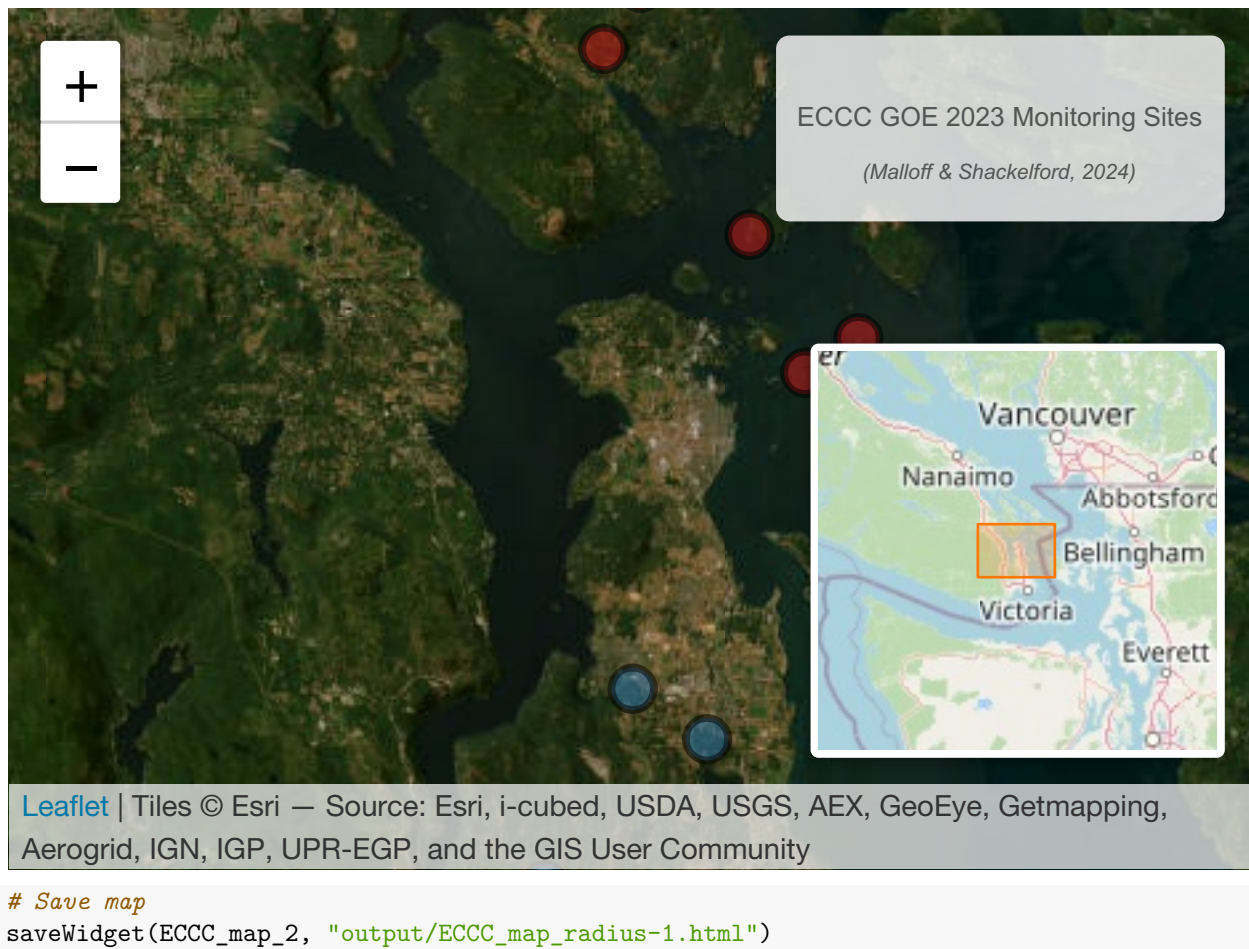
# Title control
# Malloff & Shackelford. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak mead
title <- '<p style="text-align: center; height: 30px; ">
<span style="font-size:10px;font-weight:bold; background-color: rgba(255, 255, 255, 0.9);">ECCC GOE 20
<span style="font-size:8px;font-style:italic; background-color: rgba(255, 255, 255, 0.9);">(Malloff & Sha

# https://rstudio.github.io/leaflet/articles/markers.html
# Create a palette that maps factor levels to colors
pal <- colorFactor(c("#d7191c", "#2c7bb6"), domain = c("Saanich Peninsula", "Gulf Islands"))

ECCC_map_2 <- leaflet(goe_eccc_sites_1) %>%
  addProviderTiles("Esri.WorldImagery") %>%
  addCircleMarkers(
    ~ Lng, ~ Lat,
    # https://stackoverflow.com/questions/40137212/variable-marker-size-feature-in-leaflet-r
    #
    radius = ~goe_eccc_sites_1$Composite_Index * 5,
    # radius = ~goe_eccc_sites_1$Area.ha,
    # radius = ~sqrt(goe_eccc_sites_1$Area.ha),
    # size smaller is Saanich, else larger if Gulf
    radius = ~ifelse(Subregion == "Saanich Peninsula", 8, 8),
    fillColor = ~pal(Subregion),
    color = "black",
    weight = 3, # size of circle border
    stroke = TRUE, fillOpacity = 0.5,
    popup = paste0("<b>ECCC GOE 2023 Monitoring Data</b>", "<br>", "<i>(Malloff & Shackelford,
      "<b>Site:</b> ", "<b>", goe_eccc_sites_1$Site, "</b>", "</b>", "<br><br>", "<b>Subregion
        "<b>Latitude:</b> ", goe_eccc_sites_1$Lat, "<br>", "<b>Longitude:</b> ", goe_eccc
        "<b>Land Manager:</b> ", goe_eccc_sites_1$Land_Manager, "<br>",
        "<b>Main Restoration:</b> ", goe_eccc_sites_1$Main_Restoration_Type, "<br>",
        "<b>Restoration Intensity:</b> ", goe_eccc_sites_1$Restoration_Intensity, "<br>
        "<b> Area:</b> ", goe_eccc_sites_1$Area_ha, " ha",
        "<br><br>",
        "<b>Proportion of native species:</b> ", goe_eccc_sites_1$Proportion_of_native_s
        "<b>Cultural species richness:</b> ", goe_eccc_sites_1$Cultural_species_richnes
        "<b>Exotic species:</b> ", goe_eccc_sites_1$Exotic_species, "<br>",
        "<b>Trampling:</b> ", goe_eccc_sites_1$Trampling, "<br>",
        "<b>Herbivory:</b> ", goe_eccc_sites_1$Herbivory, "<br>",
        "<b>Composite Index:</b> ", goe_eccc_sites_1$Composite_Index, "<br>")) %>%
  setView(-123.44799, 48.64919, 10) %>%
  # add controls
  addMiniMap(width = 150, height = 150, zoomLevelOffset = -4) %>%
  addControl(title, position = "topright")

# Display map
ECCC_map_2

```

Mermaid Methods Flow Chart

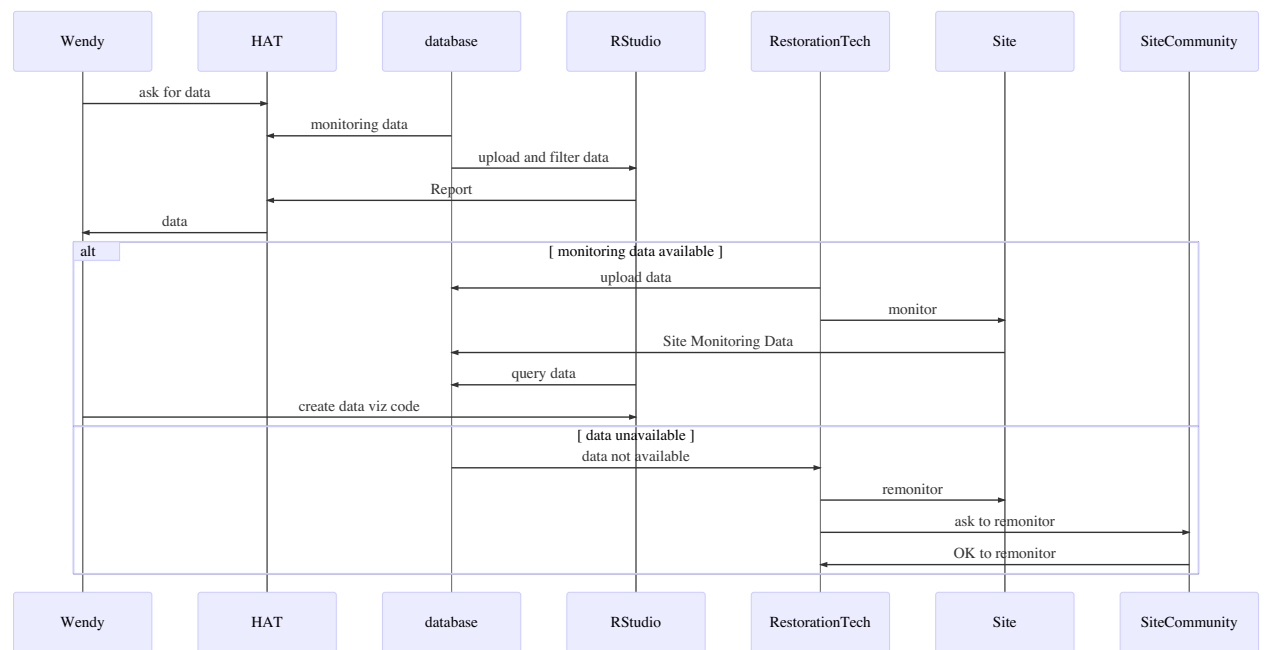
Flow for ER390 Monitoring Data

```
Diagrammer("
sequenceDiagram;
    Wendy->>HAT: ask for data;
    database->>HAT: monitoring data;
    database->>RStudio: upload and filter data;
    RStudio->>HAT: Report;
    HAT->>Wendy: data;
    alt monitoring data available
        RestorationTech->>database: upload data;
        RestorationTech->>Site: monitor;
        Site->>database: Site Monitoring Data;
        RStudio->>database: query data;
        Wendy->>RStudio: create data viz code;
    else data unavailable
        database->>RestorationTech: data not available;
        RestorationTech->>Site: remonitor;
        RestorationTech->>SiteCommunity: ask to remonitor;
    end
")
```

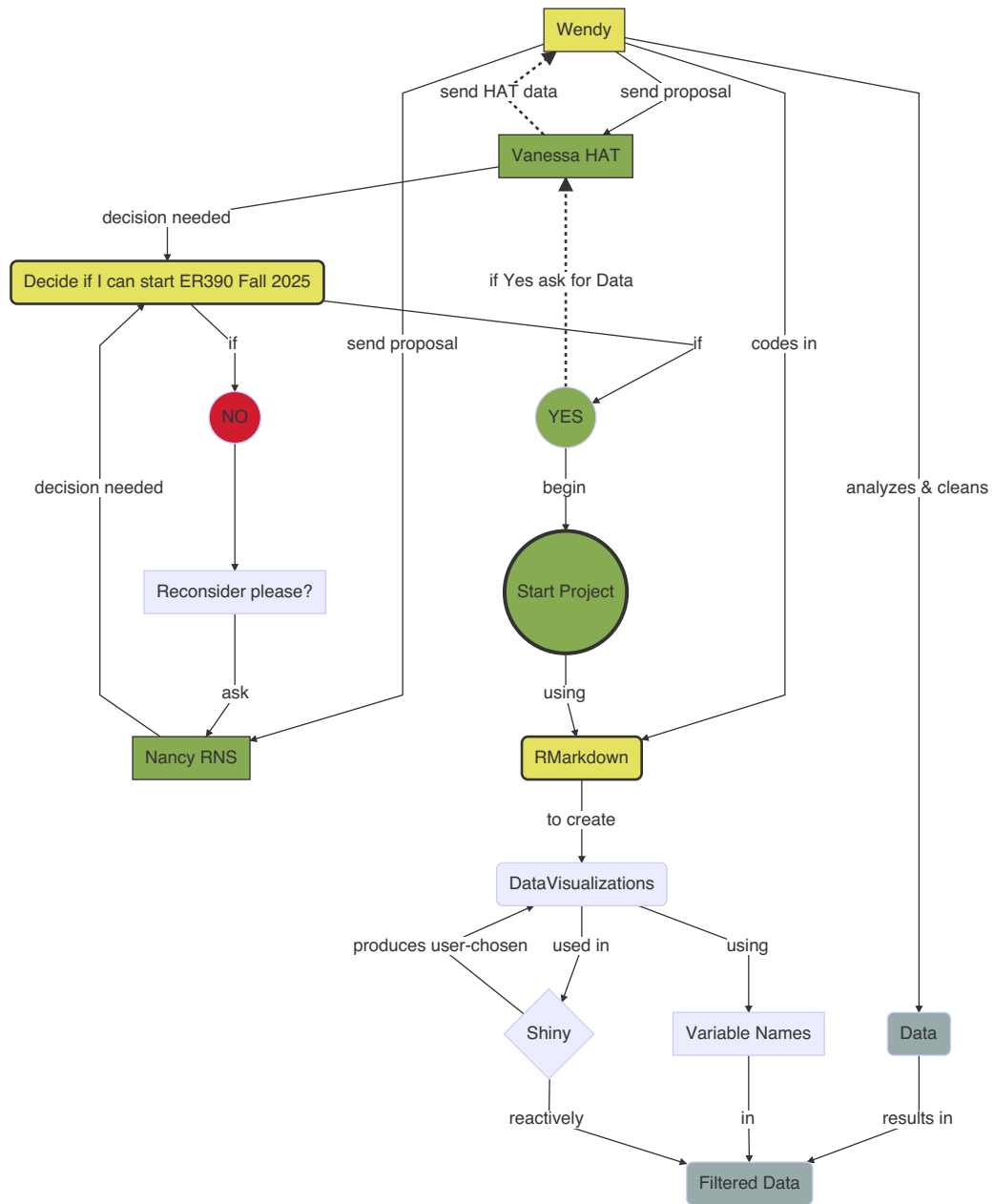
```

SiteCommunity->>RestorationTech: OK to remonitor;
end
")

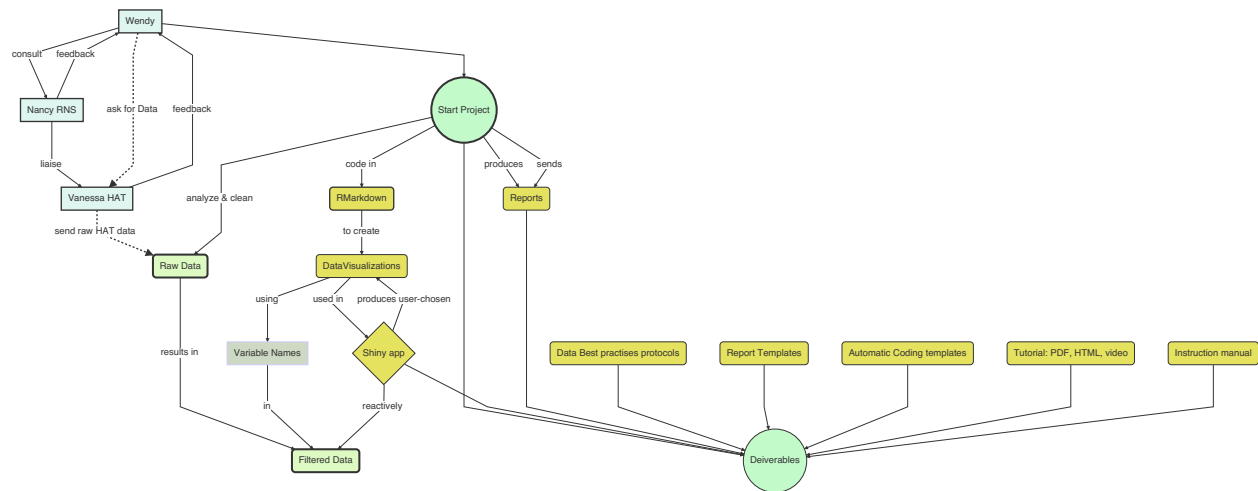
```



Graph for ER390 Proposal Methods Workflow



Workflow Graph



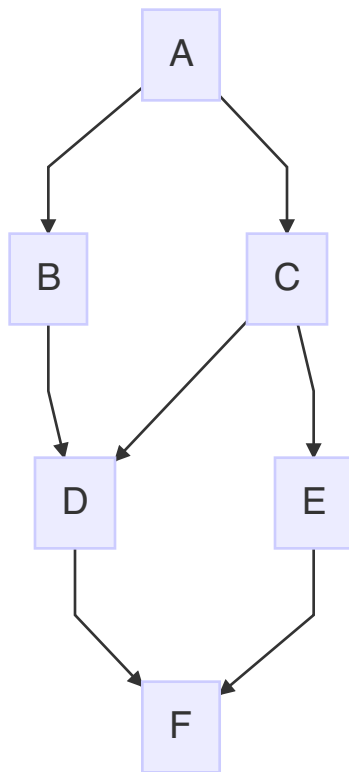
Examples

Example graph

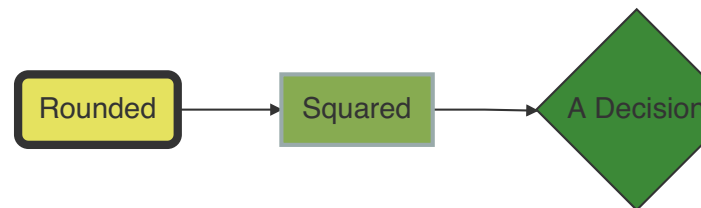
```

DiagrammeR("
graph TB
  A--->B
  A--->C
  C--->E
  B--->D
  C--->D
  D--->F
  E--->F
")

```

example Diagrammer shapes



example Mermaid shapes

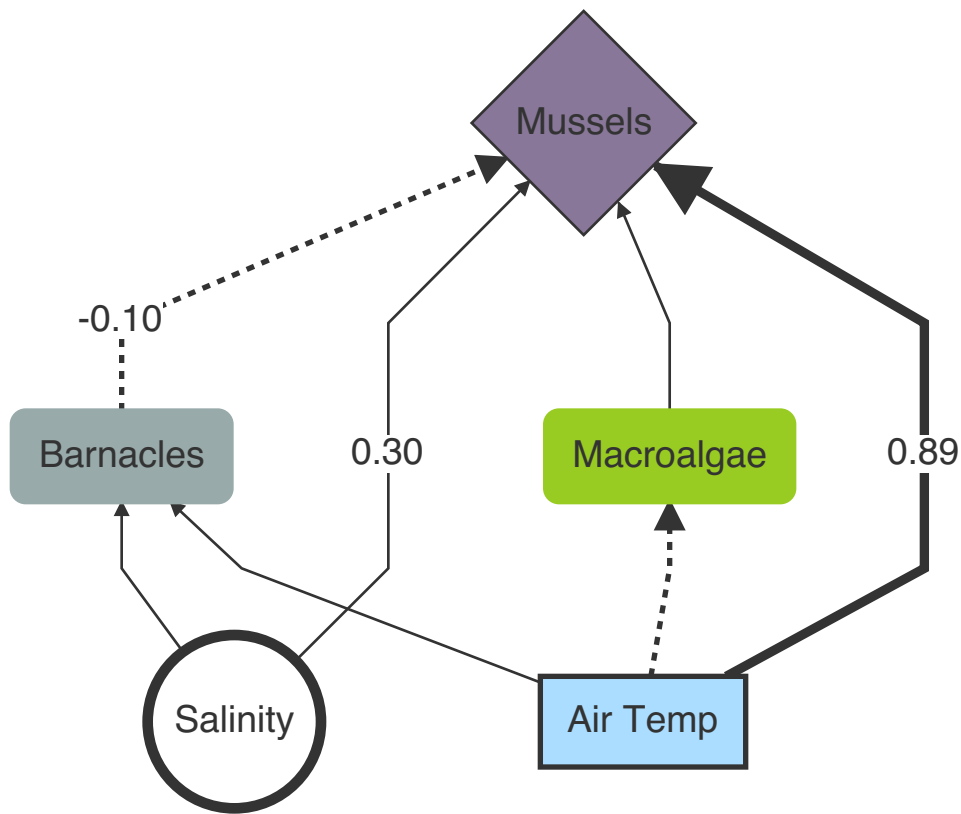
```

mermaid("
  graph BT
    A((Salinity))
    A-->B[Barnacles]
    B-.->| -0.10 | B1[Mussels]
    A-- 0.30 -->B1

    C[Air Temp]
    C-->B
    C-.->E[Macroalgae]
    E-->B1
    C== 0.89 ==>B1

    style A fill:#FFF, stroke:#333, stroke-width:4px
    style B fill:#9AA, stroke:#9AA, stroke-width:2px
    style B1 fill:#879, stroke:#333, stroke-width:1px
    style C fill:#ADF, stroke:#333, stroke-width:2px
    style E fill:#9C2, stroke:#9C2, stroke-width:2px
")

```

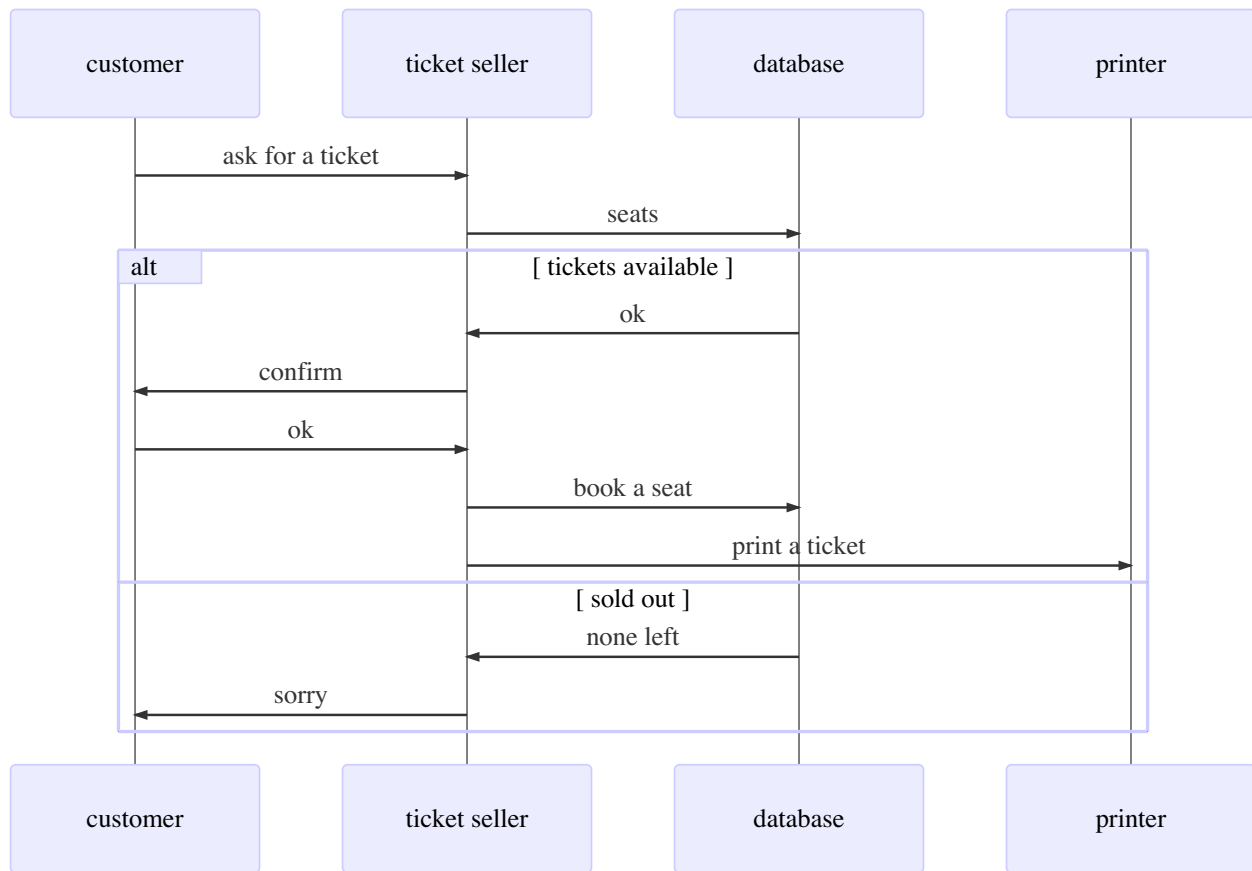


Example sequenceDiagram

```

DiagrammeR("
sequenceDiagram;
customer->>ticket seller: ask for a ticket;
ticket seller->>database: seats;
alt tickets available
database->>ticket seller: ok;
ticket seller->>customer: confirm;
customer->>ticket seller: ok;
ticket seller->>database: book a seat;
ticket seller->>printer: print a ticket;
else sold out
database->>ticket seller: none left;
ticket seller->>customer: sorry;
end
")

```



Gantt Charts for Timelines

Style Gantt code chunk

- supposed to widen Gantt chart, but doesn't

Gantt Chart ER390

```

mermaid(
gantt
dateFormat YYYY-MM-DD
title ER390 HAT Data Viz Project Timeline

section ER390 Project
  Nancy suggested Topic      :crit, done, first_7, 2025-07-22, 2025-07-23
  Talked to Vanessa at HAT   :crit, done, first_8, 2025-08-27, 2025-08-28
  Writing Proposal           :crit, active, first_9, 2025-08-27, 5d
  Start Project              :active, first_10, 2025-09-03, 356d

section Important Things
  Applied For RNS Program     :crit, done, 2025-08-21, 24h
  Created Gantt Chart        :active, 2025-08-31, 24h

section Basic Tasks
  Creating RMarkdown code     :done, a1, 2025-08-04, 2025-08-31
  Writing Proposal            :active, after a1, 2025-08-31, 3d
  
```

```

Email Nancy-RNS & Vanessa-HAT :active, 2025-09-02

section Shiny App
  Created Shiny App           :done, first_1, 2025-08-18, 2025-08-28
  Refining Shiny App          :active, first_5, 2025-08-18, 130d
  Interactive Data Table      :done, 2025-08-18, 2025-08-19

section RMarkdown
  Created Test Document       :done, first_6, 2025-08-04, 2025-08-05

section Shiny Clone
  Create Shiny Clone          :active, first_11, 2025-09-30, 150d

section Data Collection
  Collecting Example Data     :done, first_2, 2025-08-04, 10d
  Finish Collecting Data      : first_3, after first_2, 5d
  Do this after that          : first_4, after first_3, 5d

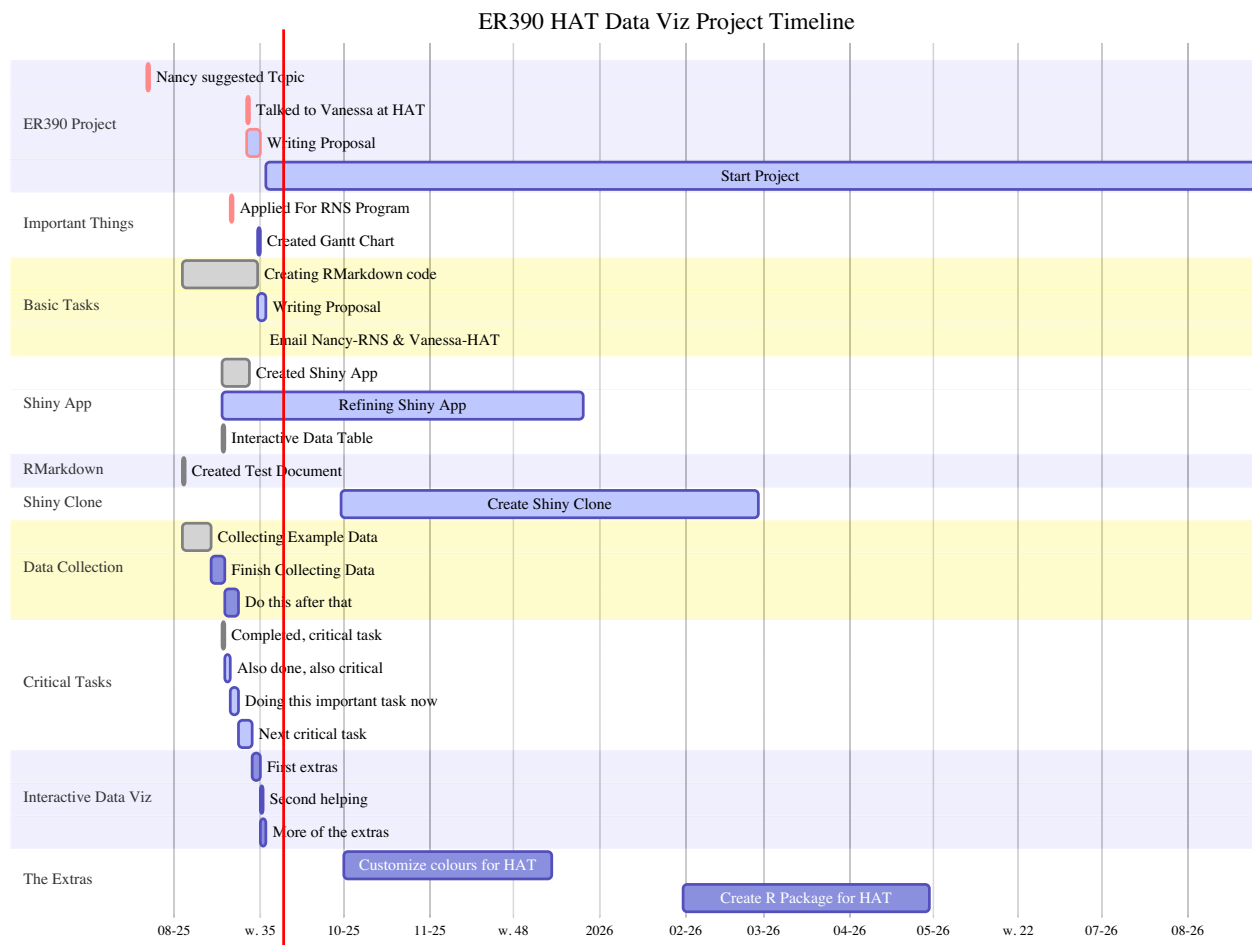
section Critical Tasks
  Completed, critical task    :done, done, import_1, 2025-08-18, 24h
  Also done, also critical    :active, done, import_2, after import_1, 2d
  Doing this important task now :active, active, import_3, after import_2, 3d
  Next critical task          :active, import_4, after import_3, 5d

section Interactive Data Viz
  First extras                : extras_1, after import_4, 3d
  Second helping              : extras_2, after extras_1, 20h
  More of the extras          : extras_3, after extras_1, 48h

section The Extras
  Customize colours for HAT   : 2025-10-01, 2025-12-15
  Create R Package for HAT    : 2026-01-31, 2026-04-30

")

```

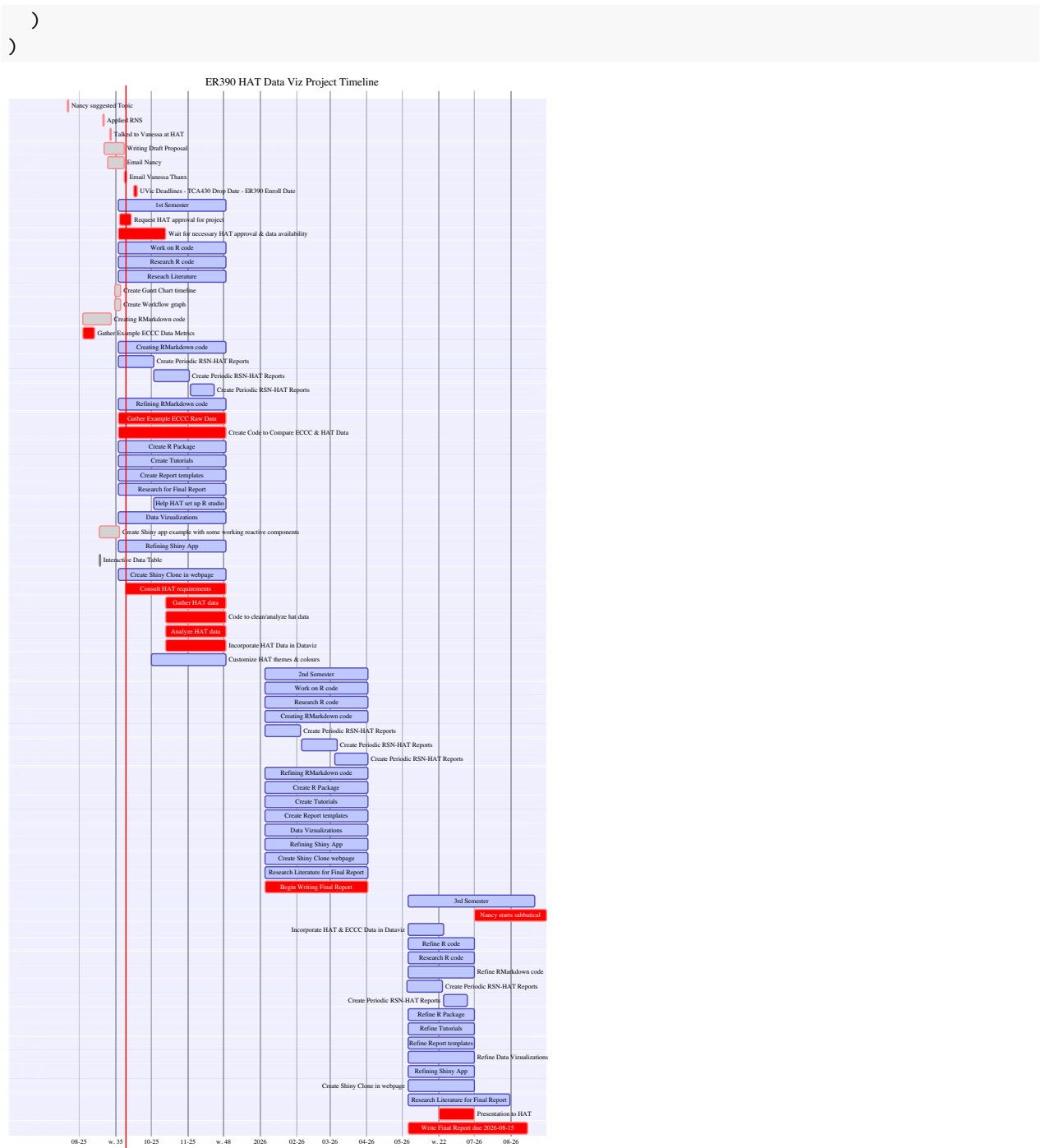


Gantt ER390 dataframe

```
ER390_gantt <- read.csv("data/ER390-Gantt-Tasks.csv", header = TRUE, sep = ",")

library(tidyr)
library(dplyr)
library(Diagrammer)

mermaid(
  paste0(
    # mermaid "header", each component separated with "\n" (line break)
    "gantt", "\n",
    "dateFormat YYYY-MM-DD", "\n",
    "title ER390 HAT Data Viz Project Timeline", "\n",
    # unite the first two columns (task & status) and separate them with ":"
    # then, unite the other columns and separate them with ","
    # this will create the required mermaid "body"
    paste(ER390_gantt %>%
      unite(i, task, status, sep = ":") %>%
      unite(j, i, pos, start, end, sep = ",") %>%
      .$j,
      collapse = "\n"
    ), "\n"
```



Gantt ggplot 1

```
library(reshape2)
```

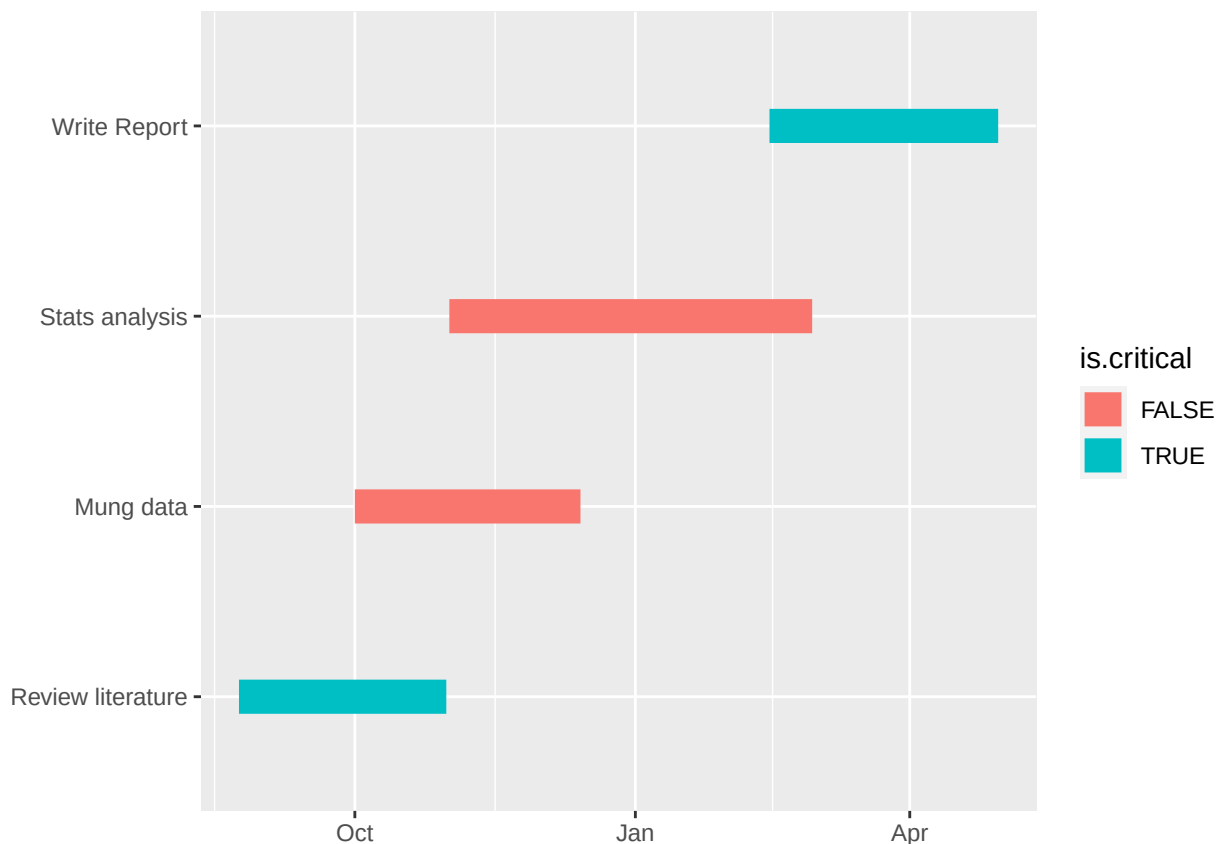
```
##
## Attaching package: 'reshape2'

## The following object is masked from 'package:tidyr':
##
```

```
## smiths
library(ggplot2)

tasks <- c("Review literature", "Mung data", "Stats analysis", "Write Report")
dfr <- data.frame(
  name = factor(tasks, levels = tasks),
  start.date = as.Date(c("2010-08-24", "2010-10-01", "2010-11-01", "2011-02-14")),
  end.date = as.Date(c("2010-10-31", "2010-12-14", "2011-02-28", "2011-04-30")),
  is.critical = c(TRUE, FALSE, FALSE, TRUE)
)
mdfr <- melt(dfr, measure.vars = c("start.date", "end.date"))

gantt_ggplot <- ggplot(mdfr, aes(value, name, colour = is.critical)) +
  # Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
  geom_line(linewidth = 6) +
  xlab(NULL) +
  ylab(NULL)
gantt_ggplot
```



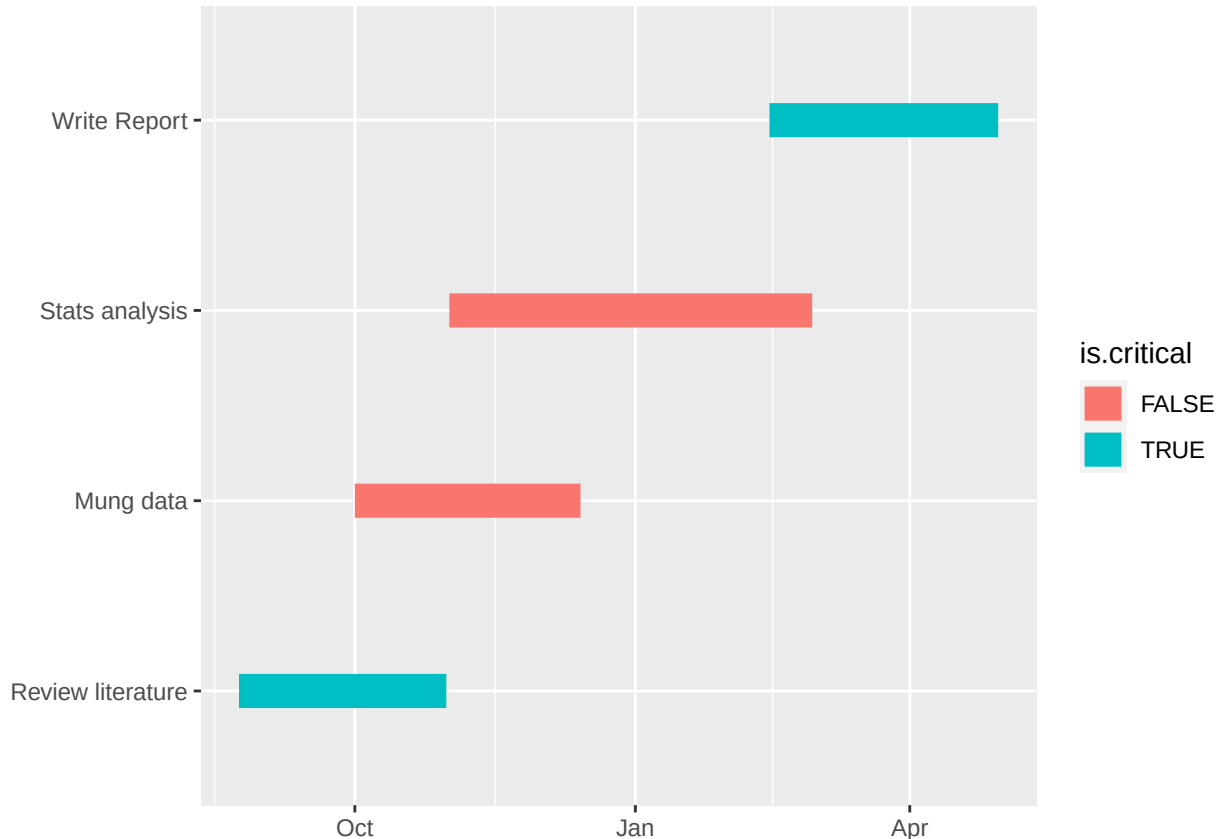
<https://stackoverflow.com/questions/3550341/gantt-charts-with-r>

```
tasks <- c("Review literature", "Mung data", "Stats analysis", "Write Report")
dfr <- data.frame(
  name = factor(tasks, levels = tasks),
  start.date = as.Date(c("2010-08-24", "2010-10-01", "2010-11-01", "2011-02-14")),
  end.date = as.Date(c("2010-10-31", "2010-12-14", "2011-02-28", "2011-04-30")),
  is.critical = c(TRUE, FALSE, FALSE, TRUE)
)
```

```
)

ggplot(dfr, aes(x =start.date, xend= end.date, y=name, yend = name, color=is.critical)) +
  geom_segment(size = 6) +
  xlab(NULL) + ylab(NULL)
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
```



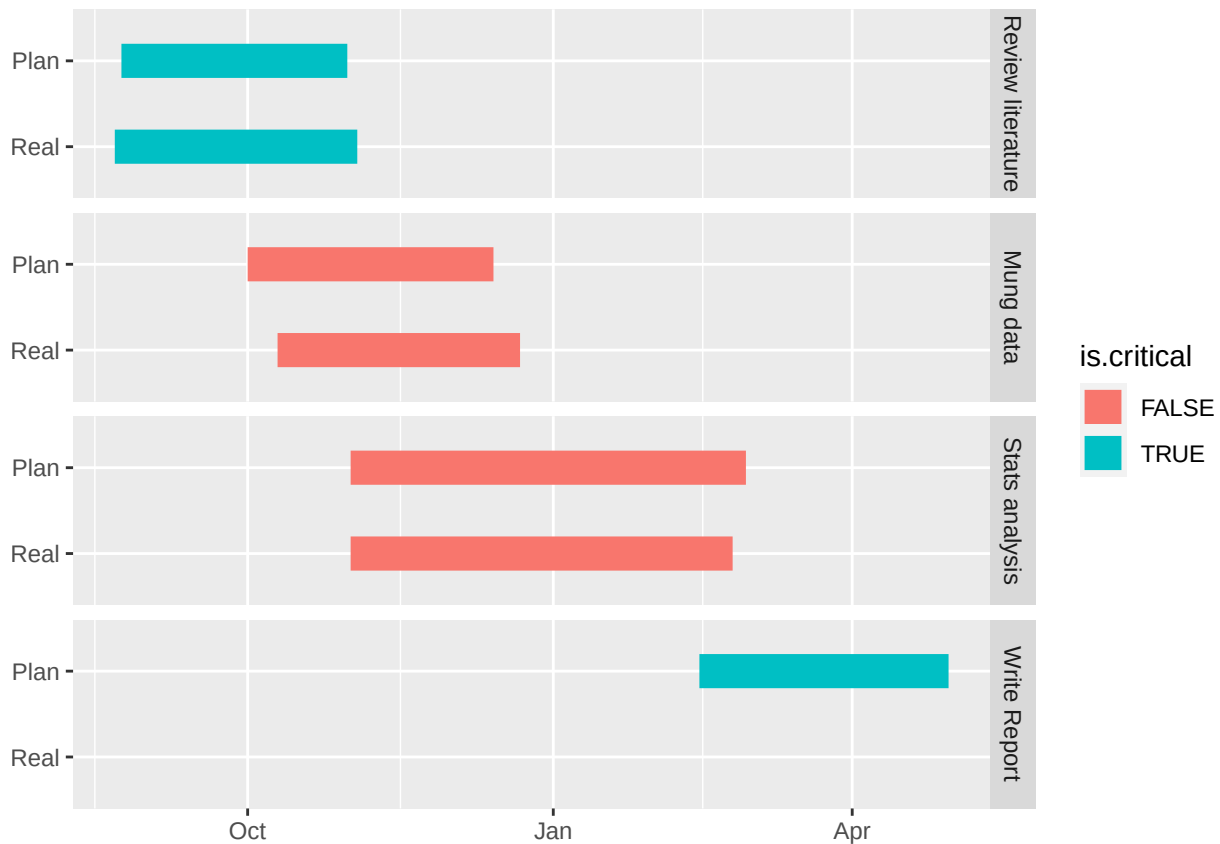
Gantt ggplot 2

```
library(reshape2)
tasks <- c("Review literature", "Mung data", "Stats analysis", "Write Report")
dfrP <- data.frame(
  name      = factor(tasks, levels = tasks),
  start.date = as.Date(c("2010-08-24", "2010-10-01", "2010-11-01", "2011-02-14")),
  end.date   = as.Date(c("2010-10-31", "2010-12-14", "2011-02-28", "2011-04-30")),
  is.critical = c(TRUE, FALSE, FALSE, TRUE)
)
dfrR <- data.frame(
  name      = factor(tasks, levels = tasks),
  start.date = as.Date(c("2010-08-22", "2010-10-10", "2010-11-01", NA)),
  end.date   = as.Date(c("2010-11-03", "2010-12-22", "2011-02-24", NA)),
  is.critical = c(TRUE, FALSE, FALSE, TRUE)
)
mdfr <- merge(data.frame(type="Plan", melt(dfrP, measure.vars = c("start.date", "end.date"))),
  data.frame(type="Real", melt(dfrR, measure.vars = c("start.date", "end.date"))), all=T)
```



```
library(ggplot2)
ggplot(mdf, aes(x=value, y=type, color=is.critical))+
  geom_line(linewidth=6) +
  facet_grid(name ~ .) +
  scale_y_discrete(limits=c("Real", "Plan")) +
  xlab(NULL) + ylab(NULL)
```

```
## Warning: Removed 2 rows containing missing values (`geom_line()`).
```



Gantt timevis interactive

```
# https://stackoverflow.com/questions/3550341/gantt-charts-with-r
library(timevis)
```

```
##
## Attaching package: 'timevis'
## The following object is masked from 'package:shiny':
##
## runExample
data <- data.frame(
  id      = 1:4,
  content = c("Item one" , "Item two" ,"Ranged item", "Item four"),
  start   = c("2016-01-10", "2016-01-11", "2016-01-20", "2016-02-14 15:00:00"),
  end     = c(NA           ,          NA, "2016-02-04", NA)
)
```

```
timevis(data)
```

Item four	+ -
Ranged item	
2016	

Gantt Proposed Timeline

- red coloured items are considered critical
- grey items are completed
- Note: Gantt Chart coded with DiagrammeR using imported csv file of data, but no colour or size

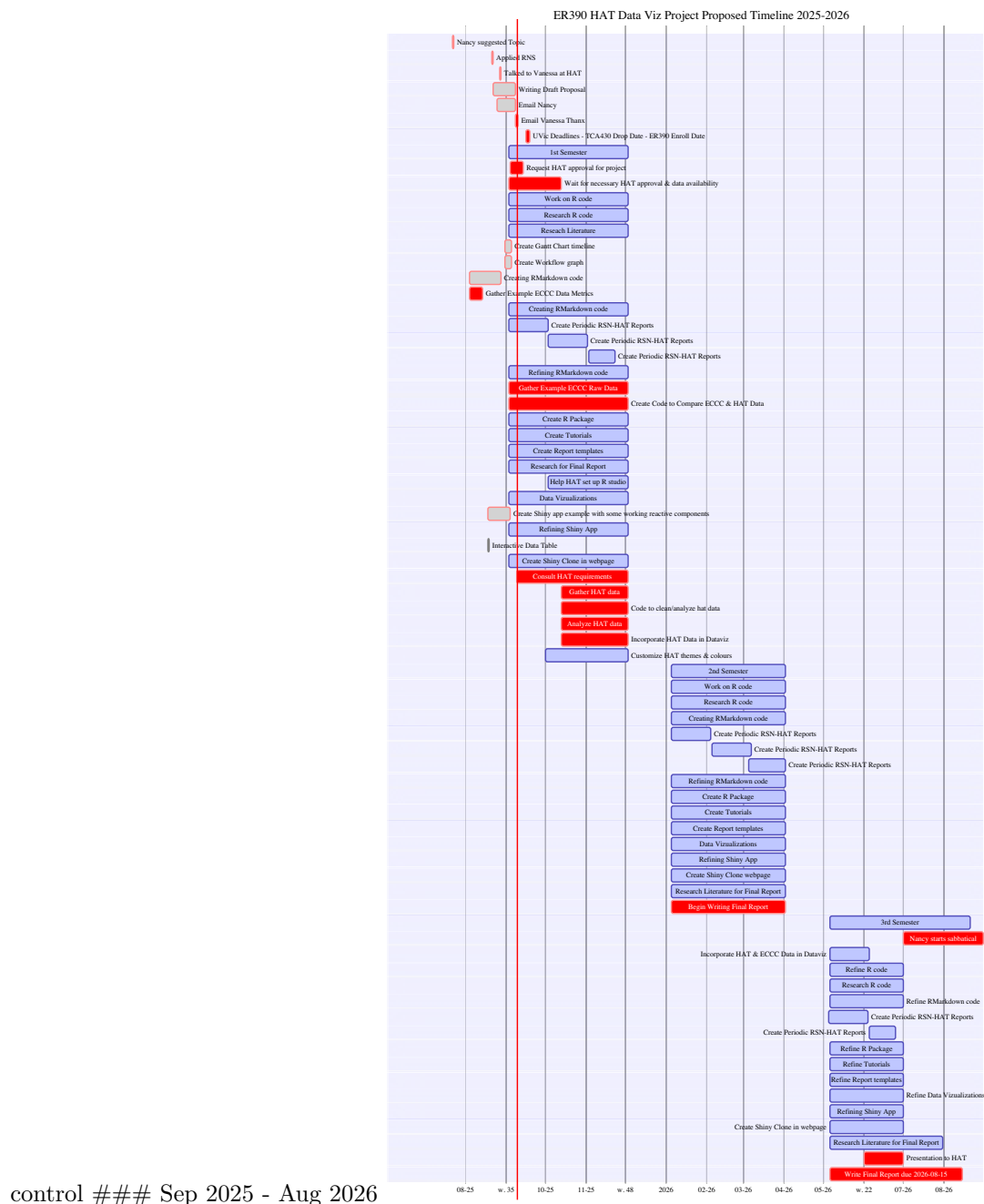


Figure 2: Gantt Proposed ER390 Timeline, Sep 2025-Jul 2026

Sep-Dec, 2025

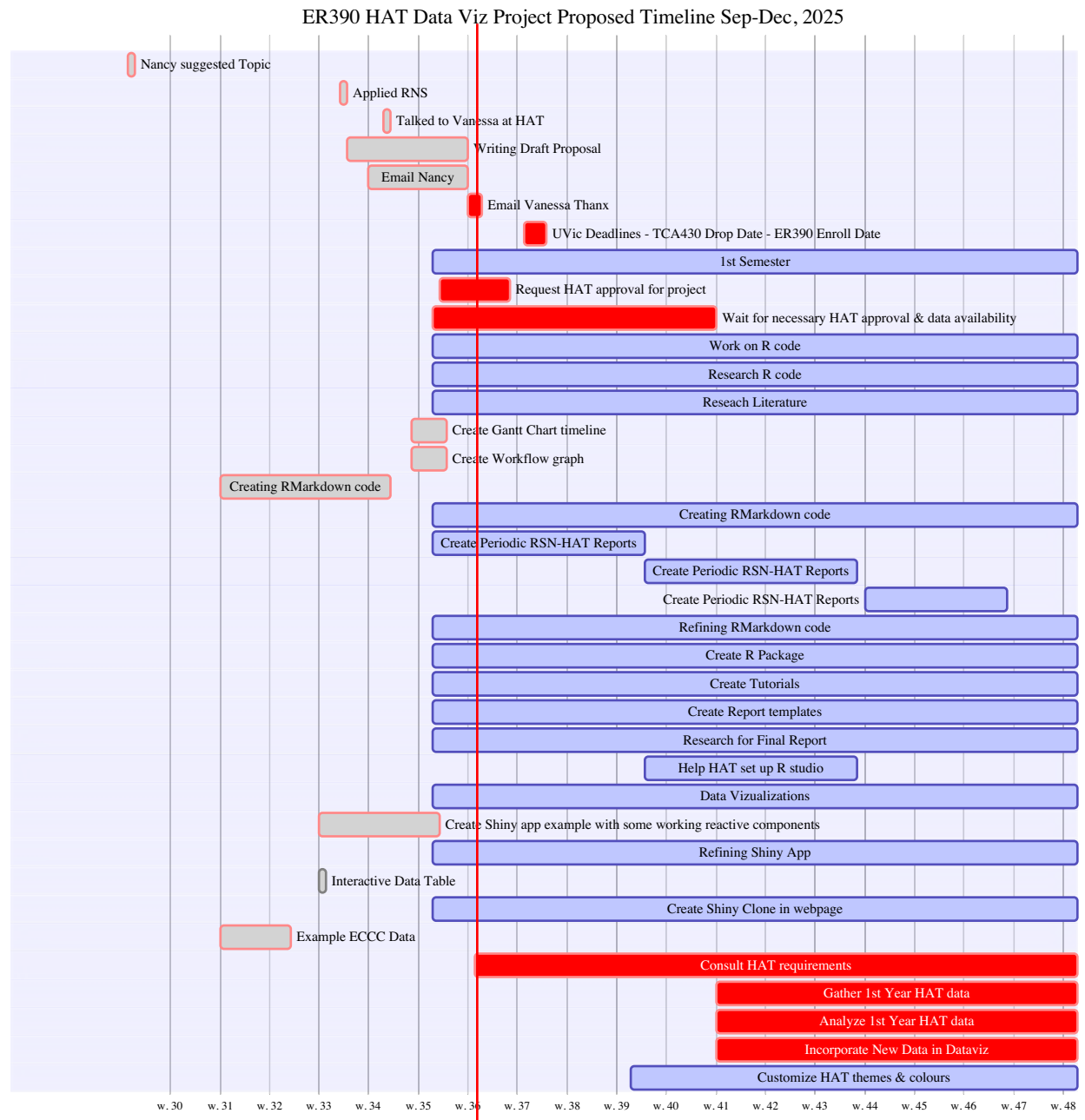


Figure 2.1: Gantt Proposed ER390 Timeline Sep-Dec, 2025

Jan-Apr 2026

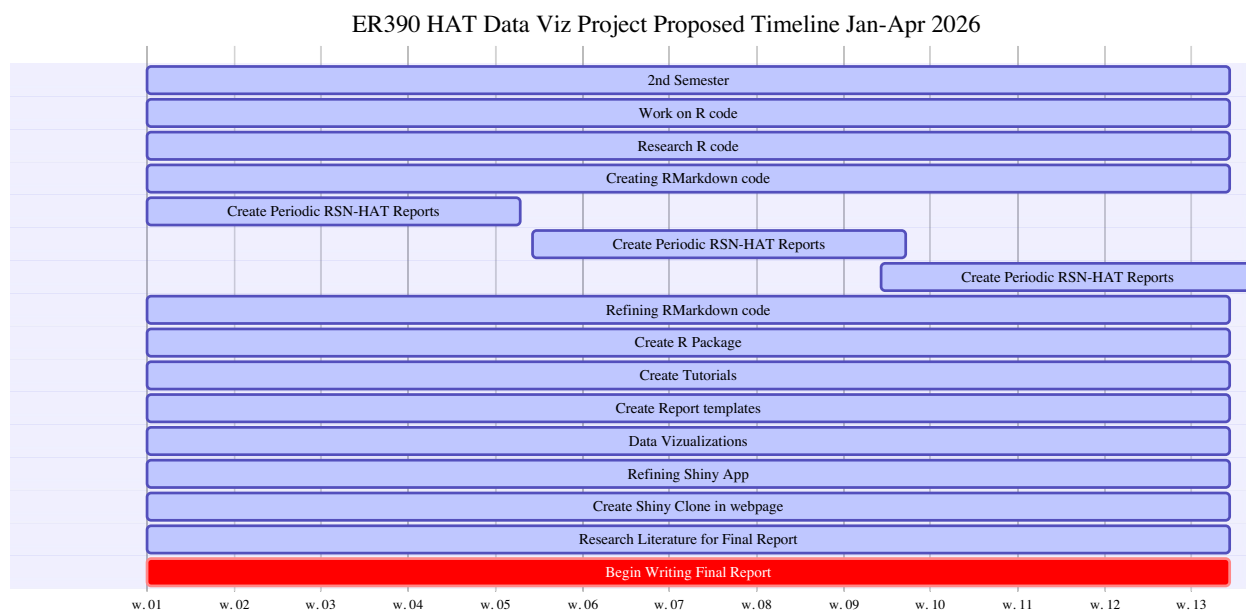


Figure 2.2: Gantt Proposed ER390 Timeline, Jan-Apr 2026

May-Jul, 2026

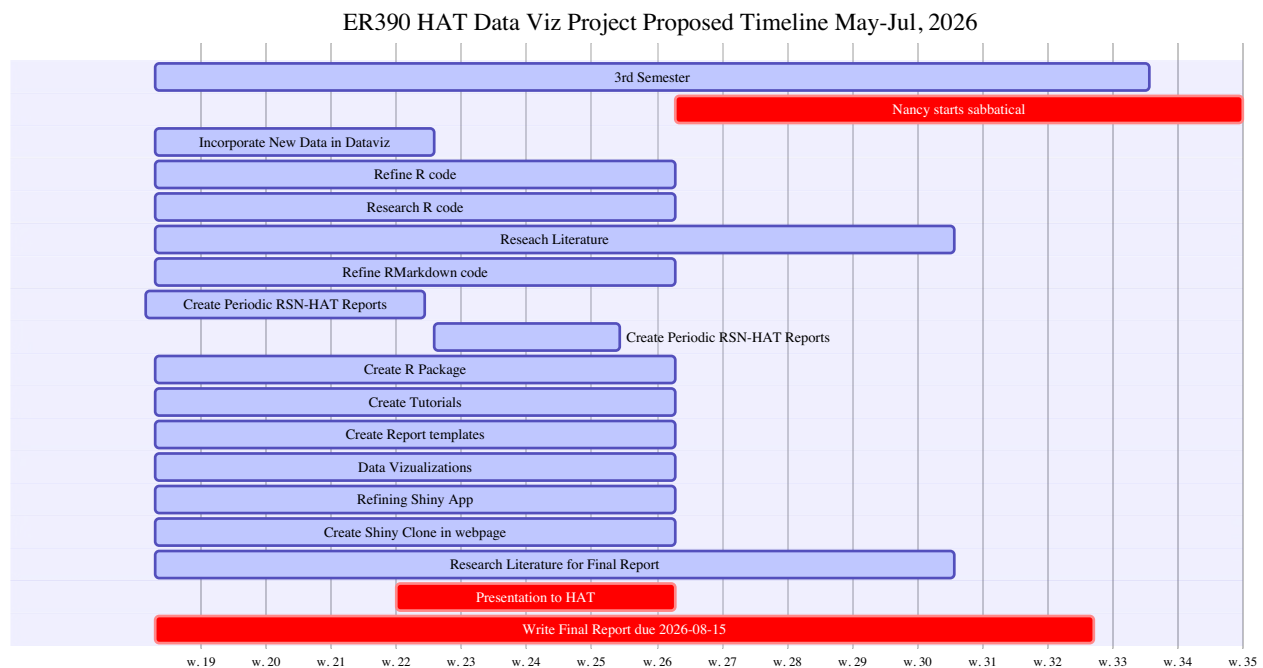


Figure 2.3: Gantt Proposed ER390 Timeline May-Jul, 2026

Images Include when knitting



Descriptive Summary Stats

```
summary(goe_eccc)
```

## Subregion	Site	Proportion_of_native_species
## Length:24	Length:24	Min. :0.3200

```
## Class :character    Class :character    1st Qu.:0.4475
## Mode  :character    Mode  :character    Median :0.5450
##                                         Mean  :0.5504
##                                         3rd Qu.:0.6425
##                                         Max.   :0.9400
## Cultural_species_richness Exotic_species    Trampling    Herbivory
## Min.      :0.420          Min.       : 0.17    Min.       : 1.640    Min.       : 0.000
## 1st Qu.:1.242          1st Qu.:22.38    1st Qu.: 4.950    1st Qu.: 0.425
## Median :1.730          Median :32.26    Median : 8.700    Median : 1.300
## Mean    :2.042          Mean    :37.67    Mean     : 9.558    Mean     : 8.975
## 3rd Qu.:2.732          3rd Qu.:55.81    3rd Qu.:11.300    3rd Qu.: 8.375
## Max.    :4.400          Max.     :83.16    Max.     :32.330    Max.     :70.800
## Composite_Index      Year
## Min.      :0.000    Min.      :2023
## 1st Qu.:0.600    1st Qu.:2023
## Median :1.000    Median :2023
## Mean    :1.008    Mean    :2023
## 3rd Qu.:1.450    3rd Qu.:2023
## Max.    :2.000    Max.    :2023
```

Shiny

Shiny Gadget

lmGadget all sites

- code works but can't knit html file unless adding "echo=TRUE, eval = FALSE"

Inline Shiny app

Shiny Showcase

app-WORKS-TabsetLayout-choose-Reactive-Table-Plot-BoxNames-comments-showcase.R

```
##
## Listening on http://127.0.0.1:7724
```

Exploring GOE Monitoring Dataviz

Choose a Subregion

All ▼

Choose a Site

All ▼

Using App: Choose all or one Subregion and/or Site, then click tab to view results:
Table, Comparing Subregions, and Comparing Sites

About the Data: GOE Data from report tables: Malloff, J., & Shackelford, N. (2024). Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems. Restoration Futures Lab at the University of Victoria.

About This App

Data Table

Comparing Subregions

Comparing Sites

Proposed Interactive Data Viz Tool

for HAT's GOE Monitoring Data

Here is what I propose to offer ...

I. Final Products

° Shiny Interactive Dataviz Tool, RMarkdown Code Document that cleans data, and creates dataviz

II. Data Preparation

° Clean data, ensure consistent naming and data formats, saving outputs, storing data

III. Interactive Data Tables

° Filter by Subregion Group or Site; Search; Save Results

IV. Plots

° Dot plots, bar charts, violin plots, Individual Sites, Subregion Group, save results

V. Maps

° With markers sized equally, or by a variable value, Interactive Webpage Leaflet Map

Reactive choose

```
## [1] "Subregion"          "Site"
## [3] "Proportion_of_native_species" "Cultural_species_richness"
## [5] "Exotic_species"      "Trampling"
## [7] "Herbivory"           "Composite_Index"
## [9] "Year"

## 'data.frame':  24 obs. of  9 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Gulf Islands" "Gulf Islands" ..
## $ Site           : chr  "Anniversary Island" "AVNR" "Brackman Island" "Crows Nest" ...
## $ Proportion_of_native_species: num  0.94 0.55 0.77 0.66 0.75 0.53 0.45 0.65 0.62 0.58 ...
## $ Cultural_species_richness   : num  2.71 1.3 2.8 1.07 3.6 0.8 0.42 2 3.6 2.33 ...
## $ Exotic_species             : num  0.17 57.19 11.17 31.51 8.8 ...
## $ Trampling                  : num  5 4.7 5.4 1.64 12.2 ...
## $ Herbivory                  : num  1.14 36.5 0 26.71 0.1 ...
## $ Composite_Index            : num  1.8 0.6 2 1 1.6 0.8 0.2 1.4 1 1 ...
## $ Year                        : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...

##
## Listening on http://127.0.0.1:5662
```

Choose a Subregion

All ▼

Choose an Site

All ▼

Subregion	Site	Proportion_of_native_species	Cultural_species_richness	Exotic_species	Trampling	Herbivory	Composite_Index	Year
Gulf Islands	Anniversary Island	0.94	2.71	0.17	5.00	1.14	1.80	2023
Gulf Islands	AVNR	0.55	1.30	57.19	4.70	36.50	0.60	2023
Gulf Islands	Brackman Island	0.77	2.80	11.17	5.40	0.00	2.00	2023
Gulf Islands	Crows Nest	0.66	1.07	31.51	1.64	26.71	1.00	2023
Gulf Islands	Dock Islet N	0.75	3.60	8.80	12.20	0.10	1.60	2023
Gulf Islands	Eagle Islet	0.53	0.80	28.25	16.60	0.00	0.80	2023
Gulf Islands	Mt Maxwell	0.45	0.42	64.30	10.00	6.00	0.20	2023
Gulf Islands	Owl Islet	0.65	2.00	27.72	10.67	0.67	1.40	2023
Gulf Islands	Reay Islet	0.62	3.60	11.51	11.00	8.00	1.00	2023
Gulf Islands	Retreat Island	0.58	2.33	33.00	32.33	0.00	1.00	2023
Gulf Islands	Rum Islet	0.45	1.00	73.21	22.20	70.80	0.00	2023
Gulf Islands	SISCENEM	0.67	3.40	16.12	12.80	0.50	1.60	2023
Saanich Peninsula	Bear Hill Park	0.62	1.86	24.46	7.57	9.50	1.00	2023
Saanich Peninsula	Camas Hill	0.61	2.43	27.19	7.71	4.71	1.40	2023
Saanich Peninsula	Gonzales Hill Park	0.32	1.00	83.16	10.40	1.20	0.40	2023
Saanich Peninsula	Gore Park	0.64	4.40	7.35	6.20	1.60	1.60	2023
Saanich Peninsula	Highrock Park	0.49	1.60	37.10	8.20	1.20	1.00	2023
Saanich Peninsula	Mill Hill Park	0.41	1.56	46.33	14.33	25.67	0.40	2023
Saanich Peninsula	Oak Haven Park	0.47	1.50	34.60	1.88	12.75	1.00	2023
Saanich Peninsula	Scafe Hill	0.34	1.40	55.35	10.20	1.40	0.40	2023
Saanich Peninsula	Seymour Hill	0.34	2.20	65.99	9.20	0.20	0.80	2023
Saanich Peninsula	Stewart Hill	0.44	1.00	79.56	4.80	0.80	0.60	2023
Saanich Peninsula	Trial Island	0.54	3.57	30.73	2.00	0.14	1.60	2023
Saanich Peninsula	Uplands Park	0.37	1.45	49.38	2.36	5.82	1.00	2023

Leaflet Reactive Site Details

app-leaflet-reactive-AllSiteDetails-WORKING

```
##
## Attaching package: 'bslib'

## The following object is masked from 'package:utils':
##
##      page

## [1] "Subregion"          "Site"
## [3] "Land_Manager"       "Main_Restoration_Type"
## [5] "Restoration_Intensity" "Area_ha"
## [7] "Lat"                "Lng"
## [9] "Proportion_of_native_species" "Cultural_species_richness"
## [11] "Exotic_species"      "Trampling"
## [13] "Herbivory"           "Composite_Index"
## [15] "Year"                "Percentage_ns"

## 'data.frame':   24 obs. of  16 variables:
## $ Subregion      : chr  "Gulf Islands" "Gulf Islands" "Saanich Peninsula" "Gulf Islands" ...
## $ Site           : chr  "Anniversary Island" "AVNR" "Bear Hill Park" "Brackman Island" ...
## $ Land_Manager    : chr  "GINPR" "Saltspring conservancy" "CRD" "GINPR" ...
## $ Main_Restoration_Type : chr  "herbivore reduction" "herbivore reduction" "invasive removal" ...
## $ Restoration_Intensity : chr  "high" "minimal" "low" "high" ...
## $ Area_ha        : num  4.39 20.54 3.8 4.41 10.1 ...
## $ Lat            : num  48.8 48.8 48.5 48.7 48.4 ...
## $ Lng            : num  -123 -123 -123 -123 -124 ...
## $ Proportion_of_native_species: num  0.94 0.55 0.62 0.77 0.61 0.66 0.75 0.53 0.32 0.64 ...
## $ Cultural_species_richness : num  2.71 1.3 1.86 2.8 2.43 1.07 3.6 0.8 1 4.4 ...
## $ Exotic_species : num  0.17 57.19 24.46 11.17 27.19 ...
## $ Trampling      : num  5 4.7 7.57 5.4 7.71 1.64 12.2 16.6 10.4 6.2 ...
## $ Herbivory      : num  1.14 36.5 9.5 0 4.71 ...
## $ Composite_Index : num  1.8 0.6 1 2 1.4 1 1.6 0.8 0.4 1.6 ...
## $ Year           : int  2023 2023 2023 2023 2023 2023 2023 2023 2023 2023 ...
## $ Percentage_ns  : int  94 55 62 77 61 66 75 53 32 64 ...

##
## Listening on http://127.0.0.1:6168
```

ECCC GOE Monitoring Sites Interactive Map

Source: Malloff & Shackelford. (2024). *Feeling the Pulse: Monitoring methods and initial outcomes in oak meadow ecosystems*. Restoration Futures Lab at the University of Victoria.

Note: Marker size is based on Composite Index.

Select Subregion:

All ▼

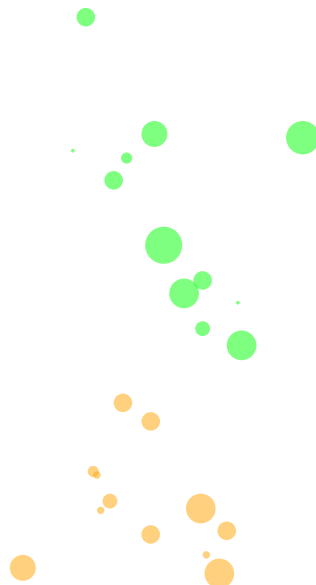
Note: If selecting by Subregion, ensure Sites are All selected.

Select Site:

All ▼

+

-



Datatable search plot

app-datable-search-plot-update-ECCC-TEST <https://stackoverflow.com/questions/52880657/in-an-shiny-app-i-want-a-plot-to-update-based-on-the-search-results-in-a-datat>

```
##
## Attaching package: 'DT'

## The following objects are masked from 'package:shiny':
##
##   dataTableOutput, renderDataTable

##
## Listening on http://127.0.0.1:5941
```

ECCC GOE Monitoring Data

Subregion	Site	Land_Manager	Main_Restoration_Type	Restoration_Intensity	Area_ha	Lat	Lng	Proportion_of_native_species	Cultural_species_richness	Exotic_species	Trampling	Herbivory	Composite_Index	Year	Percentage_ns
All		All	All					All	All	All	A	#	All		All

Download filtered table data ()

Plot Select Axis Variables app4-SelectAxisVariables-WORKS.R - this shiny app won't knit - loses connection ???

```
##
## Attaching package: 'shinyscreenshot'

## The following object is masked from 'package:timevis':
##
##   runExample

## The following object is masked from 'package:shiny':
##
##   runExample

##
## Listening on http://127.0.0.1:7410

## [1] "Exotic_species"

## Warning in file(file, "rt"): cannot open file 'data/TEST-GOE-Monitor.csv': No
## such file or directory

## Warning: Error in file: cannot open the connection

## [1] "Proportion_of_native_species"

## Warning in file(file, "rt"): cannot open file 'data/TEST-GOE-Monitor.csv': No
## such file or directory

## Warning in file(file, "rt"): Error in file: cannot open the connection
```

Plotting ECCC GOE Monitoring Variables

Input sidebar panel selections

Select x-axis

Exotic_species ▼

Select y-axis

Proportion of native species ▼

Take Screenshots
(Note: Screenshots are saved to same location as Shiny app)

📷 Entire page

📷 Input panel

📷 Plot 1

📷 Plot 2

Main panel Input Selections

X axis:

Exotic_species

Y axis:

Proportion_of_native_species

Plot1: Resulting ggplot by Site

Error: cannot open the connection

Plot2: Resulting ggplot by Subregion

Error: cannot open the connection

TO DO

- leaflet map jumps when clicking popup (in RStudio, not in html file)

DONE

to combine dataframes csv files to include site info

1. make sure they each have same length number of unique rows
2. arrange alphabetic order
3. cbind
4. remove duplicate columns

Session info

```
# to document specific packages used to run script
sessionInfo()

## R version 4.2.1 (2022-06-23)
## Platform: x86_64-apple-darwin17.0 (64-bit)
## Running under: macOS Mojave 10.14.6
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.2/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.2/Resources/lib/libRlapack.dylib
##
## locale:
## [1] en_CA.UTF-8/en_CA.UTF-8/en_CA.UTF-8/C/en_CA.UTF-8/en_CA.UTF-8
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] shinysscreenshot_0.2.0 DT_0.26          bslib_0.4.0
## [4] timevis_2.1.0          reshape2_1.4.4    htmlwidgets_1.5.4
## [7] leaflet_2.1.1          cowplot_1.1.1     DiagrammeR_1.0.9
## [10] plotly_4.10.1          forcats_0.5.2     stringr_1.4.1
## [13] dplyr_1.0.10           purrr_0.3.5       readr_2.1.3
## [16] tidyr_1.2.1            tibble_3.1.8      ggplot2_3.4.0
## [19] tidyverse_1.3.2        shiny_1.7.3
##
## loaded via a namespace (and not attached):
## [1] fs_1.5.2                fontawesome_0.4.0  lubridate_1.8.0
## [4] webshot_0.5.4           RColorBrewer_1.1-3 httr_1.4.4
## [7] tools_4.2.1            backports_1.4.1    utf8_1.2.2
## [10] R6_2.5.1               DBI_1.1.3          lazyeval_0.2.2
## [13] colorspace_2.0-3        withr_2.5.0        tidyselect_1.2.0
## [16] processx_3.7.0          compiler_4.2.1     textshaping_0.3.6
## [19] cli_3.4.1              rvest_1.0.3        xml2_1.3.3
## [22] sass_0.4.2             labeling_0.4.2     scales_1.2.1
## [25] callr_3.7.2            systemfonts_1.0.4  digest_0.6.30
```

## [28] rmarkdown_2.17	pkgconfig_2.0.3	htmltools_0.5.3
## [31] dbplyr_2.2.1	fastmap_1.1.0	highr_0.9
## [34] rlang_1.0.6	readxl_1.4.1	rstudioapi_0.14
## [37] jquerylib_0.1.4	visNetwork_2.1.2	farver_2.1.1
## [40] generics_0.1.3	jsonlite_1.8.2	crosstalk_1.2.0
## [43] googlesheets4_1.0.1	magrittr_2.0.3	Rcpp_1.0.9
## [46] munsell_0.5.0	fansi_1.0.3	lifecycle_1.0.3
## [49] stringi_1.7.8	yaml_2.3.5	plyr_1.8.7
## [52] grid_4.2.1	promises_1.2.0.1	crayon_1.5.2
## [55] haven_2.5.1	hms_1.1.2	knitr_1.40
## [58] ps_1.7.1	pillar_1.8.1	uuid_1.1-0
## [61] markdown_1.1	reprex_2.0.2	glue_1.6.2
## [64] evaluate_0.17	leaflet.providers_1.9.0	data.table_1.14.2
## [67] modelr_0.1.9	png_0.1-7	vctrs_0.5.0
## [70] tzdb_0.3.0	httpuv_1.6.6	cellranger_1.1.0
## [73] gtable_0.3.1	assertthat_0.2.1	cachem_1.0.6
## [76] xfun_0.37	mime_0.12	xtable_1.8-4
## [79] broom_1.0.1	later_1.3.0	ragg_1.2.3
## [82] googledrive_2.0.0	viridisLite_0.4.1	gargle_1.2.1
## [85] memoise_2.0.1	ellipsis_0.3.2	